

Smart City Surveillance in Africa: Mapping Chinese AI Surveillance Across 11 Countries

March 2026

Wairagala Wakabi and Tony Roberts

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Executive summary

A massive expansion of digital surveillance is under way across the continent of Africa, which violates the fundamental human right of citizens to privacy.

This report provides the most comprehensive account to date of smart city surveillance in Africa. Expert researchers draw on their contextual experience of their own countries in detailed reports on Algeria, Egypt, Kenya, Mauritius, Mozambique, Rwanda, Senegal, Uganda, Zambia, and Zimbabwe.

African governments are spending billions of dollars on 'smart' surveillance of public space. We document over US\$2 billion expenditure on facial recognition and car tracking technologies in 11 countries. The real total is certainly higher because (1) surveillance spending is often secret; (2) no figures were available for two of the 11 countries studies; (3) the public accounts for the other nine countries were incomplete; and (4) this study included only 11 of Africa's 55 countries.

Nigeria alone has spent over US\$470 million on AI-enabled facial recognition and automatic car number plate recognition (ANPR), making it Africa's largest buyer of smart city surveillance technologies.

China is by far the largest supplier of AI surveillance technologies to African governments.

However, companies from many other countries are actively competing for the same contracts.

The Chinese safe city surveillance package is typically financed by soft loans from Chinese banks.

A typical package involves a loan of US\$250 million from Eximbank tied to the purchase of surveillance cameras from Hikvision and a command and control centre built and serviced by Huawei or ZTE.

The Chinese safe city package typically includes installing thousands of smart closed-circuit television (CCTV) surveillance cameras, which transmit geo-located facial recognition and car number plate data in real time for analysis using artificial intelligence at dedicated data centres that serve as command and control facilities for police and security operatives.

China supplied smart city surveillance technologies to all 11 countries studied.

South Korea and Russia supplied smart city surveillance technologies to three of the countries studied and the United Arab Emirates supplied two. The USA, Israel, France, Spain, Iran, and Japan each supplied smart city surveillance technologies to one country, as illustrated in Table 1.

Table 1: Countries that supplied smart city surveillance technology to the countries studied

Supplier	China	Korea	Russia	UAE	Israel	France	Spain	USA	Japan	Iran
Algeria	X		X			X				
Egypt	X	X					X	X		
Kenya	X								X	
Mauritius	X			X	X					
Mozambique	X	X								
Nigeria	X									
Rwanda	X	X		X						
Senegal	X									
Uganda	X	X								
Zambia	X									
Zimbabwe	X	X								X

Source: Authors' own, based on the 11 country reports in this volume.

Digital surveillance of terrorists and the most serious criminals is justified in the public interest, but the mass surveillance of all citizens – suspected of no crime – violates human rights. However, our research shows that the rapid growth of digital surveillance is occurring in the absence of adequate legal regulation or oversight.

The rapid expansion of smart city surveillance is having a 'chilling effect' causing citizens and journalists to self-censor, reducing effective freedoms to speak truth to power and hold governments accountable, and inhibiting freedom of movement, association, and affiliation.

The scale and invasive nature of mass surveillance is not 'legal, necessary or proportionate' to the legitimate aim of providing security. Mass surveillance of public space is even carried out in countries that have no terrorist threat or serious crime challenges.

There is little evidence that the expansion of digital surveillance reduces overall crime. We found no compelling evidence that smart surveillance had led to a commensurate number of prosecutions secured with surveillance footage, or to a reduction in terrorism or serious crime.

This gives rise to concern from human rights organisations that smart surveillance systems are really designed to conduct surveillance on opposition leaders, peaceful dissidents, and marginalised groups to repress opposition and reinforce incumbent hold on power.

Surveillance is a tool of powerholders. The reports detail a century of pre-digital surveillance to illustrate how each new government retrains surveillance on targets it perceives threaten its power.

Table 2 summarises the scale of smart city deployments in the 11 countries studied and their evaluation of existing legal protections.

Table 2: Smart city surveillance comparison

	Alg	Egy	Ken	Mau	Moz	Nig	Rwa	Sen	Uga	Zam	Zim
Date first CCTV cameras deployed	2003	unknown	2014	2011	2016	2010	unknown	2019	2014	unknown	2014
Date smart/safe city project announced	2015	2015	2014	2016	2016	2010	2017	2019	2017	2021	2021
Date first smart CCTV cameras deployed	2015	2015	2014	2019	2016	2019	2019	2020	2018	2022	2018
Total Number of Smart Cameras by 2025	5,000	6,000	2,000	4,143	450	10,000	849	500	5,000	1,600	unknown
Total Cost of Safe City Project by 2025 (USD\$)	unknown	\$58m	219m	456m	147m	470m	unknown	167m	189m	210m	10m
Date Data Protection Law Passed	2018	2020	2019	2004	none	2023	2021	2008	2019	2021	2021
Was a human rights risk assessment completed on safe city project?	no	no	no	no	no	no	no	no	no	no	no
Is there a validly enacted law on smart/safe city covering public space surveillance?	yes	partial	no	partial	no	no	no	no	no	no	no

	Alg	Egy	Ken	Mau	Moz	Nig	Rwa	Sen	Uga	Zam	Zim
Is the 'legitimate aim' of public space surveillance defined/limited in law?	yes	partial	yes	yes	no	no	no	no	no	no	no
Does the law limit what data can be collected and which actors can use it for what purposes?	partial	no	partial	yes	no	no	no	no	no	no	no
Does the law provide citizen redress mechanisms?	no	no	partial	partial	no	partial	no	no	no	no	no
Is 'smart' public space surveillance legal, necessary and proportionate?	no	no	no	no	no	partial	no	no	no	no	no
Are controls in place to restrict access to data?	partial	no	partial	partial	no	yes	no	no	no	no	no
Does an independent oversight body exist?	no	no	yes	partial	no	partial	no	no	no	no	no

Source: Authors' own.

It is possible to balance the need for warranted surveillance with citizens' privacy.

Governments can conduct narrowly focused surveillance on specific targets at the same time as fulfilling their existing legal commitments to uphold human rights, including the right to privacy, and freedom of association and assembly. This can be achieved by the following four mechanisms:

- **A human rights risk assessment should be carried out** prior to any deployment of smart city surveillance and recommendations implemented to mitigate risks.
- **A dedicated law on smart surveillance of public space is necessary** to define and limit which actors are allowed to conduct public space surveillance.

- **Surveillance should be limited to instances warranted by the court** to ensure that any surveillance is legal, necessary, and proportionate.
- **A genuinely independent oversight body needs to be adequately resourced.** Oversight must be independent of the government, police, and judiciary to ensure that surveillance is carried out as envisioned in the law, provides remedy and redress, and publishes transparency reports to support good governance and secure public confidence.

Synthesis report: an analysis of the 11 country reports

Wairagala Wakabi and Tony Roberts

1. Introduction

State surveillance is rapidly expanding across Africa. States are investing billions of dollars in technologies to conduct surveillance of internet traffic, mobile communications, social media posts, financial transactions, and citizens' movements in public space (CIPESA 2022; Roberts *et al.* 2023). This has led to growing concerns about the impact of surveillance on human rights and freedoms. Whereas surveillance is conducted by governments all over the world, it becomes a concern when it is implemented without a strong supportive law, implemented subjectively to target perceived state opponents, or conducted in the absence of independent judicial oversight. In many African countries, surveillance has historically, and increasingly, been used as a tool to monitor and silence human rights defenders, the political opposition, and critical voices (Munoriyarwa and Mare 2022; Roberts and Mare 2025).

Smart city surveillance of public space in Africa violates citizens' right to privacy. Thousands of cameras are being installed in major towns and cities that feature constant facial recognitions and ANPR technologies using artificial intelligence (AI). The emergence of these sophisticated new surveillance capabilities enables mass state surveillance of citizens who are not suspected of any crime. As the country reports in this publication evidence, across the ten countries studied, government agencies are running video surveillance systems that collect real-time data 24 hours a day, 365 days a year. The facial recognition and ANPR functionalities both extend the nature of the states' surveillance capacity and the scope for it to be abused unless it is conducted with sufficient regulation and oversight – which this report shows is rarely the case.

China's safe city version of smart city surveillance is now widespread across the African continent. The safe city project is part of the Chinese government's Digital Silk Road programme, which itself is part of their global Belt and Road initiative to fund large infrastructure projects strengthening links to China. The safe city project is marketed as a means for city governments to monitor roads and reduce crime. Safe city projects typically provide several thousand closed-circuit television (CCTV) cameras (and sometimes drones) equipped with facial-recognition and car number plate recognition AI technology, which send real-time images to the national data centre that provides police and intelligence services with a command and control centre to monitor surveillance on citizens and vehicles.

The safe city model grew out of state surveillance projects developed in mainland China. The safe city model can be traced back to

China's Golden Shield Project, launched in the 1990s to create a nationwide CCTV surveillance network that minimised human labour and centralised data control. Jili (2022) argues that safe cities represent a computational model of urban planning, intended to improve governance efficiency through integrated information and communication technology systems. At least 16 countries in Africa have since rolled out the safe cities programmes (Feldstein 2019), with most of them relying on technology and funding from China (Woodhams 2022). The systems typically feature high-definition CCTV cameras, facial recognition, ANPR, and centralised command and control centres.

Mass surveillance by the state raises concerns about the impact on human rights. While governments claim these technologies are necessary to enhance public safety, counter terrorism, and modernise security operations, serious concerns have been raised about their implications for privacy, freedom of expression, and democratic accountability (Access Now 2021; CIPESA 2022; Jili 2023). In many cases, the surveillance tools have been deployed to entrench political control and target perceived enemies of the state. In countries with growing levels of authoritarianism (Akbari 2022), including suppression of the digital civic space, this remains a live concern. Indeed, United Nations experts warned of a growing trend in which surveillance technology is exported to repressive regimes with minimal controls, enabling abuses such as the targeting of journalists, activists, and opposition leaders (United Nations Human Rights Council 2019).

States have good reason to conduct narrow and warranted surveillance on terrorists. This can be balanced with government's existing commitment to human rights with appropriate regulation and oversight. However, in countries with weak democratic institutions and limited legal safeguards, these digital surveillance tools often enable opaque state practices that mirror the repressive mechanisms of the past (Roberts *et al.* 2021). As such, they often fail to meet the standards outlined in the Declaration of Principles on Freedom of Expression and Access to Information in Africa (2019), as well as the Necessary and Proportionate Principles that guide the application of surveillance technologies in a rights-respecting manner.

This research provides the most comprehensive account to date of public space surveillance in Africa. Detailed country reports are included in this publication on 11 countries: Algeria, Egypt, Kenya, Mauritius, Mozambique, Rwanda, Senegal, Uganda, Zambia, and Zimbabwe. The research focuses on the use of smart surveillance technologies in public spaces – including facial recognition and vehicle number plate recognition – and the analysis of this data, often using AI, at centralised control centres. The research traces the evolution of surveillance from colonial-era

intelligence networks to today's digitally enabled public spaces monitoring systems. It identifies the key actors, including government departments and agencies, foreign technology companies, and local private sector actors involved in the supply and implementation of Smart Cities.

The country reports document the scale of smart city surveillance in each country. This study set out to address a series of related research questions about the extent of smart city surveillance of public space across Africa. We aimed to document the scale and scope of public space surveillance and the main actors and technologies involved. The research asks what impacts the rapid expansion of surveillance has on citizens' rights and whether existing legislation succeeds in balancing the state's need to conduct warranted surveillance with existing legal commitments to human rights. Among the issues explored are: when and how smart city and safe city projects were launched; contract sizes, timelines, the locations and capabilities of the surveillance systems; concerns expressed by citizens and civil society action to defend human rights. Importantly, the research also assesses the legal and policy frameworks governing the operations of these systems, using a series of eight tests. The research findings will be valuable for civil society advocates, government policymakers, regulators and international development and human rights actors.

1.1 Research approach

The studies employed a research methodology grounded in a comprehensive review of diverse literature that included government policy documents (such as those produced by Parliament, Office of the Auditor General, Police and intelligence services...), media reports, as well as academic and independent research reports related to smart city surveillance. Additionally, the research entailed an analysis of laws related to surveillance and data protection. These exercises aimed to establish concrete details on procurement for smart city systems, functionalities of the systems, operational practices, the human rights implications, and public responses to their establishment and implementation.

The studies then evaluated the surveillance systems through the dual lenses of rule-of-law legality and rights-based compliance. To do this, the study adapted the Centre for Internet and Society (CIS) Digital ID Evaluation Framework (Bhandari, Trikanad and Sinha 2020), which provides a systematic framework for assessing the human rights credentials of government digital governance systems and services. This framework was developed to address concerns about privacy, surveillance, and exclusion harms caused by state-issued digital IDs, but we found it to be directly applicable to smart city surveillance systems, as similar human

rights concerns manifest in the two systems. This framework informed a consistent analysis across the 11 countries, which enables the kind of cross-country comparison illustrated in Table 2. The analysis applied the four legal tests and the four rights-based tests outlined below.

The Legal Tests

Is it legal?: Is public space surveillance in the country backed by a validly enacted law that defines the purpose and scope of surveillance?

Legitimate aim: Does the law have a 'legitimate aim'? Does the law clearly define and limit the purposes for which public space surveillance can be used (for instance crime prevention and public safety) in order to guard against 'mission creep' or the gradual drift away from originally agreed objectives?

Actors and purposes: Does the law governing public space surveillance clearly define and limit the actors (e.g. police and intelligence services) permitted to access or use the surveillance data? Does the law clearly define and limit the nature of data that can be collected?

Accountability and redress: Does the law provide an independent and adequately resourced regulatory mechanism to ensure accountability of the administrator of the data? Does the law provide for adequate redress mechanisms against actors who use the surveillance data?

The Rights-Based Tests

Necessary and proportionate: Are the privacy violations arising from the use of surveillance data necessary and proportionate to achieve the legitimate aim?

Access control: Are there protections in place to limit access to the digital trail of personally identifiable information created through the use of surveillance data by both state and private actors?

Oversight mechanism: Does the law provide for an adequately funded and fully independent oversight body to ensure that public space surveillance is operated in accordance with the law and consistent with national commitments to human rights?

Human rights impact assessment: Was a human right impact assessment carried out prior to implementation?

Every country report in this publication includes a detailed qualitative evaluation of each smart city surveillance project against these eight tests. Scoring is narrative-based, highlighting strengths, gaps, and deviations from legally and ethically sound surveillance practices. Particular attention is paid to how smart city systems may

deviate from their stated objectives, and the extent to which legal and rights-based safeguards are neglected or implemented.

1.2 Research limitations

This research was limited by the secrecy about surveillance scope, practices, and procurement contracts. We found the availability of public expenditure records to be uneven across countries as well as between departments of state within the same country. In two of the countries studied, no public records were available on the value of smart city surveillance contracts. In other countries the financial records were incomplete.

Although there is considerable scholarship about surveillance, there is little granular data about public space surveillance in Africa, such as the precise number of cameras installed and the value of procurement contracts. Getting access to reliable and complete data remains a challenge. In many cases, surveillance infrastructure is acquired through opaque contracts, often involving bilateral agreements with foreign technology providers, notably Chinese firms such as Huawei and ZTE (Hillman and McCalpin 2019; Woodhams 2022). For example, Kenya and Uganda's safe city projects were developed with little public consultation or disclosure of terms, raising concerns about vendor influence, data sovereignty, and oversight mechanisms (Kafeero 2022; Wanjama 2018). In Mozambique, the local business partner in the safe cities project is Msumbiji Investment Limited, which is thought to be connected to the son of the former president of that country. Finally, conducting research on surveillance can also present security challenges. This has implications for which sources can be safely accessed and what methods can be employed without placing researchers at risk, which was our primary concern.

2. Background to the country reports

Each country report begins by situating smart city surveillance in the context of pre-digital surveillance practices in that country (see for example the account of colonial and post-independence surveillance in the Kenya and Mauritius country reports in this publication). A brief history of surveillance practices from the colonial period through to the post-independence governments is detailed. The reports then document the current state of play of public space surveillance under the so-called smart or safe cities, which are marked by high levels of digitisation, collection of large amounts of data (including biometric data), capabilities such as facial recognition, and the use of AI by the government-run systems.

Smart city surveillance practices amplify the impact on privacy rights.

By locating smart city surveillance within the context of pre-digital surveillance practice, country report authors show that surveillance is not caused by digital technologies and has strong continuities with analogue surveillance. However, automating surveillance and analysing data using AI significantly extends the scope and depth of surveillance. Smart surveillance cameras enable the automation of surveillance around the clock, the transmission of real-time, geolocated facial recognition images to data centres where AI is used to cross reference facial recognition records with biometric digital ID and car licence records, as well as other surveillance data from internet, mobile phone, and financial transaction surveillance.

We found no evidence that smart surveillance in Africa has reduced terrorism or serious crime.

While the rapid increase in state surveillance of citizens is commonly justified as a security measure, our research found no evidence that surveillance has increased convictions of terrorists or serious crime. Civil society organisations are concerned that the motivation for increased surveillance includes surveillance of peaceful opposition and a desire to retain political control (Hilman and McCalpin 2019; Woodhams 2022). Each country report provides the reader with a clear timeline of key milestones in the announcement of the smart city initiatives and details on the implementation of these systems.

Our research follows the money in Africa's multibillion dollar surveillance economy.

The country reports document which companies from which countries are providing which technologies for public space surveillance, and the dollar value of the contracts. They also explore the infrastructure that enables the operations of the smart cities, which often includes a command and control centre. These centres are staffed by police and

security agencies, often working in consort with the military and intelligence services and in partnership with foreign technology companies.

Is mass surveillance of public space legal, necessary, or proportionate? The country reports submit the smart city surveillance projects to a series of eight tests to establish whether the routine digital surveillance of public spaces meets the accepted threshold in international human rights law of being legal, necessary, and proportionate. The reports capture the motivation for the establishment of smart city surveillance initiatives, pointing to the stated objectives (where countries stated them), but also showing that in several countries the entities that run the smart city programmes have profoundly deviated from the original objectives and turned these programmes into infrastructures for conducting mass surveillance against citizens.

Finally, the reports each provide actionable recommendations for smart cities to be transparent in operations and procurement, to respect human rights, to provide for robust independent oversight and redress mechanisms, and to be backed by clear legislation that specifies the necessity and proportionality of public space surveillance.

2.1 History of public space surveillance in Africa

Institutionalised state surveillance was often introduced to African countries by colonial governments to monitor people, places, and movements that were perceived as a threat to colonial power. During this period, intelligence gathering relied heavily on human informants, local chiefs who compiled reports on political activities, and rudimentary interception methods, such as telephone tapping (Kalyegira 2005). For instance, the colonial Special Branch, an intelligence unit of the British government, monitored nationalist movements, political dissidents, and community leaders, employing infiltration tactics and covert observation to maintain political control (West 2012). These practices created a precedent for viewing surveillance primarily as a tool for state security and political stability rather than purely for public safety.

Colonial regimes across the continent institutionalised these practices and intensified them through formal policing structures, intelligence services, and bureaucratic record keeping. In colonial Egypt, for instance, surveillance infrastructures such as the 'City Eye' relied on networks of informants drawn from the working classes – beggars, porters, drivers, and vendors – who were incentivised to report suspicious activity (Kandil 2012). British authorities also established formal intelligence divisions such as the Special Section within the Ministry of Interior in the early 20th century to expand domestic surveillance (Kandil 2012). In

southern and central Africa, colonial administrations used surveillance to control labour, suppress resistance, and extract resources. Informant networks and administrative data systems tracked the movements and activities of working-age African men and nationalist organisers, particularly in strategic regions such as the Copperbelt (Phiri 2006).

Under Portuguese colonialism, surveillance served as a counterinsurgency strategy aimed at crushing anticolonial mobilisations. Security services like *Polícia Internacional e de Defesa do Estado* (PIDE) and *Serviços de Coordenação de Informações Militares* (SCCIM) infiltrated nationalist movements such as FRELIMO in Mozambique and tracked resistance networks across borders (Araújo n.d.). After independence in 1975, post-liberation governments often inherited and adapted these surveillance infrastructures. In Mozambique, for example, the ruling FRELIMO party justified pervasive social monitoring during the civil war (1976–1992) as essential to defending the revolutionary project (Bussotti 2015). These postcolonial surveillance practices were often legitimised through discourses of national unity, security, and stability, even as they curtailed dissent and political pluralism.

Colonial surveillance functions were retained by the post-liberation governments, who then used the surveillance apparatus against those perceived as threats to public safety – or to the new rulers' hold on power. Today, governments across the region increasingly rely on digital technologies to conduct surveillance, with the safe cities infrastructure providing a notable ramp up of the governments' capacities to monitor the activities of citizens and to listen in to their communications. As the Africa Digital Rights Network (ADRN) has argued, current digital surveillance practices must be understood within a broader historical continuum in which states have sought to accumulate and exercise power through various mechanisms of population monitoring and control (Roberts and Mare 2025).

Physical surveillance – the 'direct observation of people, vehicles, or activity taking place at some given location for the purpose of obtaining information' (Raymond and Fredrickson 2016) was the prevalent form of surveillance from colonial times right to the early 2000s. It often involves watching individuals or organisations, including their movements and activities. While it may not require sophisticated skills or a lot of resources to conduct, it definitely requires a lot of manpower and sometimes several vehicles if there are multiple targets of surveillance. On the other hand, video surveillance enables a few individuals seated hundreds of kilometres from their targets to contemporaneously collect intelligence on numerous individuals, to store that information for prolonged periods, and to speedily analyse that information, in order to build profiles and cases against the surveilled individuals.

3. Country case studies

This section provides a summary of findings about the establishment of digital surveillance systems as part of smart city (or safe city) projects derived from the 11 country reports.

3.1 Smart surveillance in Africa

Kenya was the first African country to adopt Huawei's safe city surveillance package in 2014. The government of Kenya launched its Integrated Public Safety Communication and Surveillance System (IPSCSS) in collaboration with Safaricom, the country's largest telecom provider (Wangari 2023). Whereas the government had been gradually increasing its surveillance capacity over the years, rising insecurity and the massive public outcry following the Westgate Shopping Mall attack in 2013, provided a strong impetus and justification for investment in the IPSCSS (Howden 2013). The system was designed to assist law enforcement agencies to monitor high-crime areas, track criminal activities, and respond to security threats through a centrally controlled system of urban surveillance (JEPA Africa 2024). However, like elsewhere on the continent where such systems have been utilised, the Kenya government has been accused of using the IPSCSS to surveil protest leaders, human rights activists, and opposition politicians (Kapiyo 2025).

Kenya installed the first smart CCTV cameras in 2014, with Egypt (2015), Algeria (2015) and Mozambique (2016) also being early adopters. Next to join the fray were Uganda (2018), Zimbabwe (2018), Mauritius, Nigeria, and Rwanda (all 2019). Senegal (2019) and Zambia (2022) are the latest entrants to the club. Most countries have since boosted the capabilities of their systems. Mozambique, which first installed surveillance cameras in 2016, had by 2025 mounted an estimated 450 cameras in the capital Maputo and Matola city, and infused facial recognition and car number plate recognition capabilities. Senegal, which launched smart CCTV in 2019, has installed about 500 cameras in Dakar and five other cities. Uganda, which first installed the Huawei-supplied smart CCTV in 2018, has become the archetypal public-spaces-digital-surveillance state. Its system has expanded across all major towns and cities, and has facial recognition and number plate recognition capacity as well as vehicle tracking and automated traffic penalties enforcement.

4. Key actors

This section addresses the research question: what roles do key actors play in shaping the current smart city surveillance landscape? In most countries studied, the main actor is the state through an array of different ministries, police, and security and intelligence agencies. In the case of Algeria, for example, the public space surveillance system was installed by the Department of Defence, with data access managed by the Ministry of the Interior at the national level, with access provided to police at the local level.

State control of the smart surveillance system is often in partnership with one or more foreign technology companies. In the Chinese safe city model the lead contractor is often Huawei, as is the case in Mauritius, with elements of system building and training sub-contracted to ZTE, and facial recognition and ANPR cameras provided by Hikvision. In the case of Egypt, the US company Honeywell takes the place of Huawei. In Mozambique and Rwanda companies from South Korea play key roles as technology suppliers, and in Mauritius and Rwanda the United Arab Emirates (UAE) is a key actor.

International funders have been key in financing smart city surveillance. The largest single source of financing has been private Chinese banks, notably Eximbank. The programme is part of the Chinese government's Digital Silk Road project, which is part of the wider Belt and Road Initiative that involves major infrastructural development around the world. Chinese banks have provided the huge amount of funding necessary to African governments to build and maintain the digital infrastructure of surveillance cameras and command and control centres. These huge loans are conditional on the purchase of Chinese technology and services needed to build and transfer the 'safe city' systems.

Civil society organisations are key actors in smart city surveillance, playing a watchdog role, raising awareness, advocating for rights-compliant systems, and using strategic litigation in some cases to hold government accountable to its human rights commitments. In the case of Kenya, Privacy International, KICTANet and CIPESA have all played important roles in conducting research on how smart surveillance impacts privacy rights.

5. Regulatory landscape: the legal tests

This section summarises key findings from the country reports with respect to the eight tests derived from the CIS framework and as outlined in the introduction above.

Test 1. Are there validly enacted laws governing public space surveillance?

A key area of inquiry for the research is whether the various countries have validly enacted laws that guide the rollout and implementation of public space surveillance. Besides giving indications on whether these systems are compliant with rule of law requirements in each country, assessing this area can also help to establish whether the surveillance is run in accordance with regional and international standards such as those prescribed by the African Commission on Human and Peoples' Rights and the United Nations. According to Bhandari *et al.* (2020), a validly enacted law has three components: (1) it should be passed by the Legislature, and not the Executive; (2) it should be accessible and foreseeable – this is to ensure the 'quality of law'; and (3) it should be clear and precise – this is to limit the scope of discretion.

In all countries studied, the constitution guarantees the right to privacy (for example article 71 for Mozambique, articles 27 for Uganda, article 13 for Senegal, article 23 for Rwanda, section 57 for Zimbabwe, and article 31 for Kenya). These privacy guarantees, as well as provisions in various national laws, also protect against unlawful surveillance in general. In turn, the various countries have passed laws that provide for lawful surveillance, describing elements such as the procedures, authorised officers, and duration of surveillance. This is generally found in laws on cybercrime, antiterrorism, electronic transactions, and those that are specific to the conduct of lawful interception and monitoring of communications.

However, the countries studied lack laws that specifically relate to video (CCTV) surveillance or to related aspects such as face recognition and the use of AI. Algeria, Egypt, and Mauritius have snippets of provisions in some laws that regulate some related aspects but they are not sufficiently specific or comprehensive to regulate the breadth of digital public space surveillance. It is also notable that across the study countries, many of the generic laws that enable state surveillance pre-date the establishment of safe city initiatives, so any relevant provisions that they have are not suitable for handling the increasing sophistication of public space surveillance.

Besides the laws governing surveillance generally, which all countries have in some form, another piece of legislation relevant to public space surveillance is a data protection law. Among the countries under study, all countries except Mozambique have these laws that were enacted by their legislatures. While this gives the laws some legitimacy in one aspect, the laws are not precise enough to limit the scope of their applicability. Notably, these laws do not explicitly authorise public space surveillance.

Senegal's data protection law of 2018, while creating a data protection authority, does not mention public space surveillance. However, the 2019 Personal Data Protection Bill (which is yet to come into law), proposes to explicitly recognise that data processing (including video) is lawful when it is 'necessary for the exercise of official authority or for the performance of a task carried out in the public interest' (CIPESA 2021). In 2019, Kenya drafted a draft national CCTV policy, yet to be finalised or implemented, which sought to 'guide installation, operation and management of all CCTV systems' for crime prevention.

Test 2. Do the laws define a legitimate aim for public space surveillance?

Is the legitimate aim of public space surveillance defined in the laws of the countries studied? Does the law clearly define and limit the purposes for which public space surveillance can be used? As noted above, the focus countries do not have laws that govern public space surveillance. Also, even where there are general laws that govern surveillance, or laws that generally provide for limits on the enjoyment of privacy and related rights, they often relate to general exemptions on the basis of undefined security and public order considerations. Indeed, for the countries that have data protection laws, national security and public order activities are exempted from several obligations related to protecting individuals' data and privacy rights.

According to European Court of Human Rights determinations, the legitimate aim needs to be identified in a valid law, and should be necessary in a democratic society, implying that it must answer a 'pressing social need' (European Union Agency for Fundamental Rights 2018). In almost all countries, while the laws under which the safe cities were introduced was not always stated, the stated objectives appeared to serve a social need, for example: enhancing public safety in Rwanda (Muvunyi and Gitinywa 2021); reducing crime in Mozambique (DW 2024) and Uganda (UBC 2025); addressing security challenges in Senegal (Sénégal Numérique 2018); enhancing public safety and urban efficiency in Zambia (Ministry of Home Affairs and Internal Security 2023).

However, as many reports show, there has been a mixed record on reducing crime – and in most instances no studies have been conducted to establish

what positive impacts the systems have had. Moreover, there is considerable evidence to show that this infrastructure has been used to surveil political opponents and critics, which would suggest that, rather than serve a social need, their aim has been largely diverted to serving selfish political interests. Notably, while objectives such as ensuring public safety, and detection and prevention of crime may be legitimate, the absence of a law to regulate public surveillance is a clear gap, which means there is no framework for oversight, thus leaving room for abuse of powers by law enforcement, including to target human rights defenders (Kapiyo *et al.* 2024).

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

The laws that govern surveillance and data protection have limits on what data can be collected and the purposes of collection. They all have data principles and provide for data rights. However, these limitations are not specifically spelt out in relation to public spaces surveillance. And since the laws and *modus operandi* of the smart cities are not entirely known, there is no telling as to whether the operators of these systems are adhering to the stipulated limits on what data is collected, who collects it, the conditions under which the data may be shared, the purposes for which it is kept, and the periods of time over which that data should be kept.

In the countries where there are data protection laws, the laws require that the processing of personal data be based on consent or a clearly defined public interest purpose. However, the laws do not sufficiently define or limit who may access data and conduct surveillance, and for what purposes. In laws pertaining to general surveillance, some limits exist in some countries (such as the requirement for authorisation by a judicial or independent judge before communications interception or monitoring; designation of the actors that have powers to conduct search and seizures or authorise surveillance; and duration of surveillance warrants). However, besides the inherent weaknesses in such laws (including shortage of oversight mechanisms, lack of transparency over the conduct of surveillance, the extensive powers bestowed upon some government officials to order surveillance without obtaining a court order, and compelled service provider assistance), these laws are not specific to public space surveillance.

Test 4. Does the law define mechanisms for citizen redress and accountability?

All 11 countries studied currently fail to provide adequate mechanisms for citizens to obtain remedy or redress in case of smart surveillance errors or abuse. If citizen's rights are violated in the collection, sharing, or use of surveillance images, existing mechanisms do not provide for accountability.

In seven of the 11 countries studied, existing legislation provided no redress mechanism. In the case of Kenya, Mauritius, and Nigeria, partial redress was possible through data protection legislation.

Country report authors argued that citizen's rights would be better protected by new legislation dedicated to regulation of public space surveillance, which stipulated redress mechanisms and established and resourced a genuinely independent oversight body. The authors of the Rwanda country report in this publication speak for other researchers when they report, 'The absence of specific surveillance oversight bodies means that citizens have limited avenues for challenging surveillance decisions or seeking redress for privacy violations'.

A key weakness of generic data protection legislation is that state actors (police and security agents) are often explicitly excluded from compliance. In order to protect the constitutional rights of citizens from unwarranted state violation of fundamental rights, it is necessary that legal protections apply to all state and non-state actors and that the legislation is explicit in providing mechanisms for remedy and redress via an independent oversight body. Without transparent institutional channels to seek redress, public trust in state surveillance will not be forthcoming and confidence in government will be eroded in the face of continually expanding public space surveillance.

6. Human rights tests

This section summarises findings from the country reports in response to the four rights-based tests.

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

In all countries studied but one (Nigeria), the reports cast strong doubt on whether smart public space surveillance is legal, necessary, and proportionate. The legal framework surrounding public space surveillance includes broad statutory exemptions that allow for extensive surveillance activities yet it fails to provide stringent judicial oversight. Coupled with the absence of clarity in transparency and accountability in both procurement and operation of the systems, concerns about potential abuses of power and the erosion of digital rights are widespread. The legality of public space surveillance comes in doubt right from the absence of legislation that specifically provides for it. But since there is no transparency as to how surveillance is conducted, it is often difficult to tell what laws countries are using, and to therefore assess the level of legality.

Is the surveillance necessary and proportionate? Necessity can be assessed at more than one level: the purpose for which the systems were established and the purpose they are serving. In almost all countries, there is insufficient independent and rigorous study done to ascertain how the systems have contributed to fighting crime, terrorism, streamlining transport management, and generally improving public safety. On the other hand, across several countries, including Mozambique, Kenya, Uganda, and Zimbabwe, there are reports of surveillance infrastructure being utilised to surveil opposition figures and government critics (Paradigm Initiative 2024; Kapiyo *et al.* 2024), intercept their communications, and monitor the movements of protesters and identify key organisers who were then arrested (Daily Monitor 2022). The effect of these measures is creating an atmosphere of fear that hinders their mobilisation efforts and hampers their freedom of expression and assembly and association. There have also been reports of surveillance without court warrants (Privacy International 2017).

Test 6. Does the law control who has access to surveillance data?

The countries that have data protection laws have some limits on access to data in general, but those laws do not specifically speak to public space surveillance data. As such, these controls are not robust since they lack specificity in terms of relating to this data and do not institute measures specifically intended to limit access to it. Mozambique, which

has not yet enacted a specific data protection law, provides for limits on access to and exchange of data under the Electronic Transactions Law.

Overall, countries such as Nigeria, Algeria, Kenya, and Mauritius have more controls on data access, although these protections are not robust or specific enough to public space surveillance. The insufficient safeguards and low transparency around how data is stored and who has access to it raise fears of undue surveillance and have a chilling effect on citizens. In some instances, insufficient safeguards over data access and authorisation for surveillance raise similar concerns. For instance, under Rwanda's law 60/2013 regulating the interception of communications, the prosecutor can issue a 'verbal warrant' to the national security and intelligence service to watch, intercept, and decrypt any information generated, transmitted, received, or stored in any computer resource for reasons of national security (Gitinywa 2020).

Test 7. Does the law create an independent oversight body?

The entities responsible for independent oversight over data are data protection authorities (DPAs) – in some countries known as data protection commissions or offices. All the countries that have passed data protection laws provide for these entities. However, many of them exist only on paper, with only the likes of Kenya, Mauritius, and Nigeria having well-established DPAs. For the rest of the countries, the DPAs, even where they have been established, are poorly resourced and thinly staffed; some lack independence and are given limited powers, such as not being able to fine entities that commit offences related to data protection. A recent example is Uganda's data protection regulator who found Google in breach of the country's laws for exporting Ugandans' data without authorisation and local registration, but did not penalise the entity because it lacks statutory powers to do so.

The inherent weaknesses of the DPAs, contained both in the laws that give them a narrow mandate and in the practical realities of their poor-resourced existence, mean that they have limited enforcement power, particularly over powerful actors that run public city surveillance. As such, independent oversight is largely lacking, and this extends beyond the DPAs into the realm of surveillance being conducted without court warrants even where they are needed. It is further compounded by the challenges faced by other oversight bodies, such as National Human Rights Institutes and the judiciary, as they face a shortage of human, financial, and technical resources (Kapiyo *et al.* 2024), as well as lacking true independence from political interference.

Test 8. Has a human rights impact assessment been carried out?

Digitalisation initiatives such as national IDs and smart city programmes can have many negative effects on human rights. That is why it is essential to conduct human rights risk assessments before those programmes are implemented. Such assessments will then find ways to proactively mitigate these human rights risks, such as by changing programme design, or requiring specific and timely interventions to assess and remedy any negative impacts. The human rights impact assessments (HRIAs) are akin to environmental impact assessments, whereby entities commonly known as national environment management authorities require projects like farms or factories to evaluate the potential environmental and social consequences of the proposed projects before they are approved, to ensure that any negative impacts are addressed for sustainable development.

In the arena of data protection, some laws require a data protection impact assessment (DPIA) to be conducted to identify risks arising out of the processing of personal data in order to minimise these risks as quickly and as extensively as possible. An example is section 31 of Kenya's Data Protection Act of 2019, which states that a DPIA is needed 'where a processing operation is likely to result in a high risk to the rights and freedoms of a data subject, by virtue of its nature, context, scope or purpose'. The assessment can then help to identify the least privacy intrusive way of achieving a legitimate aim of processing the data (OPDC 2024).

The criteria for HRIAs include meaningful participation, non-discrimination, empowerment, transparency, accountability, use of human rights standards, analysis of actual and potential impacts (caused, contributed to, and directly linked to), assessment of severity of human rights impacts, inclusion of impact mitigation measures, and a focus on access to remedy (DIHR 2020).

In none of the country reports do we see evidence of an impact assessment having been conducted before or during the implementation of the smart city initiatives. It is apparent that governments and the departments that implemented these systems did not think of human rights impacts as a key element in planning and rolling out these initiatives. In the absence of such assessments, these governments are guilty of paying lip service to the human rights implications of public space surveillance.

Similarly, while we cannot say for certain what the full human implications of these initiatives are, we can confidently assert that they have fallen short on some crucial elements, such as transparency, meaningful participation, inclusion of mitigation measures, and access to remedy. Some observers, including the Senegalese Association of Users of Digital Technologies (ASUTIC), say the introduction of the programme before doing an assessment on its effects on privacy, freedom of movement, and expression means that it

is fraught with threats to human rights online, including 'questions about the protection of personal data and the risks of mass surveillance' (ASUTIC 2020).

7. Discussion

The 11 country reports in this publication document a massive expansion of public space surveillance across Africa, using automated digital systems that employ AI, biometrics, and centralised population registries to monitor the movement of citizens.

It is generally accepted that state surveillance of terrorists and of the most serious criminals is justified in the public interest. At the same time, African governments have made a series of constitutional and legal commitments to protect fundamental human rights including the right to privacy, and freedom of movement, expression, association and assembly.

Mass surveillance of public space is not consistent with the right to privacy. It is universally accepted in international law that any violation of fundamental rights must meet the test of being legal, necessary, and proportionate.

Our research documents the rapid expansion of public space surveillance in the absence of a legal framework to both enable and regulate such surveillance. This report documents the expansion of constant mass surveillance in countries with no terrorist threat, in locations with no incidents of serious crime, and of citizens who are suspected of no crime. This clearly fails to meet the test of being legal, necessary or proportionate.

The countries studied have each spent an average of US\$240 million on smart city surveillance. Most of the countries studied have now had smart surveillance of public space in place for between 5 and 10 years. However, none have put in place a legal framework able to balance the state's need to conduct warranted surveillance with their existing commitments to protect and promote fundamental human rights.

When citizens seek to hold the government to account for unwarranted surveillance and violation of privacy rights, the absence of a legal basis for surveillance prevents access to justice. The absence of any mechanisms for transparency about which actors can legally carry out surveillance of citizens, what data can be collected, how long it is stored, and who it can be shared with are obstacles to accountability. The lack of any mechanisms for remedy, redress, or reporting make it impossible to hold power to account.

The advent of data protection legislation offers some hope of legal protection about how surveillance data is captured and managed. However, in many of the countries studied, data protection legislation exempts the police and security forces from compliance. In other countries, state of emergency or claims of 'national interest' that are not defined in law

are used to evade compliance or transparency. The authors of several country reports therefore call for a dedicated law to regulate public space surveillance so that the state can conduct legal surveillance with oversight and citizens can enjoy the right to privacy.

The Chinese safe city model

It is clear that China is the preferred supplier of smart surveillance technologies in Africa. China supplied public space surveillance technology to all of the 11 countries studied. The value of the Chinese contracts was greater than all of the other surveillance contracts that we had access to.

However, smart surveillance technologies are also provided to African governments by companies from Korea, Russia, UAE, France, Spain, USA, Japan, Israel, and Iran. Although China is currently winning the competitive battle to supply smart surveillance technologies to African governments, they are by no means the only country keen to supply these technologies.

One of the reasons that China is beating off the competition is because of the safe city surveillance package that they offer. Private Chinese banks (notably Eximbank) provide loans in the order of US\$250 million to African governments to purchase a package of thousands of surveillance cameras equipped with facial recognition technology and ANPR (often from Hikvision) alongside a build-and-transfer project to establish a national 'data centre'. As part of this, Chinese technicians and trainers (often from ZTE or Huawei) establish local capacity in AI systems to analyse the surveillance streams in a dedicated command and control centre for police and security operatives.

This rapid expansion of smart city surveillance of public space in Africa is occurring alongside similar expansion in the capacity of other forms of digital surveillance including internet surveillance, mobile phone surveillance, social media surveillance, mandatory SIM-card registration and biometric digital IDs (often linked to financial transaction monitoring (CIPESA 2022; Munoriyarwa and Mare 2022; Roberts *et al.* 2023; Roberts and Mare 2025).

Our researchers document how this rapid expansion of unwarranted surveillance violates the right to privacy and gives rise to a 'chilling effect' that reduces civic space and tightens the grip of incumbents on power. The massive expansion of surveillance – especially when unrelated to terrorism threats or to any proportional rise in serious crime – gives rise to concern from human rights organisations that smart surveillance systems are really designed to conduct surveillance on opposition leaders, peaceful dissidents, and marginalised groups to repress opposition and reinforce incumbent hold on power.

8. Conclusion

This report set out to improve our level of understanding about the use of 'smart' surveillance systems to monitor citizens in public spaces. It achieved this by providing the most comprehensive account to date of the scale and specifications of digital surveillance of public space in 11 detailed country reports by local researchers in Algeria, Egypt, Kenya, Mauritius, Mozambique, Nigeria, Rwanda, Senegal, Uganda, Zambia, and Zimbabwe.

Each country report contains in-depth analysis of the current state of the art in smart surveillance systems, including the most current data on the number of cameras and dollar value of known procurement contracts. In two countries no public reporting on the value of contracts exists. In most countries studied the value of surveillance contracts was partial and incomplete. Despite these limitations we were able to document US\$2 billion spent by African governments on smart surveillance of public space.

This level of expenditure translates into an average spend in the region of US\$240 million per country. Nigeria alone has documented public expenditure of US\$470 million on AI-enabled facial recognition and ANPR, making it the continent's largest buyer of smart city surveillance technologies. In all cases, we know that the real total is significantly higher because (1) surveillance spending is often secret; (2) no figures were available for two of the 11 countries studied; (3) the public accounts for the other nine countries were incomplete; and (4) this study included only 11 of Africa's 55 countries.

The country reports clearly show that the expansion of public space surveillance is an unwarranted violation of citizens' right to privacy which fails to meet the test of being legal, necessary, or proportionate. Our team of researchers argue that expanding surveillance damages citizens' rights and freedoms and has a 'chilling effect' causing citizens and journalists to self-censor, shrinks civic space, and limits citizens' ability to hold governments to account. This extension of state surveillance is particularly troubling because it is occurring in the absence of adequate legal regulation or oversight. It would be possible to balance the state's legitimate aim of conducting surveillance on the most serious criminals if there was an adequate legal framework that detailed protections for citizens' rights and provided truly independent oversight.

Our team of researchers found no compelling evidence that the expansion of digital surveillance reduces overall crime. Court records did not show a significant number of prosecutions secured with surveillance footage, or any attributable reduction in terrorism or serious crime.

9. Recommendations

No country studied had in place a sufficiently robust legal framework to enable government to conduct warranted surveillance at the same time as maintaining their existing legal commitments to uphold human rights, including the right to privacy, and freedom of association and assembly.

A dedicated law on smart surveillance of public space is recommended that defines and limits which actors are allowed in law to conduct collection of which digital surveillance images and that requires a prior court warrant so that a judge can test whether the surveillance is legal, necessary, and proportionate.

An adequately resourced oversight body is needed that is genuinely independent of the government, police, and judiciary to ensure surveillance is carried out as envisioned in the law, to provide remedy and redress, and to publish transparency reports to support good governance and secure public confidence.

This is the end of the introductory section in which we have provided a synthesis of findings drawn from across all of the 11 countries studied. What follows is each of the 11 country reports from Algeria, Egypt, Kenya, Mauritius, Mozambique, Nigeria, Rwanda, Senegal, Uganda, Zambia, and Zimbabwe – in that order.

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Smart city surveillance in Algeria

Yosr Jouini Roberts

1. Introduction

This report critically examines the current state of public space surveillance practices in Algeria, tracing their trajectory from colonial informant networks and repressive legal frameworks to the contemporary integration of advanced digital surveillance technologies, notably within the smart city projects and their associated initiatives.

This report addresses the following principal research questions: To what extent have digital surveillance technologies been integrated into Algeria's public spaces, particularly through smart city projects? How does Algeria's current legal and regulatory framework govern the acquisition, deployment, and oversight of these surveillance systems? What roles do key actors play in shaping the smart city surveillance landscape? And lastly, how do these surveillance practices, and their underlying governance framework affect citizens' digital rights?

By examining these questions, the report aims to present a digital-rights focused understanding of the implications of public space surveillance practices in Algeria, offering insights and recommendations to relevant stakeholders. This report serves as the most comprehensive and detailed account of smart city surveillance in Algeria to date.

2. Background

Algeria's history of surveillance is shaped by its colonial past and turbulent political landscape.

During French colonial rule in Algeria, surveillance was a central aspect of governance. The early 'Bureaux Arabes' (Arab Affairs Bureaus) utilised a network of informants and local notables to monitor the locals and gather intelligence (ANOM n.d.). This system was later institutionalised through the 'Code de l'indigénat' (1881), a legal framework that legalised constant monitoring and identity checks, and prohibited local subjects from holding meetings without authorisation or travelling outside their municipality without a permit (Laïssaoui 2009). During the Algerian War (1954–1962), which claimed 1.5 million Algerian lives, surveillance was further systematised through the creation of Sections Administratives Spécialisées, military and administrative units tasked with exercising surveillance and administration in rural areas (Frémeaux 2002). These practices reveal how colonial control relied on embedding surveillance into both law and everyday life.

The post-independence era (post 1962), initially marked by the National Liberation Front's single-party rule, witnessed the development of a climate where dissent and opposition were met with state surveillance and repression. The political instability following Ahmed Ben Bella's election in 1963 and subsequent 1965 coup d'état by Colonel Houari Boumediene, whose own rule was marked by a 1967 attempted coup and a 1968 assassination attempt, further solidified the military's prioritisation of security and control. This shift resulted in the increased surveillance of political opponents to maintain autocratic control of power. The emphasis on internal security intensified during the civil war (1991–2002) following the military's refusal to accept the results of the 1992 legislative elections. This forced President Chadli Ben Djedid to resign, triggering a state of emergency. The 19-year-old state of emergency, in place from 1992 until 2011, provided a legal framework for extensive surveillance of the population, employing both traditional and increasingly sophisticated techniques (Berrahmoun 2023).

The impact of the civil war, combined with the historical influence of the military over civil political institutions, creates a complex background essential for understanding the context for public space surveillance in Algeria. The state's repressive control had the effect of shrinking civic space, effectively limiting avenues for dissent and public protests. The violent suppression of the protests on 5 October 1988 foreshadowed later crackdowns (Berrahmoun 2023). The 'Hirak' ('movement' in Arabic), erupting in February 2019, witnessed weekly mass marches that were halted

in 2021 due to the Covid-19 pandemic. The closure of public gathering spaces in Algiers in 2021, coupled with the excessive use of force and detention of activists, demonstrated the state's determination to prevent any resurgence of the Hirak movement (Human Rights Watch 2022). Overall, the authorities employed a variety of measures to suppress public dissent, including intimidation, surveillance, bans on protests, and violent police responses to gatherings and protests (Freedom House 2024).

With regard to digital innovation, the Algerian state was slow to expand the internet but activists were quick to see the potential of online activism. The initial slow uptake of internet technology can be attributed to several factors, including protectionist economic policies and a significant decline in investment in technological infrastructure during the 1990s civil war (Rabah 2022).

Driven by major reforms in the telecommunications sector, significant growth in internet usage was observed in the last two decades, rising from approximately 150,000 users in the early 2000s (ITU 2004) to 36.2 million (approximately 75 per cent of the population) in January 2025 (Datareportal 2025).

Algerian authorities have consistently exercised a high degree of control over online activity. The legal obligation for internet service providers to monitor and police online content in Algeria dates back to decree 98-257 of 25 August 1998. This decree specifically obligated internet service providers to maintain 'constant surveillance' to prevent access to material deemed 'contrary to public order and morality' (People's Democratic Republic of Algeria 1998). In 2004, a presidential election year, journalists reported significant delays in email communications, with some messages taking up to two days to be delivered, leading to suspicions of state interception (RSF 2004). Security forces frequently raided internet cafes following terrorist attacks in 2007, inspecting browsing histories and requiring cafes to log customer names and ID numbers, and report any suspicious activity to the police (OpenNet Initiative 2009).

In recent years, the Algerian government has continued its monitoring of internet activity under the justification of national security, concurrently adopting surveillance-enabling legislation. A notable example is Ordinance 21-09 of 2021, which permits authorities in specific cases to deploy invasive surveillance tools on individuals without prior judicial authorisation (People's Democratic Republic of Algeria 2021). Furthermore, the lack of transparency surrounding these surveillance programs, including their scope and specific targets, has fuelled concerns (Freedom House 2024). Local associations consistently report frequent state interference and electronic surveillance of communications (ICNL 2024).

3. Case study description

The concept of smart cities has gained interest in Algeria since the launch of the Algiers smart city project in 2017 (Hattab 2018). This project, part of a broader urban development and planning master plan for 2035, aims to integrate advanced technologies into urban management (Ait-yahia Ghidouche and Belghache 2018). In June 2017, the Wilaya of Algiers (State of Algiers) published a call for collaborations soliciting information and feedback from various stakeholders, including startups, research and development labs, solution providers, universities, and consulting firms. The call for collaboration focused on understanding the stakeholders' expertise, perspectives on project design and priorities, descriptions of technical and commercial solutions, and proposed partnership models (Benali 2017). Around 150 responses from 15 countries were received in response to this call (Cathelat 2019). The roadmap proposed by the team responsible for implementing the project prioritised the water, transport, and energy sectors and identified three key strategic priorities: (1) strengthening integration between the local ecosystem and global technology networks to reduce isolation; (2) fostering effective collaboration and technology transfer between global firms and local ventures to mitigate dependency; and (3) boosting the confidence and scalability of local startups (ATRST 2018).

In a follow-up action, a laboratory and an innovation hub were set up in the capital in order to allow innovative smart city solutions to be tested in real conditions (Ait-yahia Ghidouche and Belghache 2018). Additionally, an international conference, titled 'Smart Cities Global Technology and Investment Summit', was held in Algiers under the patronage of the president of the republic featuring local and international delegates (TMT Finance 2018).

The smart city project is concentrated in Algiers and surrounding areas, yet surveillance camera deployment has expanded nationwide including the states of Béjaïa, Blida, and Ouargla, as announced by the minister of the interior who committed to placing surveillance cameras in every street and alley of Algeria (Slimani 2025). Among the related cities is the Sidi Abdellah smart city project, a new city under construction. Located west of the capital Algiers and inaugurated in 2011, this city, promoted as 'intelligent and integrated', promised 'high-tech' infrastructure. It incorporates the first technology park established in the country (Arabeche 2022).

The first surveillance camera network in Algiers was launched in 2003, primarily for traffic control and public safety. Initial installations included about 25 cameras in highly trafficked areas in the capital city and gained

momentum through the 2000s, spreading across major cities and expanding both in scale and scope (Hamidache 2006). Another milestone occurred in 2015, when the General Directorate of National Security (DGSN) announced the plan to cover large provinces through placing high-technology surveillance cameras, but the specific technologies utilised were not disclosed (Echorouk Online 2015b, *Algerie360* 2016). Additionally in the same year, the DGSN started installing smart monitoring cameras, capable of recording environmental conditions such as temperature and humidity, in custody cells across several cities (Echorouk Online 2015a).

Although official figures on the total number of surveillance cameras in Algeria are not publicly available, aggregated evidence from media reports and official announcements indicates that deployments are likely in the low thousands in major cities. Available estimates suggest between 3,000 and 5,000 cameras in Algiers (*Algerie360* 2016, *Algerie360* 2018a) and around 1,600 in Oran (*Ouest Tribune* 2021), while broader national trends appear to point towards a wider diffusion of such technologies. The significant increase in security interventions based on camera footage in 2024 (over 500 per cent increase in one year) as reported by the Ministry of Interior officials points to a substantial network expansion, including both public and private installations (Benkeddada 2025). For example, the video surveillance centre in Blida recorded the processing of 16,893 cases during the first half of 2024 including 13,616 relating to public order and 3,277 others in the area of judicial police as reported in a press release from the communications unit of the Wilaya (provincial authority) (*Le Courrier d'Algérie* 2024).

Among the 'smart' surveillance projects in the capital city is the plan to ease traffic in the capital using an intelligent traffic light management system and the video surveillance system, which incorporates a pilot phase for the expansion of the video surveillance system with 5,592 surveillance cameras in addition to the 1,837 already operational as of April 2024 (APS 2024a). According to the General Directorate for National Security, the automated traffic light management system has helped reduce traffic jams by 20 per cent on the affected roads and the video surveillance system has contributed to a 15 per cent reduction in crime in the monitored areas (Bounabi 2024).

Additionally, several video surveillance monitoring centres were inaugurated in many Algerian cities. In Béjaïa, a province in northeast Algeria, a video surveillance centre was inaugurated in October 2024 to monitor 631 surveillance cameras including 413 fixed cameras and 192 mobile cameras installed across 197 control points set up across the different districts. These cameras are said to include car number plate recognition capabilities (Laouer 2024). Similarly, in Constantine province in northeastern Algeria,

the extension of the network of surveillance cameras has been underway since 2015 and the overall number of cameras is projected to exceed 2,200 (Ghioua 2024).

The network of cameras installed by public authorities is expected to expand as the Ministry of Defence has published national and international calls for purchase of surveillance cameras and related material for the specialised video surveillance institution. The specialised video surveillance institution, ERSV (Établissement de Réalisation des Systèmes de Vidéosurveillance), operates under Algeria’s Ministry of Defence and is responsible for designing, implementing, and maintaining video surveillance and detection systems for public establishments and sensitive zones. The publicly available information on the mentioned procurement calls on the Ministry of Defence website does not advertise the number of cameras, nor the functionalities they need to have (Ministry of Defence 2025).

Table 3: Timeline of national smart city roll-out

Date	Event
2003	First CCTV cameras installed in Algiers as a pilot programme.
2009	Presidential decree 09-337 is adopted, establishing a specialised video surveillance institution.
2011	The Sidi Abdellah smart city project is launched.
2015	Presidential decree 15-228 is adopted, governing the national video surveillance system. The first smart surveillance cameras with enhanced monitoring capabilities are deployed.
2017	The Algiers smart city project is launched, followed by a call for collaborations targeting various stakeholders.
2018	The Smart Cities Global Technology and Investment Summit takes place, and a technology lab and innovation hub are inaugurated.
2024	Huawei is contracted to establish the National Center for Digital Services, a data centre. Regional video surveillance centres are inaugurated.

Source: Author’s own.

4. Key actors

The drivers behind smart city projects and their associated initiatives have consistently been framed within a development-centric narrative (Algerie360 2018b). In the specific instance of the Algiers smart city project, the Wilaya emerged as the principal entity responsible for articulating these developmental drivers and actively advocating for the project among various stakeholders (Wilaya of Algiers 2018). However, the protection of national security, antiterrorism measures, and crime prevention have also been used extensively to justify the deployment of surveillance technologies, specifically the national video surveillance system (Algerie360 2016; Slimani 2025). Additionally, the increased installation of surveillance cameras has, at times, been justified by specific events, such as the 'Constantine Capital of Arab Culture' celebration (Echorouk Online 2015b). The Ministry of Interior and Local Authorities has consistently served as the primary institutional actor driving the accelerated expansion of public space surveillance. Its officials have actively championed the deployment of such systems, frequently articulating and promoting their claimed benefits for public safety and order (Aouadi 2025).

This securitisation driving the expansion of surveillance is operationalised through the national video surveillance system. This system is composed of: a national video surveillance centre; Wilaya (provinces/states) video surveillance centres; and video camera networks installed in public places or open to the public (People's Democratic Republic of Algeria 2015).

The national video surveillance centre is placed under the authority of the prime minister, who delegates its permanent use to the General Directorate of National Security, the national civil police force of Algeria. The Wilaya (state) video surveillance centre is placed under the authority of the Wali (the governor), who delegates its permanent use to local security. The territorial services of the National Gendarmerie Command and the Internal Security Directorate have full access to this system. Civil protection services can be connected to the local video surveillance system, upon approval by the Wali, if necessary (People's Democratic Republic of Algeria 2015).

The implementation and maintenance of video surveillance systems in public places or places open to the public, are the responsibility of the ERSV, established by presidential decree 09-337 of 2009 (People's Democratic Republic of Algeria 2015). It is responsible for the design, development, and installation of these systems, as well as providing aftersales services related to these systems, and providing qualified training for sectors using these systems. Furthermore, the institution may undertake any commercial

operation related to its objectives, provided that it does not jeopardise the training programmes assigned to it. This institution is responsible for evaluating, on behalf of the public authorities, video surveillance arrangements deployed by public and private operators within urban public spaces. Based on its expertise, it contributes to adapting the regulations governing this activity, in accordance with the aforementioned decree, which stipulates that this institution may conduct expertise and implement video surveillance systems at the headquarters of institutions and sensitive sites. It will also be responsible for integrating systems deployed by public and private operators into the overall video surveillance system for public spaces (People's Democratic Republic of Algeria 2009).

Overall, the national video surveillance system is operated and managed by agencies operating under the supervision of the Ministry of Interior. The procurement of material, and the technical deployment, maintenance, and integration of the system is handled by the ERSV, under the Ministry of Defence.

While official details are scarce, a research report linked Russian company Speech Technology Center (SpeechPro) and Huawei to supplying Algeria with technologies including audio and facial recognition systems (Weber 2019). Huawei had previously contributed to the establishment of data centres in the country and opened its first African factory in Algeria (El-Kadi 2024). An article by The Wall Street Journal alleged that the Ugandan government contracted an Algerian team after the company suggested looking at Algeria's 'Huawei's intelligent video surveillance system' (Parkinson, Bariyo and Chin 2019).

In 2024, Huawei was contracted by the Algerian High Commission for Digitization for the creation of the National Center for Digital Services, a data centre aimed to host and centralise national data within the country (APS 2024b). The financial allocations associated with these contracts have not been disclosed.

The initial installation of surveillance cameras in Algiers has evoked a spectrum of responses among legal and human rights communities. Some human rights activists and legal professionals have articulated concerns regarding the potential adverse effects of these systems on the individual freedoms of Algerian citizens, rights which are constitutionally enshrined (Echorouk Online 2011). Conversely, some legal experts, including the former president of the Algerian League for Human Rights, argue that these measures do not inherently infringe upon constitutional principles (Echorouk Online 2011). Public morality and order, thereby contributing to urban security, are the expressed motives. However, even among these proponents, there is a categorical rejection of the use of surveillance equipment equipped

with advanced capabilities for remote audio and photographic capture (Echorouk Online 2011).

Another actor to consider is the National Authority for the Protection of Personal Data (ANPDP), an independent administrative body established to enforce law 18-07 of 10 June 2018, concerning the protection of individuals in personal data processing. The ANPDP's mandate includes supervising data processing, ensuring data controller compliance with the data protection law, and investigating violations (People's Democratic Republic of Algeria 2018). However, its jurisdiction does not extend to state-run surveillance systems, a limitation that significantly restricts its oversight of surveillance technologies, including those relevant to smart city projects.

Finally, smart city infrastructures, such as surveillance networks and traffic management systems, generate digital data and operate over networks that may be vulnerable to electronic surveillance. The state actor concerned is the National Authority for the Prevention and Combating of Crimes Related to Information and Communication Technologies, a specialised agency under the Ministry of National Defence. This authority is tasked with electronic surveillance, and responsibilities for preventing and investigating ICT-related crimes (People's Democratic Republic of Algeria 2009).

Table 4: Key actors

Actor	Roles	Detail
Specialised video surveillance institution – Ministry of National Defence	Implementation and maintenance of video surveillance systems throughout the national territory.	Established by presidential decree 09-337 of 2009 and placed under the supervision of the Ministry of National Defence (People's Democratic Republic of Algeria 2009)
General Directorate of National Security – Ministry of interior	National Video Surveillance centre	Delegated to use and control National Video Surveillance centre as per presidential decree 15-228 of 22 August 2015.
The Wilaya under the authority of the Wali (the governors appointed by the Ministry of Interior)	Local authority over The Wilaya video surveillance centre. Wilaya of Algiers is the primary leader of the 'Algiers, smart city' project.	Delegated to use and control The Wilaya video surveillance centre as per presidential decree 15-228 of 22 August 2015.

Actor	Roles	Detail
The local territorial services of the National Gendarmerie Command and the Internal Security Directorate – Ministry of interior	Full access to the local system	As per presidential decree 15-228 of 22 August 2015.
Civil protection services	Access to the local video surveillance system, upon approval by the Wali (the governor), if necessary	As per presidential decree 15-228 of 22 August 2015.
National Authority for the Prevention and Combating of ICT Crimes – the Ministry of National Defence	Conducts preventive surveillance of electronic communications under judicial authority. Assists the judicial authorities and judicial police services in their investigations of crimes related to information and communication technology Cooperates with national institutions dealing with crimes related to information and communication technologies.	As per law 09-04 of 5 August 2009 and amending laws
National Authority for the Protection of Personal Data (ANPDP)	Grants and withdraws authorisations for the processing of personal data. Addresses complaints falling within its jurisdiction, and accesses personal data undergoing processing for verification purposes. Develops codes of conduct and specific regulations in relation to data processing activities, including video surveillance.	As per law 18-07 on the protection of individuals with regard to the processing of personal data, state surveillance systems in Algeria are generally excluded from the jurisdiction of the ANPDP
Huawei	Data centre installation	

Source: Author's own.

5. Regulatory landscape

Algeria's national video surveillance system is regulated primarily by presidential decree 15-228 of 22 August 2015, together with decree 09-337 of 2009, which establish the legal basis for the deployment and management of cameras in public spaces for security and public order purposes (People's Democratic Republic of Algeria 2015).

In 2018, Algeria adopted a data protection law. Law 18-07 of 2018 on the protection of individuals with regard to the processing of personal data, stipulates the conditions and limitations of the collection and processing of data, including images captured by surveillance cameras. Video surveillance manages animated images of individuals identifiable by their physical, or biometric, characteristics. Thus, video recordings fall within the legal definition of personal data by law 18-07. However, this legislation governs personal data processing by both public and private entities, but it explicitly excludes data collected for national defence, security, and the prevention or prosecution of offenses (People's Democratic Republic of Algeria 2018). This exclusion means surveillance carried out under decree 15-228 (for security/public order) is outside Law 18-07's scope. In practice, therefore, the data protection law applies only to private surveillance systems and public surveillance activities not directly linked to security functions.

Additionally, for private actors, the acquisition and use of surveillance cameras is subject to legal provisions following the executive decree 16-61 of 2016 establishing the security rules applicable to activities involving sensitive equipment and the joint ministerial decree of October 2011 setting out the conditions and procedures for the acquisition, possession, operation, use, and transfer of sensitive equipment (People's Democratic Republic of Algeria 2016).

This section examines Algeria's regulatory framework for public space surveillance through four guiding questions: whether public space surveillance is backed by valid law, pursues a legitimate aim, clearly defines actors and data, and provides independent oversight and redress mechanisms.

Test 1. Are there validly enacted laws governing public space surveillance?

In Algeria, the legal framework for public space surveillance primarily consists of executive decrees, with the notable exception of the data protection law. This dependence on executive decrees raises concerns regarding the potential for executive overreach.

Presidential decree 15-228 of 22 August 2015 governing the organisation and functioning of Algeria's national video surveillance system defines the system as a tool for intelligence and anticipation, composed of: a national video surveillance centre; Wilaya (provinces/states) video surveillance centres; and video camera networks installed in public places or open to the public. Private surveillance systems are not covered unless they impact public order or safety (People's Democratic Republic of Algeria 2015). The decree also outlines specific locations that will be subject to surveillance. These include urban and suburban areas, major roads (especially those with heavy traffic), and public places such as ports, airports, and large sports arenas. The scope also extends to major economic enterprises (People's Democratic Republic of Algeria 2015).

Ultimately, decree 15-228 defines the purpose of the system as to support and assist on-the-ground security teams by providing real-time information. This support aims to optimise their actions by helping to prevent crimes and identify perpetrators, improving public security measures, and increasing the effectiveness of field operations (People's Democratic Republic of Algeria 2015).

Presidential decree 15-228 references decree 09-337 of 2009 which established the ERSV, a public entity responsible for the implementation and maintenance of video surveillance systems in public places. Decree 09-337 defines the ERSV's mission and scope of work as to provide expertise, implement video surveillance systems on behalf of public authorities, and integrate systems deployed by public and private operators into the overall public space video surveillance system (People's Democratic Republic of Algeria 2009).

Test 2. Do the laws define a legitimate aim for public space surveillance?

The legal frameworks governing the national video surveillance system in Algeria lists an explicit legitimate aim. Nevertheless, the framework lacks key safeguards, including provisions for data retention and utilisation limits, independent oversight, and explicit restrictions against extending its application beyond its defined purposes.

Decree 15-228 outlines the aims of the national video surveillance system as to: preserve public security, assist in the prevention and detection of crimes, and support emergency interventions and traffic monitoring. It applies to video surveillance installed in public spaces, government facilities, and critical infrastructure, particularly in areas considered sensitive for public safety (People's Democratic Republic of Algeria 2015).

Article 17 of the decree stipulates that the retention period for video recordings made by the national video surveillance system and the

procedures for destroying these recordings shall be specified in a specific legal text (People's Democratic Republic of Algeria 2015). No regulation had yet been adopted in this regard. The absence of clear rules on data retention is a significant legal void for such a system collecting vast amounts of personal data.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

Decree 15-228 clearly defines who may access surveillance data. Access to the system's data is granted to security agencies including the General Directorate of National Security and the National Gendarmerie Command in addition to civil protection services which necessitate the approval by the Wali (the governor) (People's Democratic Republic of Algeria 2015). However, it does not institute independent oversight or judicial access, or restrict data sharing beyond those actors.

The decree does not include any provisions detailing the nature of data that can be collected, offering no restrictions or definitions regarding video types, audio, recognition technologies, metadata, or sensitive data constraints.

Test 4. Does the law define mechanisms for citizen redress and accountability?

Accountability

The governing legal framework does not provide an independent and adequate resourced regulatory mechanism overseeing the administration of surveillance data.

The national video surveillance system is managed through a hierarchical structure: the national surveillance centre is under the supervision of the prime minister, with operational control delegated to the director general of national security. Wilaya-level centres are overseen by the Wali, with local operations handled by the security services (People's Democratic Republic of Algeria 2015). Both the director general of national security and the security services are subordinate to the Ministry of Interior. Overall, the decree centralises authority within executive and security structures. It does not set up independent oversight or impose accountability mechanisms.

The data protection law 18-07, which applies only to private video surveillance data and public non-security video surveillance data, stipulates in its article 18 that processing sensitive data is only permitted with explicit consent, national authority authorisation, or for essential public interest functions (CIPESA 2022) Any processing of sensitive data outside of these exceptions constitutes a criminal offense punishable by two to five years imprisonment and fines of 200,000–500,000 Algerian

dinars (approximately US\$1,465–3,654) (CIPESA 2022). Law 18-07 also established the National Authority for the Protection of Personal Data (ANPDP) as an independent administrative body to serve as the sole point of contact for personal data issues (People's Democratic Republic of Algeria 2018). Additionally, law 18-07 gives the ANPDP the possibility to investigate violations and develop codes of conduct to detail conditions such as data retention period and guarantees concerning the rights of data subjects in relation to data processing activities, including video surveillance. No such instruments had been communicated by the ANPDP to date (People's Democratic Republic of Algeria 2018).

Additionally, with regard to private actors installing and operating video surveillance cameras, prior authorisation is required in addition to the commitment of the video surveillance system operator not to connect it to an internet network. Applications for authorisation to operate video surveillance equipment must be accompanied by a site plan of the operator's buildings and their immediate perimeter, clearly indicating the precise location of the cameras and the identification details of the operator(s) responsible for the video surveillance system, its operation, and its maintenance (People's Democratic Republic of Algeria 2011).

Previously, security agencies have undertaken verification campaigns to ascertain the adherence of businesses, including shops, commercial centres, companies, and warehouses, to regulations concerning the installation of surveillance systems on their premises (Bachouche 2024). Additionally, media outlets have reported in 2013 that the Algerian police force seized advanced surveillance cameras, manufactured in Israel, that were being sold in various markets in the east of Algeria. These devices were reportedly imported by individuals through European countries, circumventing the scrutiny of various oversight bodies of security and customs services across Algerian airports and ports (Echorouk Online 2013).

In conclusion, the existing accountability and oversight mechanisms are restricted to private video surveillance data and public non-security video surveillance data. These mechanisms do not extend to security-led public space surveillance conducted under decree 15-228.

Redress

Decree 15-228 governing the national video surveillance system does not include any provisions for citizen complaints, remedies, or sanctions against misuse of surveillance data.

Law 18-07, which applies to private video surveillance data and public non-security video surveillance data, grants data subjects rights to information, access, objection, and rectification. In cases of infringement, they may seek

interim measures or compensation through the competent jurisdiction. Correspondingly, the data controllers must implement suitable technical and organisational safeguards (People's Democratic Republic of Algeria 2018).

6. Digital rights

Algeria is party to several international human rights instruments that provide for the protection and promotion of human rights, such as the Universal Declaration of Human Rights, the International Covenant on Civil and Political Rights, and the African Charter on Human and Peoples' Rights. Moreover, the 2020 constitution establishes protections for related rights, including the right to privacy, and the freedom of assembly (People's Democratic Republic of Algeria 2020).

The technologies used in smart city projects worldwide have demonstrated the considerable implications these technologies have on the exercise of human rights (Article 19 2021). The primary rights infringed include the right to privacy, the right to freedom of expression, and the right to freedom of assembly. For instance, CCTV cameras collect personal data that brings up a host of data protection concerns, especially worrying in jurisdictions with limited regulatory provisions about the collection, use, transfer, and retention of personal data like Algeria. When CCTV cameras include capabilities such as face recognition, the implications for human rights are amplified, because these surveillance technologies facilitate the widespread identification and tracking of individuals in public spaces, often without their explicit consent, raising risks of chilling free expression, exacerbating discrimination, and undermining their right to privacy (Article 19 2021).

This section assesses the public space surveillance system in Algeria, analysing whether it is necessary and proportionate, includes sufficient access controls, and is subject to an independent oversight mechanism.

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

No, the privacy violations arising from the use of surveillance data are not considered necessary and proportionate to achieve the stated legitimate aim.

While the governing laws identify legitimate aims, they fail to establish a robust framework to ensure that the methods used are the least intrusive necessary to achieve those aims. The absence of independent oversight, transparency, and clear data retention rules means that human rights violations could be disproportionate to the stated aims. The legal framework, therefore, appears insufficient to prevent potential abuses.

The establishment of the ANPDP is a positive development, yet its explicit exclusion from overseeing state surveillance systems significantly limits its legal reach and effectiveness. Consequently, the legal

framework's proportionality is compromised by expansive surveillance capabilities without corresponding limitations or oversight.

Additionally, there is no public record of a human rights impact assessment or other forms of risk analysis conducted prior to the Algerian state implementation of mass surveillance of citizens in public spaces including the installation of video surveillance cameras,

Overall, the current framework governing public space surveillance is limited by apparent gaps in implementation and oversight. The absence of appropriate impact assessments, incomplete legislation, and weak regulatory oversight collectively suggest that the framework does not comprehensively meet the criteria of being legal, necessary, and proportionate.

Test 6. Does the law control who has access to surveillance data?

Access to data from the national video surveillance system is granted to specific agencies under the Ministry of Interior, including the General Directorate of National Security, the National Gendarmerie Command, and civil protection services (People's Democratic Republic of Algeria 2015). Presidential decree 15-228 of 22 August 2015 addresses data retention in article 17, stating that the duration and destruction protocols for video recordings will be detailed in a future regulatory text. While this provision suggests a limitation on data retention, the absence of the promised text leaves this protection unfulfilled. Decree 15-228 establishes a legal framework for a state video surveillance system, but it does not provide robust and explicit protections for the personal data of the individuals being filmed and ensuring public accountability. The crucial questions of access, use, retention, and supervision are either left unanswered or deferred to future regulatory texts.

Test 7. Does the law create an independent oversight body?

The decrees governing the national video surveillance system create a system of surveillance with no mention of an independent body to oversee these operations and ensure they don't violate human rights. This concentration of power without external checks raises concerns about potential abuse.

Additionally, the 2020 constitution states that laws must be amended to comply with the constitution within a 'reasonable time' (People's Democratic Republic of Algeria 2020). However, civil society organisations reported that years since the adoption of the constitution, several national laws that contradict and undermine constitutional human rights protections remain in force (Voule 2024). For instance, decree 15-228 of 22 August 2015 has not

been amended to address the lack of an independent oversight body. This decree, which governs Algeria's national video surveillance system, lacks a mechanism to ensure that surveillance operations are consistent with national human rights commitments.

While the National Authority for the Protection of Personal Data was established as a positive development, its effectiveness as an oversight body is significantly limited. The law explicitly excludes this body from overseeing state surveillance systems, thereby compromising its authority. Consequently, the legal framework lacks a fully independent and adequately funded oversight mechanism to ensure that public space surveillance operates in accordance with the law and aligns with national human rights commitments.

7. Conclusion

This analysis of Algeria's smart city plans highlights critical shortcomings in governance and implementation. The research examined the available evidence on the procurement, application, and oversight of technologies like CCTV, whether deployed under smart city projects or for general urban development. While the principal smart city initiative was restricted to the capital and promoted as a project of urban modernisation, the widespread deployment of surveillance cameras has been predominantly justified by security and public order imperatives.

Furthermore, the legal framework governing public space surveillance exhibits excessive delegation of authority to the executive, lacking clarity in limitations. The absence of robust safeguards against mission creep and insufficient redress mechanisms raise serious concerns about data management and the protection of individual rights. Additionally, the procurement of surveillance technologies is characterised by a notable lack of transparency, with minimal public information available concerning vendors, usage protocols, and system management. This approach poses a significant threat to citizens' digital rights. Overall, the system is found to be neither necessary nor proportionate, as the lack of transparency, clear regulations, and independent oversight creates a high risk of disproportionate privacy violations.

In conclusion, these findings underscore the urgent need for enhanced legal clarity, stronger independent oversight, and robust risk management to protect individual rights and freedoms.

8. Recommendations

State actors should:

- Provide for public consultations on issues such as the human rights implications of the deployment of surveillance technologies in public spaces,
- Ensure the highest level of transparency and public oversight on public procurement processes for the acquisition, development, and deployment of these technologies,
- Perform and publicly disseminate adequate human rights assessments for smart city projects,
- Adopt an adequate legislative framework with specific provisions to safeguard human rights to ensure that any surveillance is legal, necessary and proportionate.

Private actors should:

- Ensure the protection and respect of standards on business and human rights when designing, developing, and deploying smart city infrastructures,
- Regularly publish their human rights impact assessments and risk assessment reports.

Civil society organisations should:

- Strengthen their monitoring and documentation of evidence of harm experienced by citizens as a result of the deployment of these surveillance technologies,
- Promote public awareness regarding the human rights implications of smart city and related technologies deployment.

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Smart city surveillance in Egypt

Afef Abrougui

1. Introduction

Egypt has a long history of state surveillance that predates the digital age. However, it is under the Sisi regime that Egypt is witnessing some of the most significant changes in public space surveillance. Since rising to power in mid-2013, the Sisi regime has taken steps to expand deployment of surveillance cameras in law and in practice. Most remarkably, the government has embarked on an ambitious smart cities programme that would make the surveillance of public spaces and tracking of Egyptians much easier. In the development of smart cities, the government has mainly adopted a 'largely idealistic' narrative, completely disregarding the potentially negative impacts not only on human rights, but also on the economy, urbanisation, and the environment (Rowaq Arabi 2024). In reality, however, developments in public space surveillance under Sisi, including the new smart cities his regime is building, is centred around control to ensure regime survival and the preservation of its political and economic interests. The military regime, still reeling from the 25 January uprising in 2011 that toppled Hosni Mubarak, the protests opposing its removal of Morsi from power in 2013, and a subsequent deadly response by the security forces, has for over a decade expanded its control of public spaces and people, including through surveillance, to avoid a repeat scenario.

This report documents developments in policies and practices of digital surveillance of public space in Egypt under the presidency of Sisi with the purpose of improving understanding of their impacts on human rights. In particular, the report aims to answer the following questions:

- What changes has Sisi introduced to expand digital public surveillance? And what role does smart city surveillance of public space play in Egypt under his presidency?
- To what extent do existing regulations provide protections against the abuse of public space surveillance?
- How do these developments affect human rights and who is set to benefit?

2. Background

Public space surveillance in Egypt is a practice that predates the modern state, when the rulers of Ottoman Egypt (1517–1798) deployed secret spies (el-Hamalawy 2023). Under British colonial rule, authorities modernised the public surveillance system, and established City Eye, ‘an expansive network of informers, or more accurately, common folk reporting any suspicious activities in return for modest rewards; these included beggars, porters, vendors, cab drivers, telephone operators, and scores of other people’ (Kandil 2012: 30). They also set up the Interior Ministry in March 1895, and in 1911, the Special Section tasked with domestic surveillance (Kandil 2012). The Special Section was expanded and renamed the General Investigations Directorate (GID), before it was once again renamed to State Security Investigations Sector (SSIS) in 1971 (Kandil 2012).

After Egypt became a republic in 1953 following a 1952 military coup led by a group of military officers, known as the Free Officers, the surveillance infrastructure was once again kept and expanded, creating a climate of constant and rife surveillance (Kandil 2012). In the new republic, surveillance relied heavily on informants and agents (el-Hamalawy 2023), but increasingly authorities began acquiring surveillance and police equipment from foreign governments, mainly the CIA in the US, the KGB, and the Eastern German Stasi intelligence service (Kandil 2012).

Public sphere surveillance continued to expand under the presidency of Anwar Sadat (1971–1980) whose policies and increased cooperation with the US turned Egypt into a police state (Kandil 2012). By making military and political concessions to the US, which most notably included a peace treaty with Israel, the US supplied the regime of Sadat with protection and surveillance technologies to help him stay in power. This included mobile listening posts, surveillance cameras, and telephone-tapping technologies (Kandil 2012).

Under the thirty-year-rule of Hosni Mubarak (1981–2011), the SSIS became increasingly feared and notorious for serious human rights abuses and keeping dissent in check, targeting dissidents with detentions, torture, and arbitrary arrests. A state of emergency that lasted his entire rule allowed for unchecked surveillance (BBC 2011).

Following a popular uprising that toppled Mubarak, the SSIS was dissolved in 2011 and replaced with a new agency called the National Security Agency (NSA) (Amnesty International 2021). However, its old leadership largely remained to serve at the NSA (Kandil 2012; Amnesty International 2021). Since

the 2013 military coup led by Abdelfattah al-Sisi against the country's first civil president Mohamed Morsi and the Muslim brotherhood, the agency has played a key role in rounding up thousands of people perceived to be a threat to the regime, including human rights defenders, members of the political opposition, journalists, protesters, and academics (Amnesty International 2021).

The Sisi regime deploys a range of tactics to surveil the public sphere. NSA monitors human rights defenders and political activists through regular 'check-ins'¹, summons, and questioning, placing people under police probation (Amnesty International 2021). The government still relies on an informal network of informants and agents to monitor the streets (Al Jazeera 2020).

Surveillance cameras are also 'widely used in public places' (Privacy International 2019). Their deployment dates back to at least the 1980s, under the presidency of Anwar Sadat, when the US provided him with surveillance cameras 'to monitor major thoroughfares and public squares in Cairo' (Kandil 2012: 224). However, under Sisi the deployment of surveillance cameras accelerated, including at the premises of businesses such as cafes, restaurants, shops, and entertainment venues, in addition to religious institutions (El-Hamalawy 2023), universities, and metro stations (Masaar 2023). It is unclear when exactly facial recognition cameras were first deployed in the country, however, authorities reportedly signed a deal to acquire the technology for the first time in 2013, following the military coup (Masaar 2023). Additionally, the government has been building smart cities that feature the deployment of facial recognition cameras further intensifying public space surveillance (Waisová 2022).

¹ It is a common practice to subject those perceived to be a threat to police probation where they are required to report to NSA offices on a regular basis or risk arrest.

3. Case study description

Since Sisi took over in a military coup in 2013, his regime has sought to expand public space surveillance. In 2015, the religious endowments minister announced a national plan to install CCTV cameras in all the country's mosques but also inside to 'monitor the Imams, workers, as well as the ideas, religious teachings, and Friday sermons preached, to ensure they were in accordance with the ministry's thought and unified sermon' (Abdelaal 2015). That same year, the prime minister established the Permanent National Committee for Security Coordination of the Video Surveillance Camera System to coordinate CCTV surveillance across the country and linking the visual feed of all CCTV cameras to a national grid (El-Hamalawy 2023). It is unclear if these cameras are equipped with facial recognition, however, authorities have increasingly been acquiring and deploying facial recognition technologies since 2013 (Masaar 2023). In 2019, a new law that required businesses including restaurants and shop owners, to install surveillance cameras outside and inside their premises, came into effect (law 154 of 2019).

Surveillance cameras are also deployed at universities. In fact, since the 2013 coup, and to prevent student protests opposing the regime, the government has required universities to up their security, collaborating together to launch an 'integrated security system that will blend the use of internal and external security operators, monitoring devices and even bombs' (Al-Fanar Media 2014). Since then, universities have invested in a range of security measures, including CCTV cameras (*ibid.*). In 2017, Spanish company Sacti was selected to install an internet protocol CCTV security system of 349 cameras at Egypt's South Valley University, which has its main campus in Qena, a city located 600 km from Cairo (asmag 2017). Sacti's cameras are equipped with facial recognition capabilities that allow for intelligent video analytics so that they 'proactively' and 'automatically' alert operators when they detect suspicious activities (asmag 2017). In one incident from 2018, Alexandria University's pharmacy faculty went as far as installing surveillance cameras in the premises of students' restrooms before deciding to take them down as a result of backlash from students (Nabbout 2018).

The most significant public surveillance development initiated by the Sisi regime is an urban transformation strategy that aims to see a significant part of the population living in smart cities. The drive for new smart cities has its roots in a series of political, economic, and social reforms spearheaded

by Sisi, whose presidency has been characterised by the launch of several megaprojects (Embassy of Egypt 2018).

In 2016 Sisi launched Egypt Vision 2030, updated in 2023 (Ministry of Planning and Economic Development 2023), which listed ‘establishing new cities on a new development axis’ as one of its programmes (Green Policy Platform n.d.). The government states that smart cities will improve the quality of life, ease overcrowding and traffic congestion, meet the housing demands of a growing population and create jobs, however, the ubiquity of smart city tech in these cities further exacerbates public space surveillance and encroaches on fundamental freedoms (Abrougui 2025).

Figure 1: Expansion of public space surveillance under Sisi: key milestones

Year	Key milestones
2014	Implementation and maintenance of video surveillance systems throughout the national territory
2015	Permanent National Committee for Security Coordination of the Video surveillance Camera System established
2016	Launch of Egypt vision 2030 and smart cities programme and construction of the NAC started
2019	A law (Law No. 154 of 2019) that requires all businesses to install surveillance cameras inside and outside their premises came into effect
2019	Honeywell struck a deal to supply NAC's surveillance system.
2020	The Data Protection law (Law No. 151 of 2020) adopted
2025	As of early 2025, 24 smart cities were being developed in Egypt and the NAC had a population of 80,000 people

Source: Author's own.

As of early 2025, there were 24 new smart cities being developed on greenfield sites in the country (Abou Shtet 2025). Additionally, authorities are working on transforming dozens of existing cities into smart ones; this was reported to be 60 cities as of late 2024 (Amwal Alghad 2024). The website of the Ministry of Housing, Utilities and Urban Communities lists nine cities currently under development that each will accommodate at least 1 million people.

Table 5: New smart cities at more advanced stages and with the plan to accommodate more than 1 million people

Project/city name	Start date and key milestones	Description
The New Administrative Capital (NAC) – in progress	Project announced in March 2015 (Arab Urban Development Institute n.d.a). Construction began in 2016 (Lewis 2024). As of 2023, all ministries and government entities relocated to the NAC (State Information Service 2023).	This is the biggest of the smart city projects in the country and it is located 45 kilometres east of Cairo. It has a target population of 6.5 million residents. As of early 2025, 80,000 people lived in the city (El-Huti 2025). It houses an administrative district with 30 ministries, a new presidential palace and the new headquarters of the parliament (Ravenscroft 2025). It has a network of 6,000 surveillance cameras (Walsh and Yee 2022).
New Alamein City – in progress	The new city was established in 2014 by presidential decree (Abougabal 2017) and was officially inaugurated in 2018 (Arab Urban Development Institute n.d.b).	Located on the northwest coast, it has a targeted population of 3 million residents and accommodated 10,000 people as of August 2024 (HAPI Journal 2024). The city was planned based on a technical study by UN-Habitat (Abougabal 2017). It features a smart network of surveillance cameras (Abdel-Ghani 2021) and a command centre that centralises all city information (Giza Systems n.d).
New Mansoura City – in progress (Presidency of the Arab Republic of Egypt 2022)	Established by presidential decree in 2017. First phase completed (New Mansoura 2022) and inaugurated in 2022 (Ahram Online 2022). The second phase is underway (Deeb 2025).	Located on the Mediterranean coast, this city has a target population of 1.5 million residents (Ahram Online 2022). The city has a video surveillance network (New Mansoura 2022) and a control and command centre (Ministry of Housing, Utilities and Urban Communities n.d.).

Source: Author’s own

In addition to the cities discussed above, the Egyptian government is developing other cities that are still at initial stages such as the new October 6 city, built as an extension of the existing October 6 city and expected to accommodate 4.7 million people once completed (Ministry of Housing, Utilities and Urban Communities 2019). The government is also working on transforming existing cities into smart cities. This is for instance the case for Badr City, established in 1982 and located in the Cairo Governorate. The city is currently undergoing urban developments into a smart city

with its population, estimated to be above 31,000 people in 2017, planned to increase to 840,000 residents (Mahmoud 2025). The project is partially funded by Korea and with technical support from UN-Habitat (El-Din 2024a).

Egypt's smart cities rely on Integrated Security Systems or ISS (Masaar 2024), 'a comprehensive, interconnected network of diverse security components that work together to provide a cohesive and unified approach to security management' (Isarsoft 2024b). These systems rely on the use of advanced technologies to gather and analyse data such as facial recognition, access control systems, and motion and intrusion detection systems to secure cities. Data collected from multiple sources feed into control and command centres that monitor and manage critical operations and infrastructures, including by detecting and analysing incidents and coordinating responses to them (Isarsoft 2024a).

The NAC smart city, for instance, features a surveillance system of more than 6,000 surveillance cameras, manufactured by US company Honeywell, feeding into a command and control centre that will run 'sophisticated video analytics to monitor crowds and traffic congestion, detect incidents of theft, observe suspicious people or objects, and trigger automated alarms in emergency situations' (Thomson Reuters Foundation 2023). The Honeywell systems are reportedly worth US\$31 million (Enterprise 2018). Lamp posts will be fitted with Wi-Fi capabilities (Thomson Reuters Foundation 2023) and the city will manage waste and the use of gas, electricity, and water with AI (Lewis 2024).

Similarly, in the New Alamein smart city, a command centre will gather all city data (Giza Systems n.d) 'centralizing information to enable smarter decision-making and enhanced management of city resources' (El-Din 2024b). One of the neighbourhoods called the Latin District will 'use a smart network to operate security cameras and smart police services' (Abdel-Ghani 2021). In Badr City, one neighbourhood development project includes smart systems for managing traffic, lighting, and emergency services (Arab City Group 2024).

4. Key actors

A range of actors are involved in the deployment of public surveillance and development of smart cities in Egypt, mainly government agencies, and the military, local and international companies, real estate developers, foreign funders (including foreign governments), and United Nations agencies.

4.1 The Egyptian government and military

Entities engaged in public space surveillance

There are several entities in the country that are responsible for public space surveillance. There is the Permanent National Committee for Security Coordination of the Video Surveillance System, established in 2015 by the prime minister (Arab Republic of Egypt 2015) with the mandate to coordinate CCTV surveillance across the country and linking the visual feed of all CCTV cameras to a national grid. It is chaired by the assistant interior minister for communication systems and information technology, and includes in its membership representatives from Egypt's three intelligence agencies, the General Intelligence Service (GIS), NSA, and Military Intelligence Services and Reconnaissance Administration (Arab Republic of Egypt 2015).

Egypt has three main agencies tasked with surveillance, the GIS, NSA, and the Military Intelligence Services and Reconnaissance Administration. The GIS is Egypt's most prominent intelligence agency, responsible for domestic and foreign national security intelligence. It works directly under the authority of the president, and is only accountable to him. He also appoints its head (Privacy International 2016). Previous research from Privacy International uncovered a secret unit called the Technical Research Department (TRD), believed to be affiliated with the GIS, that has been acquiring surveillance technologies, including CCTV cameras (Privacy International 2016). GIS has also been previously involved in 'handling the CCTV installations in the education sector – university campuses, dorms and high school classrooms during final exams – and public squares' (el-Hamalawy 2023).

The NSA is a specialised police force tasked with policing terrorism and other perceived national security threats (Amnesty International 2021). It is a key entity in the regime's repressive apparatus, responsible for surveilling and cracking down on dissent (Amnesty International 2021). Under the authority of the Ministry of Defence, the Military Intelligence Services and Reconnaissance Administration handles military-related intelligence. The army also conducts public space surveillance in sensitive areas. For instance, it deploys security cameras, and possibly drones, on the border

with Gaza, with collected data feeding into a control and command centre. Additionally, the military's own Arab Organization for Industrialization, through its electronic factory, has been collaborating with a local company, ZeroTech, to produce Egyptian-made surveillance cameras (Sobhi 2022). On its website, ZeroTech boasts its cameras are equipped with facial and number plate recognition technologies (ZeroTech Systems n.d.).

Government entities involved in development of smart cities

For decades, the military in Egypt has engaged in economic activity, and under Sisi its economic role grew even more influential (Sayigh 2015). Military agencies and companies affiliated with it are active in construction, public infrastructure, and housing projects, including smart cities (Sayigh 2015).

Key military agencies and companies include the Engineering Authority of the Armed Forces, the Armed Forces Land Projects Agency (Sayigh 2015), and the National Service Projects Organization (NSPO). The Engineering Authority of the Armed Forces is an agency at the Ministry of Defence that plays key roles in 'the establishment of major development projects and infrastructure works' (Engineering Authority of the Armed Forces on LinkedIn n.d.). The authority is implementing the construction of the NAC along with a number of other companies and agencies (Presidency of the Arab Republic of Egypt 2017). The Armed Forces Land Projects Agency (Sawaf 2016) was first set up by presidential decree in 1981 to manage the sale of desert land previously in use by the armed forces. Amendments introduced by Sisi, expanded the agency's mandate allowing it to engage in commercial activity (Sayigh 2015). The agency is tasked with overseeing construction of the NAC and the extension of Sheikh Zeyad City (Sawaf 2016). Affiliated with the Ministry of Defence, the NSPO supports 'national development plans' and owns several companies involved in construction, infrastructure, and urban development projects (National Service Projects Organization n.d.a) through companies like the National Company for roads building and development (National Service Projects Organization n.d.b) and the National Company for General Construction & supplies (National Service Projects Organization n.d.c).

Other key government entities include the Ministry of Housing, Utilities and Urban Areas and the New Urban Communities Authority affiliated with the ministry. Article 2 of presidential decree 164 of 1996, the internal system regulating the Ministry of Housing and Utilities states:

the jurisdiction of the Ministry of Housing, which includes but is not limited to designing the housing, utilities, urban communities and construction projects; supervising infrastructure programmes for drinking water and sanitation; studying and preparing plans for urban development of cities, villages and new communities and deserts and developing

standards, models and rates in the field of housing. In order to carry such responsibilities, several authorities, organisations and centres fall under the management of the Ministry of Housing" (UN-Habitat 2015). The New Urban Communities Authority (NUCA), affiliated with the ministry, was established in 1979 to implement the government's New City initiative with the aim to ease overcrowding and transportation problems in Cairo and decentralise facilities from the capital (Tipple 1986). It is playing an active role in the development of smart cities and along with the National Service Projects Organization and the Armed Forces Engineering Authority, it established the Administrative Capital for Urban Development (ACUDO, the owner and developer of the Administrative Capital. (Economic and Commercial Office of the Egyptian Embassy in Berlin n.d.)

4.2 Companies in the ICT sector

Technology and telecommunication companies, local and foreign, play a key role in the supply of public surveillance technologies.

In terms of foreign companies, Honeywell, a US company that provides various services to a range of industries globally, mainly in aerospace, building automation, energy and sustainability, and industrial automation (Honeywell n.d.a), has emerged as a key player in Egypt's smart city infrastructure. It is developing and providing the NAC's surveillance system, which includes a network of 6,000 surveillance cameras and a command and control centre (Thomson Reuters Foundation 2023).

The authorities also deploy surveillance cameras manufactured by Chinese companies Hikvision and Dahua (Masaar 2023). Global provider of ICT infrastructure and smart devices, Huawei, which is headquartered in Shenzhen, China (Huawei n.d.a), is involved in key infrastructure projects, supporting the deployment of the country's 5G network in collaboration with state-owned Telecom Egypt (Ranjan 2025), and supporting the country's digital transformation (ITIDA 2024). It is also supporting smart city development through its 'smart city solution', which consists of 'a high-speed, reliable, and secure network that can support a wide range of smart city applications' that include smart transportation, smart lighting, and sensors (Huawei n.d.b).

Additionally, with the government's focus on smart cities, companies that provide technologies necessary to power these cities, for instance, through power storage, have also emerged as key players.

For instance, French multinational provider of telecommunication and digital services, Orange (Orange n.d.), through Orange Business Services and Orange Egypt, is the provider of NAC's data centre (Barton 2021). In addition, e& Egypt (formerly Etisalat Misr) a subsidiary of the state-owned Emirati global technology and telecommunication group e& (etisalat and n.d.), worked with Honeywell on the NAC's City Operations Center (Honeywell n.d.b). It also signed an agreement with Modon Holding, the developer of Ras El Hekma mega project, for the 'design, implementation, and management of an integrated smart infrastructure', which will include 'digital communication networks, optical fibers, 5G networks, and smart life applications' (Hesham 2024).

There are also a number of smaller local companies involved in the provision of smart city systems such as Hassan Allam Technologies (Hassan Allam Holding 2022) and Advanced Computer Technology (ACT 2022), which in 2022 won a contract to provide an ICT system integration for the four high-rise New Alamein Towers in New Alamein City, including security systems. Another company, Giza Systems, won the contract for the development of the Command Operations Control (COC) data centre of the city to 'enable the centralization of all data linked to Al Alamein City, offering real-time data to enhance decision-making' (Giza Systems n.d.).

In addition to companies supplying technologies for smart city systems, other companies have also been involved in the supply of surveillance cameras not necessarily destined for smart cities, such as Spanish company Scati, which provided 349 cameras for Egypt's South Valley University (asmag 2017) and local company ZeroTech, which has been working with the military to produce Egyptian-made CCTV cameras (Sobhi 2022). Both companies supply AI-equipped cameras such as those that enable recognition of faces and the number plates of vehicles.

4.3 Other actors

In addition to the Egyptian government and the private sector, there are also other actors that are shaping the development of smart cities in the country. In particular, UN-Habitat has been engaged as a key actor, supporting the Egyptian government with establishing standards for smart cities (UN-Habitat 2019) and conceptualising a plan for the Ras El-Hekma smart city mega project (UN-Habitat n.d.).

In terms of foreign governments, the UAE is a key investor for Ras EL-Hekma, funding and developing the mega project through the Abu Dhabi Developmental Holding Company, in collaboration with the New Urban Communities Authority (Arab Urban Development Institute n.d.c). There is

also involvement from Korea in Badr City through the K-City Network Global Cooperation Program by designing the city's master plan (Smart City Korea 2023). China's state-owned China State Construction Engineering Corporation (CSCEC) is involved in construction, for instance, in the NAC (Masaar 2024).

Finally, strong participation from civil society in shaping the development of urban areas, including smart cities, is lacking (Salama 2024). A number of organisations and researchers have, however, been attempting to scrutinise the development of such cities and the deployment of public space surveillance and their implications for fundamental rights through research and writing, such as Masaar (Masaar 2024).

5. Regulatory landscape

The right to privacy is enshrined in the Egyptian constitution, which states in article 151 that 'Egypt is obliged by all international human rights conventions that it has ratified, and they have the same power as the law once published' (Roberts *et al.* 2021). The country is also a party to several international human rights instruments that enshrine the right to privacy, such as the Universal Declaration of Human Rights (UDHR) of 1948 and the International Covenant on Civil and Political Rights (ICCPR) of 1966 as well as the African Charter on Human and Peoples' Rights of 1980. In practice, however, state agencies constantly infringe on this right and domestic laws that allow for surveillance, including the deployment of smart city technology, are not adequate in ensuring the legality, necessity, and proportionality of surveillance practices (Roberts *et al.* 2021).

The remainder of this section responds to a series of tests concerning the legal framework and human rights alignment of Egypt's smart surveillance of public space. These tests are derived from the CIS framework of Bhandari *et al.* (2020).

Test 1. Are there validly enacted laws governing public space surveillance?

Law 154 of 2019 (Arab Republic of Egypt 2019) requires businesses and shops such as cafes, restaurants, and entertainment venues to install cameras outside and inside their premises. Egyptian law also requires surveillance cameras to be installed at hotels and touristic venues under law 8 of 2022 (Arab Republic of Egypt 2022).

There is also the prime minister's decision 1032 of 2015 (Arab Republic of Egypt 2015), which established the Permanent National Committee for Security Coordination of the Video Surveillance Camera System (decree 1032 of 2015). The committee is tasked, among other responsibilities, with 'assessing the current situation to secure the Republic's governorates with surveillance cameras and linking them to the national security surveillance network' and 'putting in place an effective and elaborate plan to secure vital areas, buildings and facilities, traffic axes, roads and major squares by installing surveillance cameras'. However, none of these laws define the purpose and scope of public space surveillance.

There is also the Emergency Law which provides broad surveillance powers to the executive and security agencies, allowing in article 3 for searches of people, spaces, and correspondence (Arab Republic of Egypt 1958) without adequate safeguards and protections (Abdelaal 2017).

There is a legal void concerning the deployment of smart city technologies that may infringe on residents' right to privacy in public spaces. The Egyptian government published guidelines to organise smart cities, specifically the Egyptian Code for Management, Operation and Sustainability of Smart Cities (Housing and Building National Research Center n.d.). The code lays out guidelines for smart city infrastructure and management such as sustainability of smart buildings, smart governance and economy, smart transportation systems and water and waste management, and smart recycling (Ousoul Misr Magazine 2021). However, it does not address management of privacy and human rights risks related to smart cities or data protection requirements and limits (Ousoul Misr Magazine 2021).

Test 2. Do the laws define a legitimate aim for public space surveillance?

Authorities frequently justify surveillance on the grounds of national security, states of emergency, terrorism, and cybercrime (Roberts *et al.* 2021). However, law 154 of 2019 requiring businesses and shops to install surveillance cameras and decree 1032 of 2015 establishing the Permanent National Committee for Security Coordination of the Video Surveillance Camera System do not provide a legitimate aim as they fail to define the purposes of public space surveillance and their limits. Additionally, the law 162 of 1958 on the Emergency Law provides for restrictions on fundamental freedoms and rights, including public space surveillance, in case security or public order is threatened, and in cases of internal unrest, public disasters, or the spread of an epidemic.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

A range of government actors in Egypt are involved in public space surveillance. These are the GIS, NSA, the Military Intelligence Services and Reconnaissance Administration, and the Permanent National Committee for Security Coordination of the Video Surveillance System. There are no limits established by law that limit these agencies' access to surveillance data and their use of such data. The law also further fails to define the nature of data that can be collected. Further, the law does not specify limits on access, collection, and use of smart city data. For instance, the company that owns the NAC, controlled by the government and the military, has all access to city data with no oversight (Waisová 2022). Further, there is no transparency about how the NAC city management handles the data of inhabitants, and what limits (if any) are there for collection, sharing, and use of their data (Waisová 2022).

Test 4. Does the law define mechanisms for citizen redress and accountability?

The data protection authority hasn't been established yet, and in law, it lacks independence and the mandate to oversee surveillance by national security agencies, including video surveillance of public spaces (Law 151 of 2020 Promulgating the Personal Data Protection Law). Additionally, national security agencies frequently abuse their surveillance powers (Roberts *et al.* 2021), with no adequate mechanisms in place that can allow citizens to submit complaints and seek redress.

6. Digital rights

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

A human rights impact assessment is a method used by governments and companies to ensure that planned work is consistent with their existing commitments to human rights and avoids negative impacts on people's rights.

There was no evidence that the government of Egypt or any of its implementing partners conducted any human rights impact assessment before deploying smart city public space surveillance. In fact, surveillance operations in Egypt are often practised in secrecy. For example, the Technical Research Department (TRD) was previously found to acquire surveillance technologies, including CCTV cameras, in secrecy (Privacy International 2016). Additionally, the state has expressed no interest in assessing the negative impacts of its smart city projects, mainly adopting a 'largely idealistic' narrative about them (Rowaq Arabi 2024).

Test 6. Does the law control who has access to surveillance data?

Security agencies can access the data since feeds from surveillance cameras are linked to the national security surveillance network (Arab Republic of Egypt 2015). However, no clear limits are established by law. Limits established in the data protection law do not apply to security agencies (Arab Republic of Egypt 2020).

Test 7. Does the law create an independent oversight body?

In 2017, the legislative committee at the Parliament discussed a draft law on regulating surveillance cameras in public spaces, but the draft did not provide for any adequate protections, with the exception of banning installing surveillance cameras in some private places such as women's beauty salons, changing rooms, bathrooms, and showers (Mohyi 2017). In 2019, the draft was withdrawn by the government to reformulate it (Naamallah 2019) but as of August 2025, the law has not been adopted. The data protection law establishes some requirements for processing of personal data such as legitimacy and necessity of purpose, and transparency. However, the law in its entirety, including these requirements, do not apply to national security agencies. This means that surveillance of public spaces by security authorities and all surveillance video footage that feed into the national security surveillance network is conducted without adequate protection.

The laws previously mentioned do not set time limits for surveillance data collected. The draft law on regulating surveillance cameras in public spaces established three years for data retention, however it has not been adopted as of August 2025 (Naamallah 2019).

Test 8. Has a human rights impact assessment been carried out?

No independent oversight body was established to review how smart surveillance data is being used, stored, or secured as a check that state surveillance is being conducted in accordance with the law and in compliance with human rights commitments. Egypt's smart cities are largely controlled by the government and the military, leaving no room for independent oversight. For instance, in the NAC, 'all the data generated by the surveillance system and gathered and evaluated by the control centres is owned by ACUD, which has no civilian control' (Waisová 2022). The ACUD is owned by the government, and partially by the military.

7. Conclusion

Since assuming power, the Sisi regime has taken several measures to expand public space surveillance. The deployment of surveillance cameras, including those equipped with facial and number plate recognition, has increased in public spaces, including at mosques and universities. It became mandated by law for all businesses to install surveillance cameras outside and inside their businesses in order to operate. In the meantime, a permanent committee made up of representatives from the country's security agencies was launched with the mandate to coordinate CCTV surveillance across the country and linking the visual feed of all CCTV cameras to a national grid (El-Hamalawy 2023).

The most significant developments carried out by the Egyptian authorities under Sisi in public space surveillance are the result of smart cities. Dozens of cities are being built across the country, the biggest of which is the NAC. The scale of Egypt's smart city programme will mean a significant part of the population will increasingly be living under the constantly watchful eyes of the state. According to some estimates 15 million people will be living in smart cities in Egypt by 2030, which would amount to over ten per cent of the country's population (Waisová 2022).

These cities deploy integrated security systems that rely on advanced technologies like facial recognition, number plate recognition, access control systems, and motion and intrusion detection systems to gather data which feed into control and command centres that analyse the data, monitor incidents and coordination responses.

As the government moves forward with the deployment of more advanced public space surveillance technologies, it has failed to ensure proper limits to minimise abuse and protections for data collected by smart cities. In fact, guidelines established by the government to organise smart cities do not address management of privacy and human rights risks or data protection requirements and limits. Additionally, security agencies have a free hand to access data collected as part of public space surveillance, for instance, through the country's video surveillance camera system, or through their control of smart city systems, as is the case with the NAC, which is partially owned by the government. Finally, independent oversight of public space surveillance and the agencies conducting it is non-existent, leaving citizens with no accountability and remedy avenues to turn to when their privacy rights are violated.

The scale of Egypt's smart cities programme, the invasive nature of technologies deployed and the lack of a robust regulatory framework and independent oversight that places checks on public space surveillance, will place citizens under pervasive and unchecked surveillance. Already in a republic that has been witnessing one of the most severe crackdowns on human rights, civil society and independent media in its history, the public sphere has dramatically shrunk, and the Sisi regime has eliminated almost all forms of dissent to this rule (Said 2023). Public space surveillance is an essential tool at keeping dissent and any other potential threats in check. The constant and invasive surveillance enabled by smart city tech only further encroaches on what little space is left for Egyptians to freely express themselves, organise, protest, and assemble, thus consolidating the regime's power and the advancement of its agenda.

8. Recommendations

- Egyptian legislators should adopt regulations to place limits on public space surveillance and reign in access of security agencies to people's information, including data collected by smart city systems. They should also amend the data protection law to end the exemption given to security agencies from abiding by the law's provisions and ensure the independence of the data protection authority, which under the current law is made up of government representatives including ministries of interior and defence.
- Legislators should adopt a law that regulates smart cities, including risks to privacy and data collected by smart cities and requiring systematic human rights impacts assessments to assess and mitigate risks that may emanate from smart cities.
- Providers of public space surveillance technologies should refrain from doing business with the Sisi regime due to severe risks that their technologies could be abused in human rights violations.
- In its technical support provided to the Egyptian government, UN-Habitat should not refrain from bringing up and addressing risks to privacy and human rights in general. It can, for instance, offer guidance on how smart cities can ensure security of data, address data breaches, and deploy invasive technologies.
- Civil society actors including human rights groups, activists, journalists, academics, and artists need to continue working on documenting public space surveillance and think of creative ways to raise the public's awareness about its harms.

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Smart city surveillance in Kenya

Victor Kapiyo

1. Introduction

Kenya has rapidly expanded public space surveillance in recent years, deploying thousands of CCTV cameras in two major cities and implementing various measures to gather vast personal data through various initiatives, including a national digital ID system (Maisha Namba) and mandatory SIM-card registration. While there are legitimate aims that justify the mass data collection programmes, there are also widespread concerns regarding the approaches adopted, including compliance with data protection laws, and misuse of collected data. This report contains the most comprehensive analysis to date of Kenya's Integrated Public Safety Communication and Surveillance System (IPSCSS), which is one of Huawei's safe city projects implemented in Kenya between 2014 and 2016. The report begins by tracing the history of public space surveillance in Kenya, before documenting the deployment of the IPSCSS, reviewing the legal framework and human rights concerns raised before drawing conclusions and making some actionable recommendations.

2. Background

Public space surveillance in Kenya dates back to the 19th century when the British colonial administration set up the first forms of state surveillance machinery. This included intelligence gathering through informal networks of porters, missionaries, mercenaries, local chiefs, and informers and formally through the formation of the East African Police in 1892, its Criminal Investigations Division (CID) in 1926 and later the Special Branch in 1952 as the formal intelligence agency responsible for surveillance of domestic opposition groups (Wagah 2024). These institutions and their problematic working methods were retained, rebranded, formalised, and expanded by successive governments post-independence, with the CID being renamed the Directorate of Criminal Investigations (DCI) in 2011, and the Special Branch becoming the Directorate of Security Intelligence (DSI) in 1986 and later the National Security Intelligence Service (NSIS) in 1998, and now National Intelligence Service (NIS) in 2012 (Wagah 2024; Kapiyo, Oyier and Monyango 2024).

Today, public space surveillance has been significantly bolstered and expanded with investments in digital surveillance infrastructure over the past decade. The IPSCSS was implemented in 2014 and was the first Huawei safe city project in Africa (Jili 2022). Although the government had already been gradually increasing its surveillance capacity over the years, it was the massive public outcry following terror attacks such as the Westgate Shopping Mall attack in 2013 (Howden 2013) that provided the impetus and justification for investment in the IPSCSS. The system was designed to assist law enforcement agencies to monitor high-crime areas, track criminal activities, and respond to security threats through a centrally controlled system of urban surveillance (JEPA Africa 2024).

The system, which Huawei and Safaricom implemented for KES18.8 billion (US\$214 million), installed by May 2016 1,800 high-definition CCTV cameras with ANPR and 200 traffic surveillance cameras in Nairobi and Mombasa towns and on major highways (Wangari 2023). The video feeds are shared with the Integrated Control and Communication Centre (IC3), which links 195 police stations. The companies also developed a 4G network for the police, supplied 7,600 radio communication devices, and connected 600 vehicles to the command centre (Kapiyo and Githaiga 2014). The central control centre features a critical incident management system, emergency call centre, dispatch centre, and a data centre, where data from the deployed CCTV cameras are collected and analysed (Privacy International 2017).

According to media reports, proponents of the system claimed it led to a 46 per cent reduction in crime rates between 2014 and 2015 within its coverage areas, enhanced operational efficiency of the police, and enhanced citizen confidence in urban safety (Kirui *et al.* 2024). However, crime levels subsequently rose by 50 per cent between 2016 and 2017 (Francois 2020; Baraka *et al.* 2021), leading to questions over its overall effectiveness, given their limited coverage, especially in high crime areas, among other challenges (Kirui 2024). A 2022 survey also indicated that only 4.2 per cent of respondents identified the use of technologies such as CCTV as effective methods of crime reduction (Gitonga and Soi 2022). The system was also criticised as it was implemented in the absence of a data protection law, or any clear policy on public space surveillance (Wangari 2023). An audit of the Integrated Urban Surveillance System (IUSS), the predecessor to the IPSCSS, revealed massive gaps in training, maintenance, and utility of the initial 42 cameras installed (Office of the Auditor-General 2017).

Public space in Kenya remains problematic, given Kenya's chequered history of unchecked state surveillance and abuse of police powers (IJM 2025). The key challenges include weaknesses in transparency, accountability and oversight of state security actors, gaps in the policy and legal frameworks, and the abuse of surveillance powers to target civic actors and human rights defenders (Privacy International 2017; Kapiyo *et al.* 2024). With increasing digitisation of government services, such as Maisha Namba Digital ID and eCitizen, the government services online portal, and given the low regard for privacy rights, especially by state actors, as was evident during the Covid-19 pandemic period, the risks to the privacy of citizens are likely to increase if concrete measures are not taken (Mwanzia and Gichohi 2024; Mwanzia *et al.* 2021).

Recent trends do indicate growing concerns. For example, during the #AntiFinanceBill2024 protests, the government is reported to have deployed sophisticated digital tools to monitor activists, journalists, human rights defenders, and critics some of whom were arrested or abducted for their online activism (CIPESA 2025). A similar trend was noted in June 2025, when similar architecture was deployed. Also, the procurement by the Directorate of Criminal Investigations (DCI) of a KES50 million (US\$386,996.90) Optimus 3.0 social media spyware system in June 2025 indicates a clear intent by government to monitor citizen activities online (Mwangi 2025). Further, proposals to intensify surveillance are evident in provisions of the Kenya Information and Communications (Amendment) Bill, 2025, whose section 27A would require internet service providers to, among others, assign unique 'internet meter numbers' to their subscribers, monitor real-time internet usage, and collect detailed personal information (Wothaya 2025).

3. Case study description

The genesis of the IPSCSS can be traced back to May 2012, when the then prime minister Raila Odinga revealed that the government had secured a grant of KES 8.5 billion (US\$100 million) from the Chinese Ministry of Commerce to install CCTV in Nairobi, Mombasa, and Kisumu with the stated aim of countering terrorism and improving urban security (AIDDATA 2021).

By September 2012, the state-owned Chinese firm Nanjing Les Information Technology was awarded a KES464 million (US\$3.58 million) contract to supply, install, and test IUSS in Nairobi (AIDDATA 2021). That October, two milestones followed: the National Intelligence Service Act formally established the National Intelligence Service (NIS) and granted it broad powers to gather intelligence and interception communications (Republic of Kenya 2012), and the Prevention of Terrorism Act (POTA) 2012 similarly empowered security agencies to monitor and intercept communications in the name of counterterrorism (Republic of Kenya 2023).

When the Jubilee administration took office in March 2013, its manifesto included the deployment of CCTV to fight crime (Kapiyo and Githaiga 2014). By November 2015, 40 CCTV cameras in 24 locations within Nairobi and 52 vehicle number plate recognition systems in 15 locations had been installed under the IUSS project (Office of the Auditor-General 2017; Kinyanjui 2019).

With rising insecurity in the country, Safaricom Limited was in May 2014, hand-picked to implement the IPSCSS as part of the safe city project (Burt 2018). On 25 November 2014, the Ministry of Interior and Coordination of National Government signed a five-year, KES14.9 billion (US\$169.6 million) contract with Safaricom to deploy the National Surveillance, Communication and Control System (NSCCS) across Nairobi and Mombasa with an additional KES440 million (US\$5 million) per annum earmarked for maintenance and support over five years (Safaricom 2014; Kapiyo and Githaiga 2014). The Security Laws (Amendment) Act was also adopted in December 2014, expanding NIS's surveillance mandate (Republic of Kenya 2014).

Subsequently, between November 2014 and May 2016, Safaricom and its technology partner – Huawei Technologies Kenya Limited – completed the installation of 1,800 CCTV cameras across Nairobi. In October 2016, Huawei hosted its Safe City Africa Summit and Kenya Showcase in Nairobi and demonstrated the 4G and cloud-based smart safe city solution and the emerging lessons from its deployment in Kenya (Huawei 2016).

Building on these security investments, the government also adopted the National Broadband Strategy 2018–2023, which complements the implementation of infrastructure and devices to support smart cities as part of the Big 4 Agenda (Republic of Kenya 2018a). On 30 May 2018, the Computer Misuse and Cybercrimes Act came into force, providing for a range of cybercrimes, search and investigative procedures, and evidence handling, and also required service providers to assist in lawful surveillance (Republic of Kenya 2018b). Later that year, the National Police Service upgraded its Integrated Command and Control System (ICCS) by rolling out the NEC NeoFace facial recognition system for CCTV installed along major roads and highways in Nairobi and Mombasa (Burt 2018). It formed part of the Critical Incident Management Suite (CIMS) managed by the Directorate of Criminal Investigations.

In 2019, a draft National CCTV Policy was published to regulate the installation, operation, and management of all CCTV systems in public and private premises and for their use to deter, detect, and prevent crime (CIPESA 2021). In November, the Data Protection Act was enacted to regulate the collection and processing of personal data, including CCTV footage, amid concerns of expanding surveillance and data breaches (Republic of Kenya 2019). However, the drive for more surveillance continued as in August 2021, the Nairobi County Assembly adopted a motion calling on the County Executive to develop a policy for the installation of CCTV in all public areas and county facilities, and commercial and residential buildings (Nairobi County Assembly 2021).

Figure 2: Timeline of the national smart city roll-out in Kenya

Year	Key milestones
2012	PM Rella Odinga announces a us \$100 million Chinese grant to install CCTV and State-owned Nanjing Les tnfotech awarded KES 464 million contract National Intelligence service Act, 2012 and Prevention of Terrorism Act, 2012 enacted to combat terror and facilitate surveillance and interception.
2014	Kenya Information & communications Act amended to strengthen the communications Authority and mandate SIM registration. Govt signs a KES 14.9 billion contract with Safaricom and HUAWEI to install 1 BOO CCTV under the Integrated Public Safety communication & Surveillance System (iPSCSS).
2015	Security Laws (Amendment) Act, 2014 expands NIS's surveillance mandate and criminalizes undermining security operations.

Year	Key milestones
2016	Computer Misuse & Cybercrimes Act, 2016 comes into force, criminalizing cyber-offences and obliging service providers to assist in lawful surveillance.
2018	National Police service rolls out NEC NeoFace facial-recognition CCTV feeds in Nairobi & Mombasa as part of its ICT upgrade.
2025	Directorate of Criminal Investigations budgets KES 150 million for "Optimus3.0" spyware acquisition & forensic-lab upgrades-

Source: Author's own. Created in Canva.

3.1 Scale, functionality, and geographical coverage

The safe city project, a collaboration between Huawei and Safaricom, covered Nairobi and Mombasa cities. The project had the following components (AIDDATA 2021):

- The installation of 1,800 high-definition CCTV cameras with face and motor vehicle number plate recognition capabilities in strategic locations in Mombasa and Nairobi,
- Setting up a command and control centre where footage from the CCTV cameras and handheld devices will be relayed in real time,
- A video conferencing system, connecting 195 police stations with high-speed internet,
- The development of a 4G high-speed mobile network for the police with 80 base stations,
- Supplying the police with 7,600 radio communication devices with SIM cards and photo and video capability,
- Linking 600 police vehicles to the command and control centre.

The NEC NeoFace Watch software has the following features (Security Vision 2024):

- Capability to match footage from multiple camera feeds,
- Capability to provide real-time alerts from video feeds,
- Ability to process live and archived video footage and images,
- Ability to capture faces of individuals, extract them from the video feed, and match to a database in real time,
- Ability to process multiple camera feeds, extract facial data, and match them to thousands of faces from a centralised database per minute.

Essentially, this means that the footage from the CCTV cameras can be analysed in real time, to identify individuals based

on facial recognition technologies, and matched with their details as recorded from national identity databases.

3.2 Biggest contracts (US\$ values) to which companies

Table 6: The key contracts and vendors involved in Kenya’s safe city and surveillance infrastructure

Date	Company	Value	Source
2012	Nanjing Les Information Technology (China)	KES464 million (US\$5.5 million)	Government of Kenya
2016	Safaricom PLC (Kenya) Huawei Technologies Kenya (China) Shanghai DS Communication Equipment Co. Ltd (China)	KES18.8 billion (US\$214 million) and KES440 million (US\$5 million) annually	Safaricom PLC Government of Kenya
2018	NEC (Japan)	Unknown value	

Source: Author’s own.

4. Key actors

Table 7: Key actors

Actor	Roles	Detail
Ministry of Interior and Coordination of National Government	Lead agency	Project owner and signed agreement with Safaricom for the NSCCS. It is the parent ministry for the National Police Service and the National Intelligence Service.
Ministry of Information, Communication and the Digital Economy	ICT policy	Responsible for national ICT policy and is the parent ministry for the ICT Authority, Communications Authority, and the Data Protection Commission.
National Police Service (NPS)	Law enforcement agency	Operates the CCTV surveillance systems and the command centres.
Directorate of Criminal Investigations (DCI)	Law enforcement agency	Monitors the CCTV system as part of the Critical Incident Management Suite (CIMS).
Office of the Inspector General of Police (IGP)	Law enforcement agency	Responsible for overall police operations
National Intelligence Service (NIS)	Law enforcement agency	Plays a key role in surveillance and intelligence gathering to combat security threats.
National Assembly	Legislature	Allocates budgets, makes laws, and has committees, e.g. on security to oversee security agencies.
Communications Authority of Kenya (CA)	Government regulator	Regulates the telecommunications sector, and reviews data from licensees.
Office of the Data Protection Commissioner (ODPC)	Government regulator	Regulates the collection and processing of personal data.
Independent Policing Oversight Authority (IPOA)	Oversight body	Civilian oversight of the police with power to investigate complaints, monitor operations, and inspect police premises.
Kenya National Commission on Human Rights (KNCHR)	Oversight body	Independent body to promote, protect, and monitor human rights in Kenya.
Safaricom PLC (Kenya)	Private sector	Kenya's largest mobile service provider, a key partner in implementing the IPSCSS.

Actor	Roles	Detail
Huawei (China)	Private sector	A major global telecom firm and partnered with Safaricom as provider of the safe city solution implemented in the IPSCSS.
Nanjing Les Information Technology (China)	Private sector	Supplier of CCTV technology and traffic management.
Privacy International	Civil society	Research and advocacy on privacy
KICTANet	Civil society	Research and advocacy on privacy
CIPESA	Civil society	Research and advocacy on privacy

Source: Author’s own.

4.1 Government ministries and agencies

Public space surveillance in Kenya has largely been led by government security agencies, who have been the primary beneficiaries of the systems. The Ministry of Interior and Coordination of National Government championed the safe city initiative as a response to rising crime and terror attacks such as the Westgate mall terror attack (Reuters 2014). The IPSCSS is primarily used by the National Police Service, and serves the needs of other security agencies. Security agencies have benefited from the acquisitions of powerful surveillance tools with real-time feeds and improved communications thus improving their overall capacity (Wangari 2023). Government officials have managed to be perceived as taking tough action on crime, while also benefiting from a system that could be used against their political opponents.

The National Assembly was instrumental in approving the projects, and had previously demonstrated its power by suspending and later allowing the Safaricom contract in 2014 despite allegations of single sourcing (Wokabi 2014). Other relevant bodies include the Communications Authority of Kenya (CA), Office of the Data Protection Commissioner (ODPC), Independent Policing Oversight Authority (IPOA), and the Kenya National Commission on Human Rights (KNCHR), which exercise oversight of security agencies. Also, the county governments of Nairobi and Mombasa were instrumental in approval of the CCTV installations.

4.2 Telecom and technology companies

The IPSCSS was implemented by Safaricom PLC and Chinese tech giant Huawei, while earlier CCTV cameras were installed by Nanjing Les Information Technology. The Safaricom contract was single-sourced

without open tender, sparking controversy and allegations of favouritism (Reuters 2014). Safaricom benefited from its local presence and provided its infrastructure and mobile network, while Huawei supplied and installed the CCTV network and devices (Wangari 2023). In addition to reaping a financial reward, Safaricom obtained a 4G licence and Huawei got additional business in other countries (Wangari 2023).

4.3 Funders and international partners

Public surveillance initiatives in Kenya have also been supported by foreign actors. The IPSCSS was largely sponsored by the Chinese government under the Digital Silk Road Strategy, which is a component of the Belt and Road Initiative to spread its economic and political influence globally (Wangari 2023). The initial grant of US\$100 million was obtained from the Chinese Ministry of Commerce. Huawei played a strong role in the implementation and popularisation of the smart city initiatives, which have been adopted in other African countries.

4.4 Drivers of public space surveillance

There are three main drivers of public space surveillance in Kenya: securitisation, political interests, and profit. Firstly, securitisation has been the primary rationale for the rapid expansion of public space surveillance in Kenya. The government's narratives to justify public surveillance have centred on enhancing national security, ensuring public safety, and combating crimes such as terrorism (JEPA Africa 2025). The IPSCSS was seen as a compelling technological silver bullet for crime prevention in the wake of terror attacks at the time, such as the 2013 Westgate attack. This helped rally support for its urgent deployment as proponents argued that the technologies were essential for public safety and improving police response times (Wangari 2023).

Secondly, political interests also came into play as the deployment of a large-scale surveillance system provided good political optics for Uhuru Kenyatta's government, which wanted to be seen as 'modern', 'digital', and strong on crime. Thus, the project served as an example of the government taking stern action to combat crime and counter terrorism. In fact, the president in a political advertisement in June 2014, warned terrorists, criminals, and thugs to 'run and hide' as there would be 'thousands of cameras and millions of pairs of eyes watching' them (Kiberenge 2019). Concerningly, these systems have been repurposed for political control, targeting activists, journalists, and protesters (Kapiyo, Oyier and Monyango 2024). Also, Kenya intensified its ties with China as part of its 'Look East'

policy, which during the period saw increased Chinese debt, mega projects, and infrastructure investments in Kenya under the Belt and Road Initiative (Eickhoff 2022).

Lastly, the safe cities project opened up lucrative commercial and economic opportunities for Huawei, which was eager to expand its market presence in Kenya and across Africa. Equally, Safaricom benefited from a large contract and favourable 4G licensing terms, with an opportunity to build its network using Huawei technologies. These companies have often been accused of enabling surveillance states, including by quietly handing over user data or supporting location tracking (Kapiyo *et al.* 2024).

5. Regulatory landscape

Test 1. Are there validly enacted laws governing public space surveillance?

Kenya does not have a single law that authorises public space surveillance via CCTV or other mass surveillance systems. However, the country adopted various laws through which the legality could be inferred. The Constitution of Kenya 2010 guarantees the right to privacy (article 31), and since the right is not absolute, surveillance can be justified under article 24, which embodies the three-part test (legality, necessity, and proportionality) (Republic of Kenya 2010; Privacy International 2025). Article 24 provides among others that a right may only be limited by law, and only 'to the extent that the limitation is reasonable and justifiable in an open and democratic society based on human dignity, equality and freedom, taking into account all relevant factors'. These factors include the nature of the right, the importance and purpose of the limitation, the nature and extent of the limitation, the need to ensure the enjoyment of the right does not prejudice rights of others, and the relation between the limitation and whether there are less restrictive means to achieve the purpose.

The Data Protection Act, 2019, is the primary statute governing privacy and data protection. Under this law, the capturing of facial images via CCTV amounts to collection and processing personal data such as number plates, biometric data (e.g. facial images), and religious data (if recognisable), which are sensitive data. Therefore, any persons or agency that collects such sensitive personal data (i.e. CCTV footage) must register with the ODPC as per section 18, and comply with the principles of data protection under section 25. Notably, section 30 of the Act recognises various lawful bases for the processing of personal data, including the performance of a contract, compliance with a legal obligation, performance of a task in the public interest or in the exercise of official authority, performance of a task by a public authority, or for a legitimate interest that does not override individuals' rights.

Further, law enforcement agencies often derive authority from laws such as the Prevention of Terrorism Act (POTA) 2012, Mutual Legal Assistance Act 2011, Computer Misuse and Cybercrimes Act 2018, the Kenya Information and Communication Act 2011, and the Private Security Regulation Act 2016, which permit surveillance and the interception of communications. However, none of these laws specifically define or limit general CCTV surveillance of public places. In 2019, a draft national CCTV policy, which is yet to be published or implemented, sought to 'guide installation, operation and management of all CCTV systems' for crime prevention. Similarly,

in August 2021, the Nairobi County Assembly adopted a motion for the County Executive to develop a policy to compel owners of commercial and residential buildings to install CCTV, and for the county to install CCTV in all public areas and county facilities (Nairobi County Assembly 2021).

Test 2. Do the laws define a legitimate aim for public space surveillance?

From the framing of article 24 of the constitution, surveillance must pursue a legitimate aim and purpose. Therefore, the right to privacy right can only be limited by lawful, reasonable, and justifiable measures in the pursuit of legitimate objectives. As highlighted above, the Data Protection Act recognises acceptable purposes for processing personal data. The government in announcements of the deployment of public surveillance projects and in the draft CCTV policy, articulated their objectives as ensuring public safety and detection and prevention of crime, e.g. terrorism and national security (CIPESA 2021). While these aims may be legitimate, the absence of a law to regulate public surveillance is a clear gap, which means there is no framework for oversight thus leaving room for abuse of powers by law enforcement, including to target human rights defenders (Kapiyo *et al.* 2024). Kenyans have reason to be concerned in this regard as state authorities have been reported conducting other forms of surveillance 'without oversight and outside of legal procedures' (CIPESA 2021).

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

Currently, the Ministry of Interior and Coordination of National Government, National Police Service, and Nairobi county government manage the IPSCSS, although no law specifically bestows them such responsibility. Also, despite collecting personal data en masse, none of these entities are registered with the ODPC, nor have they conducted data protection impact assessments as required under section 31 of the Data Protection Act. Moreover, law enforcement agencies may seek warrants to search and seize information or property as evidence in criminal procedures or during investigations. These may include CCTV recordings held by private persons. Notably, section 51(2) (b) of the Act exempts the processing of personal data from the provisions of the Act 'if it is necessary for national security or public interest'. Neither the Ministry of Interior and National Administration or the National Police Service are registered with the ODPC as data handlers (ODPC 2025).

Notably, impunity by law enforcement remains rampant, while oversight, transparency, and accountability measures are weak and ineffective (Missing Voices Kenya 2025). While police may have lawful basis under various laws to collect personal data, these powers can be abused. Since its establishment, the ODPC provides overall oversight of the collection

and processing of personal data. Under section 9 of the Data Protection Act, it has power to conduct investigations, receive complaints, summon witnesses, require information from any person, search premises, and impose administrative fines. Under section 65, persons who suffer damage (financial loss or distress) from contravention of the Act may complain to the ODPC who in turn may order compensation. Further, under section 72, a data controller who unlawfully discloses personal data commits an offence punishable by a fine of up to KES3 million (US\$23,219.81) or an imprisonment term of up to ten years. So far, the ODPC has not made a determination against government surveillance to date. The High Court in reinforcing the right to privacy, found in 2023 that while the installation of CCTV cameras in a private property overlooking a neighbour's compound had a legitimate aim, it breached the right to privacy, as no consent was obtained and the owner was not registered by the ODPC (Judiciary of Kenya 2023).

Test 4. Does the law define mechanisms for citizen redress and accountability?

A key challenge for oversight bodies such as the Kenya National Commission on Human Rights and the Independent Police Oversight Authority is that they are plagued by deficiencies in human, financial, and technical resources (Kapiyo *et al.* 2024). They are also not truly independent as they are often subject to political interference thus limiting their impact especially in situations or cases where the government has an interest in the outcome. Moreover, others such as the Intelligence Service Complaints Board, which is required to investigate and receive complaints against National Intelligence Service security agency officers is yet to be operationalised since its establishment in 2012 (Kapiyo *et al.* 2024).

6. Digital rights

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

As previously highlighted, the government typically justifies public space surveillance as necessary to ensure national security, public safety, and to combat crimes. These legitimate aims are regulated under various laws such as the National Intelligence Service Act, National Police Service Act, and Prevention of Terrorism Act. When applied to public space surveillance, the laws in principle are meant to ensure that only surveillance necessary to the specified purpose, such as investigating a crime, is carried out, and it is proportionate to the objectives.

However, in practice, general surveillance, including public surveillance activities in the country, rarely meet the necessity and proportionality test under human rights law. According to Privacy International, Kenyan law enforcement use sophisticated surveillance tools and do not always obtain court warrants prior to conducting surveillance activities (Privacy International 2017). During #RejectFinanceBill protests in June 2024 and June 2025, several online activists and protesters were abducted and illegally detained, fuelling suspicions of illegal surveillance being deployed to target government critics (Laibuta 2024). Additionally, the exemptions under rules 54, 55, and 57 of the Data Protection (General) Regulations 2021 on national security, general situations (prevention of a serious threat to public safety, taking action on unlawful activities), and health elaborate on the window through which the Act is not applicable and could be subject to abuse. Whereas national security organs are required to apply for such exemptions, it is not clear whether they applied, and if they did, whether they were granted. The absence of independent oversight and transparency reporting of these organs perpetuates impunity and scope creep in their data collection practices and fuels their disdain and disregard for lawful procedures.

Notably, in July 2025, the High Court found that notices by the Communications Authority and the Kenya Revenue Authority requiring the mandatory collection of International Mobile Equipment Identity (IMEI) numbers for all mobile devices lacked clear legislative anchoring and were therefore unconstitutional and unlawful, and violated the right to privacy (Judiciary of Kenya 2025). This decision reinforces the supremacy of the constitution and the right to privacy, and sets a useful precedence should any litigation be brought before courts challenging the legitimacy, necessity or proportionality of public space surveillance.

Test 6. Does the law control who has access to surveillance data?

The constitution and the Data Protection Act, 2019, provide the primary framework for limiting access to the digital trail of personally identifiable information (PII) created through public space surveillance data. The Act adopts a principle-based approach to protect personal data and requires that any personal data be processed lawfully, fairly, and transparently, and collected for explicit, specified, and legitimate purposes (part V). Further, it specifies the obligations of data controllers and processors, such as entities operating CCTV systems, and requires them to register with the ODPC (section 18) and implement appropriate technical and organisational measures (section 41) to implement data protection principles, integrate safeguards, and process only data necessary for each specific purpose. The Data Protection Act under section 37 also prohibits the use of personal data for commercial purposes without express consent or specific legal authorisation. Therefore, based on the legal framework, national security organs and private sector operators of public CCTV cameras have clear obligations which they are required to comply with.

However, the effectiveness of these provisions is limited by the practical challenges in implementation. The measures if at all in place remain confidential, and thus it is difficult to verify whether any of the obligations on the state and non-state entities are complied with. As highlighted above, with more than five years since the Data Protection Act came into force, efforts to ensure national security organs accountable for their data processing activities have not yielded any fruits, and continue to be hampered by the broad exemptions, and their general impunity and disregard for their data protection obligations. Whereas Safaricom PLC and Huawei Technologies (Kenya) Company Limited are registered by the ODPC as data handlers, none of the government entities handling public space surveillance have registered (ODPC 2025). Given that some government ministries such as those responsible for agriculture and health and some county governments such as Kiambu and Mombasa are registered, there is no excuse for others not to comply.

Test 7. Does the law create an independent oversight body?

Kenya's legal framework provides for several oversight bodies, with various powers to oversee the implementation of the right to privacy and other human rights and hold various institutions, including national security organs, accountable for their actions. Key among these is the ODPC which is established under section 5 of the Data Protection Act 2019, mainly to oversee the enforcement of the Act, register data handlers, receive and investigate complaints, carry out inspections, and promote awareness and compliance with the Act.

Other oversight bodies include the Kenya National Commission on Human Rights (KNCHR), which is an independent commission established under article 59 of the constitution with a broad mandate to promote and protect human rights in Kenya. Another is the Independent Policing Oversight Authority (IPOA), established under the IPOA Act, 2011 as a civilian oversight body over the police and to receive complaints and conduct investigations relating to police misconduct. The Communications Authority of Kenya (CA) is an independent entity established under the Kenya Information and Communications Act that regulates the ICT sector, including aspects such as telecommunications, type approval, and cybersecurity management. Lastly, is the yet to be operationalised Intelligence Service Complaints Board (ISCB) established under the National Intelligence Service Act 2012 to receive and investigate complaints relating to members of the NIS.

These oversight bodies face persistent challenges in discharging their respective mandates. The main ones include the lack of real fiscal and decisional independence, limited funding to discharge their mandates (Institute for Security Studies 2024), non-cooperation of police and state agencies (Houghton 2024), and political interference. Another challenge is leadership and governance gaps due to delays in appointment of board members, the appointment of political appointees and board members who are not subject matter experts, or, as in the case of the intelligence service ISCB, non-operationalisation of the board since 2012 (Mumbi 2024), all of which adversely impact the overall effectiveness of the institutions in discharging their oversight functions. These gaps have left the judiciary as the last bastion of hope in overseeing the executive, even though courts are often bypassed in surveillance activities, and the judiciary also suffers from underfunding and political interferences, thus challenging its independence and effectiveness.

Expansion of surveillance infrastructure

Recent developments indicate the intensification of surveillance efforts and prioritisation of further acquisition of surveillance tools and equipment by the government. For example, recent protests revealed heightened use of digital surveillance against activists, bloggers, and protestors (Mutung'u 2024). Moreover, 2025 budget estimates for the Directorate of Criminal Investigations (DCI) revealed the proposed procurement of the Optimus 3.0 tool for social media monitoring and forensic labs (Wothaya 2025). Following the death of Albert Ojwang in police custody, the government announced plans to install CCTV surveillance cameras in all police stations by 2026 to enhance transparency and accountability of law enforcement actions (Biegon 2025). Also planned are laws to criminalise tampering with CCTV in the police stations (Biegon 2025). In January 2025, the Communications

Authority proposed a new licence condition for internet cafes, which, although has not been adopted, would require the cafes to install CCTV surveillance to identify persons visiting the facilities (Musau 2025). There have been concerns over government efforts around mandatory SIM-card registration, which if linked with biometric data could be misused to aid public space surveillance (Andere and Kokonya 2022). More recently, the abduction of activists and protesters following the June 2024 and 2025 protests demonstrate that triangulation of data from public CCTV, social media, and mobile SIM registration have been instrumental to security agencies in locating persons of interest.

Resistance and advocacy efforts

A robust network of local and international civil society has remained active in Kenya promoting respect for constitutional rights to freedom of expression, privacy, access to information, and media freedom and applying pressure on the government to respect them. Civil society actions need to focus on challenging the legality, necessity, and proportionality of public surveillance, demanding greater transparency and accountability of security institutions, and stronger institutional oversight of public space surveillance. Public interest litigation provides a useful opportunity to be harnessed to defend rights. However, following Albert Ojwang's death, various CCTV recordings helped reveal cover up of his death, and public sentiment leaned towards openness to CCTV installations to aid transparency and accountability. However, it is yet to be seen whether this will be implemented.

7. Conclusion

Kenya's expansion of public space surveillance, particularly through the safe city project, is driven by a complex mix of securitisation, political interests, and commercial gain. While the government has championed the project as a modern solution to crime and terrorism, its effectiveness remains questionable. At the same time, the initiative provides the state with a new tool for political control while offering a lucrative opportunity for companies like Safaricom and Huawei.

Of concern is that there is no single law governing public space surveillance. Despite having a robust Data Protection Act, its broad exemptions for national security undermine privacy protections. This regulatory gap coupled with weak oversight institutions challenged by limited independence, inadequate funding, and fragmented operations, have enabled state security agencies to continue operating with impunity, often bypassing legal procedures for data collection and surveillance.

Looking ahead, public space surveillance is likely to be more advanced as new technologies such as AI are incorporated. They are also likely to be intensified as these technologies become more affordable, digital infrastructure expands, and national security imperatives remain. If no corrective action is taken, it is likely that the human rights impacts from public space surveillance projects will increase to the detriment of the public the projects aim to protect. Therefore, Kenya must prioritise law reform, robust oversight, rule of law, and human rights-based approaches.

8. Recommendations

- **Strengthen safeguards and judicial oversight of surveillance:** Review and strengthen the Data Protection Act to remove the broad exemptions to national security and public interest, ensuring they are precise, necessary, and proportionate. Amend the Computer Misuse and Cybercrimes Act to ensure robust judicial oversight for all forms of data access and public space surveillance by state agencies. Put in place an appropriate legal framework to regulate public space surveillance, including guidance notes by the ODPC on CCTV deployment. Operationalise the civilian Oversight Board of the National Intelligence Service. Strengthen the financial, decisional, and operational independence of oversight bodies such as the ODPC, IPOA, KNCHR and the judiciary to ensure impartial investigations.
- **Protect civic space and digital rights:** Uphold constitutional guarantees in the bill of rights, including by taking measures to reform and stop the weaponisation of laws such as the Computer Misuse and Cybercrimes Act to restrict rights. Refrain from conducting arbitrary, unlawful, and unregulated public space surveillance, including targeting of activists, critics, and journalists.
- **Enhance transparency and accountability:** Strengthen the Access to Information Act by developing clear regulations for facilitating the right to information, including fast-tracking the digitisation of public records and ensuring transparency in the procurement of surveillance technologies. Hold state security agencies personally and collectively accountable for human rights violations and abuses perpetuated through or aided by public space surveillance tools.
- **Foster multistakeholder approaches:** Finalise and adopt the Public Participation Bill to institutionalise genuine and multistakeholder engagement among government, civil society, academia, and the private sector in shaping ICT-related policies and legislation and in the life cycle of digital public infrastructure projects.

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Smart city surveillance in Mauritius

Roukaya Kasenally

1. Introduction

Smart public space surveillance is a reality of the Mauritian landscape. Thousands of smart cameras with facial and number plate recognition dot the tiny island. Most citizens are oblivious of their existence and their possible infringements on their liberties and freedoms. Understanding the implications of such smart technology is imperative. The main research question explores whether the Data Protection Act 2017 offers sufficient protection and coverage when it comes to public space surveillance. There is no dedicated law that governs public space surveillance but one that covers data protection, and therefore the infrastructure that carries data in the form of smart cameras falls under no legislation. The clause pertaining to 'exceptions and restrictions' (section 44) remains problematic and potentially open to misuse or abuse. This report will chart out the presence and evolution of public space surveillance, and zoom in on one of the most recent additions – the Mauritius safe city, which is costing the Mauritian public purse an astronomical amount. Attention will also focus on the legal and digital implications brought by public space surveillance systems.

2. Background

Mauritius' colonial heritage is closely linked to one of trauma, control, and discrimination. The island is shaped by 250 years of French and British rule that used slaves and indentured labourers to toil and fructify sugar cane plantations on the island. One notable feature about Mauritius is that it is home to no indigenous people and most of its current population is of African, Indian, Chinese, and European descent. Rights and freedoms for slaves were non-existent under French rule (1715–1810) where the 'code noir' regulated their lives. Under British rule (1810–1968), which saw the advent of indentured labour from India, the island had its first constitution (1831) and subsequently new constitutions were enacted in 1885, 1948, 1958, and 1968 ultimately granting universal suffrage. The Special Branch was used in Britain and in most of its colonies to gather intelligence on political activists and Mauritius was no exception. Profiling of various ethnic groups was common practice as was the close monitoring of political and labour movements. These practices intensified as the British administration was faced with demands for political and economic emancipation by a population that understood the value of education, the ability to mobilise collectively, and the importance of self-governance. The island experienced its first state of emergency in 1938 aimed at quelling a workers' strike claiming for fairer wages and better working conditions.

After the island became independent in 1968, repressive legislation such as sedition and defamation (part of the Criminal Code 1838, the Public Order Act 1970 and the Official Secrets Act 1972) came into force. The aim of the post-independence government was to intimidate and shut down political opposition to its rule. Mauritius experienced several cycles of government crackdown on opposition groups with the first between 1968 and 1972. A new political party the Mouvement Militant Mauricien (MMM) inspired by the May Student Movement of 1968 was seen as a threat to the power of the incumbent government. National security agents kept tabs on the movement and exchanges of the party's members especially on its leader – Paul Berenger. On 25 November 1971, Berenger narrowly escaped a murder attempt by party agents close to the ruling coalition. A few weeks later a general strike was called and the then government responded by instating a state of emergency. Berenger and his MMM comrades were jailed and public gatherings were forbidden through the Public Order Act 1970. The printing press of the MMM was confiscated and other newspapers were heavily censored. In fact, the level of monitoring of the media was so much so that journalists had to bring their texts to the police headquarters to be vetted by the then police commissioner (Chan-Meetoo 2018). The

state of emergency was lifted in 1972, but the culture of surveillance persisted as general elections slated for 1972 were postponed to 1976.

The second cycle of repression was from 1984 to 1991 with state surveillance focusing on the media. In July 1984, a highly contested Newspaper and Periodical Bill was proposed 'that would have prohibited critical press reports about members of Parliament and parliamentary affairs and would have required all newspapers to post a deposit of \$40,000 to be held as bond against damages in the event of libel suits' (Bowman 1984). The proposed legislation stirred massive protest, and 44 journalists were arrested following a sit-in outside Parliament. Matters were subsequently resolved but were seen as a narrow escape from political repression and the muzzling of the media. In 1991, the government's desire for control and monitoring was evidenced when the Public Order Act 1970 became the Public Gathering Act 1991. Under this new legislation any gathering that consists of more than twelve people requires the official consent of the police commissioner, a measure that has significantly constrained the freedom of association.

The third and most recent cycle of repression occurred between 2017 and 2024, a period when the island experienced measurable democratic backsliding. During this period Mauritius lost its coveted ranking as a liberal democracy to become defined as an electoral autocracy (V DEM 2018; V DEM 2025). Public surveillance infrastructure was expanding during this period. In 2017, according to media reports, surveillance equipment from an Israeli based company, Verint (Capital 2023; *Mauritius Times* 2025), was acquired while in 2019 the safe city project was officially launched.

Online content was under surveillance by state agencies including the Information and Communication Technologies Authority (ICTA), the Independent Broadcasting Authority, and the police (IFEX 2021; York and Greene 2021; Kasenally 2022). In 2021, a highly controversial public consultation paper entitled 'Regulating the Use and Addressing the Abuse and Misuse of Social Media in Mauritius' was proposed (ICTA 2021). The aim was for social media traffic to and from Mauritius to be routed through a proxy server run by the ICTA and establish systems to break any encryption provided by social media platforms and conduct official surveillance of all this traffic. Due to major push back from local and regional civil society groups the proposals were stalled (Access Now 2021; Centre for Law and Democracy 2021). However, it was only during the election campaign of 2024, that Mauritian citizens discovered the spread and depth of state surveillance following a wiretapping leak referred to as 'Missie Moutass'. One hundred and seven audio clips exposing interference, collusion and nepotism in state matters were released to the Mauritian public (Wong 2025). In addition to the highly explosive nature of the content what

shocked was the institutionalisation of public surveillance in Mauritius. Details were subsequently released in the Mauritian Parliament that in 2022 mass surveillance equipment to the tune of US\$110 million had been purchased by the previous government (Government of Mauritius 2025a).

3. Case study description

The Mauritius safe city project is the first of its kind among the Western Indian Ocean islands. Interest in such a mega project was initially minimal. This is partly due to the fact that there was no public announcement pertaining to the project until mention was made during the presentation of the budget speech of 2016–2017:

Government is coming up with a Safe City Project on a pilot basis. In that context, smart cameras would be installed with special focus on major public areas, along main roads and motorways, pedestrian walkways and principal traffic centres. These cameras would be a powerful and effective tool to combat crimes and drugs proliferation, assist in more effective traffic and road safety management (Government of Mauritius 2016)

Although it is difficult to situate the actual genesis of the project, it started with an unsolicited bid from Huawei to a newly elected government following the 2014 general elections and the main argument was to provide 'intelligent applications' (Huawei 2018) not provided by an existing low-resolution CCTV system installed in 2011 and that had benefited from Chinese funding and technology (AidData 2011). In 2006 China's Ministry of Commerce financed a US\$3.29 million interest-free loan for a CCTV street surveillance system and digital radio system. The project consisted of the installation of 340 CCTV street surveillance cameras in key locations as well as an island-wide digital radio communication system. The project was completed in October 2011 and implemented by the China International Construction Corporation (CITCC), Zhejiang Dahu Technology Co. Ltd and ZTE (AidData 2011). The existing presence of Chinese technology might partly explain why the safe city project was pitched and possibly facilitated and ultimately operationalised in 2019. The project was managed by a small group close to then prime minister, Pravind Jugnauth, with some members of the cabinet having little or no knowledge about it. Pitched as a means to tackle crime and make the island safe using what the prime minister, Pravind Jugnauth, referred to as 'state of the art technology' (Government of Mauritius 2020b). The project was fast tracked and did not follow the public procurement process required of such big infrastructure projects. According to parliamentary records, in December 2017 following the endorsement from the Cabinet of Ministers, two lease agreements were signed between the Mauritius Police and Mauritius Telecom (a private company with a minority state-owned

share) for the provision of the Mauritius safe city project (Government of Mauritius 2018a). Details pertaining to these lease agreements are outside the remit of public scrutiny due to a non-disclosure agreement clause.

Table 8 provides a breakdown of the estimated cost of the project over a 20-year period on an operating lease basis.

Table 8: Estimated cost of the Mauritius safe city project (over 20 years)

Year	Cost in US\$
0 (2018)	13 million
1–7 (2019–2025)	132.3 million
8–20 (2026–2028)	310.7 million
Total	456 million

Source: Author's own based on data from Government of Mauritius (2018b; 2025c).

Table 9 provides details as to the technical characteristics of the equipment used and the spread and coverage of the project. With a whopping price tag of US\$456 million, the Mauritius safe city project offers 4,000 intelligent video surveillance (IVS) cameras, 300 intelligent traffic surveillance (ITS) cameras, one main command and control centre, seven sub command and control centres, an emergency response management system, and a multimedia radio communication system. This is very comprehensive for an island that is just 720 square miles and has a population of 1.3 million.

Here it is important to note that the safe city is equipped with advanced features that allow facial and number plate recognition for both IVS and ITS cameras, but currently the use of biometric data is protected by the Data Protection Act (2017). Surveillance is in real time and in the case of traffic intelligence, cameras can help count the number of vehicles at a specific time of the day (for example rush hour), and identify and track a potentially suspicious or stolen car. However, the full potential of such technology, especially in the case of traffic management and safety in public spaces, has not been actioned on. Although all the IVS cameras have been installed, less than 50 per cent of the ITS cameras have been installed to date.

Table 9: Mauritius safe city technical characteristics

Type of cameras	Characteristics	Number to be installed	Number installed	Number of sites covered
Intelligent video surveillance (IVS)	Box and drone cameras that have the capacity for facial recognition, intelligent search capacity, and real-time surveillance Video storage for 30 days Hikvision cameras (China)	4,000	4,000	2,000
Intelligent traffic surveillance (ITS)	Pan-tilt-zoom cameras Automatic number plate recognition (ANPR) Traffic counting device Speed detection Intelligent checkpoints Video recording and analysis Hikvision cameras (China)	300	143	75

Source: Author’s own based on data from Government of Mauritius (2025b,c).

4. Key actors

Who were the key protagonists in the implementation of the safe city? As mentioned earlier, the government of Mauritius promoted and justified the relevance of a safe city premised on the need to fight crime and ensure public safety. However, it is the Mauritius Police Force (who falls under the Prime Minister Office) who is the managing authority for the safe city. All police stations (72) on the island are connected to smart cameras with facial recognition and car number plate recognition capabilities classified along seven divisions, namely the Northern, Southern, Western, Eastern, Metropolitan (North), and Metropolitan (South) divisions (Mauritius Police Force 2025). The main command and control centre (MCCC) centralises all the operations pertaining to the safe city allowing it to view and monitor real-time footage from traffic flow and public spaces and the rapid deployment of police patrols. This facility was inaugurated in August 2019 by then prime minister, Pravind Jugnauth. Sub command and control centres are designed to be linked to the seven divisions of the Mauritius Police Force but to date only one is operational and there is no information as to when the remaining six will come into operation. The emergency response management system was created to enhance the Mauritius Police Force’s reactive capability in responding to the 999 calls from members of the public. As for the multimedia radio communication system it uses an eLTE (evolved long-term evolution) system, which is a high-bandwidth, secure communication network critical for real-time data transmission in public safety applications.

Table 10 summaries some of the key features of the operationalisation of the safe city.

Table 10: The Mauritius safe city project’s police command centres

Number of police stations linked to the Mauritius safe city project	72	All police stations in Mauritius are connected to the safe city cameras
Main command and control centre (MCCC)	1	Inaugurated in August 2019 it acts as the centralised coordination and monitoring point of police responses and operation pertaining to the safe city

Sub command and control centres (SCCC)	7	Only one is in operation, acting as a decentralised hub to the MCCC relaying images from the police stations
Emergency response management system	1	Launched in September 2019, this system is linked to the MCCC and SCCCs
Multimedia radio communication system	1	Launched in August 2019 it provides a comprehensive production network that integrates wireless broadband access, trunking service, SCADA data transmission, and video viewing functions
Geographical coverage of safe city cameras	100%	

Source: Author’s own based on data from Mauritius Police Force (2021).

Mauritius Telecom, a private company with the Mauritian state holding a minority stake (33.5 per cent), was chosen as the technology service provider for the safe city and is responsible for installing and maintaining the infrastructure such as the surveillance cameras and other related communication systems. The choice of Mauritius Telecom appeared appropriate due to its market leader position and the Mauritian state stake in the company.

China, through the China Eximbank and Huawei, is an important player. The relationship between China and Mauritius goes back more than 50 years and the former has funded a number of key infrastructure-related projects on the island including roads, reservoirs, and the airport. In April 2018 a preferential buyer credit to the tune of US\$73 million was signed between the China Eximbank and Mauritius Telecom to finance the commercial contract with Huawei and start the roll out of the project (AidData 2023). The loan was guaranteed by the state of Mauritius.

Table 11 outlines the key actors in the safe city project.

Table 11: Key players in the safe city project

Companies/organisations	Main function
Government of Mauritius	Promoter
Mauritius Police Force	Implementer
Mauritius Telecom	Main contractor
Huawei (China)	Project contractor and vendor

Companies/organisations	Main function
China Eximbank	Loan provider
ZTE (China)	Data centre implementation partner
Hikvision (China)	Smart CCTV camera provider
Rod Oto Ltd	Project subcontractor (local)
Powersure Ltd	Project subcontractor (local)
MEP Engineering Ltd	Project subcontractor (local)

Source: Author’s own based on data from Government of Mauritius (2025b).

Huawei was seen as the natural partner for the safe city project as it has carved a leadership position offering a turnkey solution, providing design and expertise on implementation and technical equipment in one package (Peterson 2021). It is important to emphasise that Huawei is not a new player on the island as it began its operation in 2003 and was instrumental in setting up 3G, 4G, and more recently 5G networks as well as the infrastructure for fibre coverage through its Fibre To The Home (FTTH) on the island. In 2022, Huawei was caught in a maelstrom following what came to be known as the ‘sniffing’ exercise when it was disclosed that India with the permission from the Mauritian government used special equipment to ‘sniff’ (intercept and retain) traffic on one of the submarine cables carrying internet data into and out of Mauritius. This action was justified by concerns about China carrying out digital espionage using Huawei infrastructure and that the safe city was a backdoor for snooping (*Mauritius Times* 2022; Swami 2022). The then Chinese ambassador to Mauritius took to the Mauritian media to vigorously defend Huawei’s reputation and mention that it was a deliberate ploy by the West to cast aspersion and doubt on a Chinese tech company that is a leader in its field (Defi Media 2022).

How informed was public opinion? What level of public engagement and awareness were generated by such a big-ticket infrastructure project which would cost the Mauritian public purse US\$456 million. It was the opposition members of Parliament that started to field questions through the parliamentary questions segment requesting information about the nature of the lease agreements signed, the cost of the project, the partners involved, and the type of technology used. In fact, the responses received constituted a sort of information baseline which allowed the media, researchers, and interested citizens to start to grapple with the complex layers to the Mauritius safe city. Following regime change in November 2024, the current government, headed by Navin Ramgoolam, has made a commitment to shed light on this mega project (Government of Mauritius 2025b).

5. Regulatory landscape

The Mauritius Police Force is responsible for overseeing any activity related to public spaces. This is detailed in the Police Act 1974 where the Mauritius Police Force has the duty to take all lawful measures 'to regulate processions and gatherings on public roads and in public spaces or place of public resorts' (section 9 (d)) and 'assist in preserving order in ports, harbours and airports' (section 9 (h)). There are specific issues pertaining to 'protected area' (section 13A), 'restricted area' (section 13 B) as well as 'curfew order' (section 13 C). The other legislation that relates to the management of public spaces is the Public Gathering Act 1991. Access and use of public spaces for gatherings (bigger than twelve people) require the authorisation of the commissioner of police (section 3). These two legislations are essentially governed by the principles to maintain 'public peace' and 'security and safety' (Police Act 1974, section 9) and 'public order' (Public Gathering Act 1991, section 2). National security is often evoked as a legitimate excuse to clamp down, impose restrictions and authorise searches on certain members of the public. This became more systematic from 2021 to 2024 (see Degnarain 2021; Saramandi 2021; V DEM 2022; BTI Report 2024). Covid-19 was seen as a justified starting point for mass control and surveillance on the island with the enactment of the Covid-19 (Miscellaneous Provision) Act 2020. Citizens' movements were severely restricted, quarantine measures imposed, and surveillance of public spaces was conducted in a much more rigorous manner. The use of CCTV became much more spread from October 2011 following the introduction of the 340 CCTV street surveillance cameras financed by the Chinese Ministry of Commerce. Two types of space management have developed – public and private. In the case of public, in addition to the open spaces like streets and highways, the airport and port are monitored through CCTV cameras. Private CCTV cameras became increasingly common in 2002 when a high-end real-estate scheme was launched on the island aimed at attracting foreigners to buy property. As mentioned earlier the guiding principle is to ensure security and safety of the different spaces and its users.

Test 1. Are there validly enacted laws governing public space surveillance?

The two laws mentioned above precede the use and spread of CCTV and intelligent cameras on the island. There is no specific law that deals with public space surveillance per se, but it is the personal data of subjects generated from CCTV and smart cameras that is legally protected. In fact, Mauritius was one of the earliest adoptors of a data protection law in Africa and the Data Protection Act 2004 established the Data Protection Office, a commissioner, and data controllers, and emphasised the 'rights of data

subjects' (section 41). Each of them defines how the personal data of subjects will be processed and ultimately protected. The data commissioner's main mandate is to 'exercise control on all data processing activities' but they also have the responsibility to 'create and maintain a register of all data controllers' (DPA 2004, section 5). A data controller is 'a person who either alone or jointly with any other person makes a decision with regard to the purposes for which and in the manner in which any personal data are, or are to be processed' (DPA 2004, section 2). Data controllers must imperatively register with the Data Protection Office which delivers a 'certificate' (section 40) attesting that the data controller is duly registered and has the right to process data. With the spread of CCTV cameras, the Data Protection Office deemed it necessary to issue the 'Guidelines to Regulate the Processing of Personal Data by Video Surveillance Systems' (Government of Mauritius 2010). The foreword by the data commissioner speaks of the need to exercise caution and balance in the use of such surveillance technology:

The objectives of these guidelines are to ensure that those who operate video surveillance systems including CCTV (Closed Circuit Television) and other systems such as satellite based GPS systems, adopt good practice standards. They are aimed primarily at businesses and organisations (both public and private) which regularly capture images of identified or identifiable individuals on their video surveillance equipment for viewing and /or recording the activities of individuals for specific purposes since the personal data captured may potentially be used for other incompatible purposes affecting the privacy of the individual.

One would assume that these guidelines were formulated as the CCTV network implemented by the China International Construction Corporation (CITCC), Zhejiang Dahu Technology Co. Ltd and ZTE consortium rolled out in Mauritius. The guidelines cover a range of issues such as the purpose of using CCTV systems where 'images must be processed fairly and lawfully as well as for specified, explicit and legitimate purposes'. Equally important is the storage and retention of data recorded, which must be done in a manner that safeguards its integrity and security. The Data Protection Act 2004 was repealed and replaced by the Data Protection Act 2017. This was essentially to strengthen 'the control and personal autonomy of data subjects over their personal data as well as bringing Mauritius data protection framework into line with international standards, namely the General Data Protection Regulation (GDPR)' (Aikat and Naik 2018).

Test 2. Do the laws define a legitimate aim for public space surveillance?

How does one ensure balance between a public surveillance system that in principle is supposed to promote safety and security and legislation

that protects the personal data of subjects? The Data Protection Act 2017 ensures a whole battery of clauses where the personal data of subjects are protected: 'collection of personal data' (section 23), 'conditions of consent' (section 24), 'notification of personal data breach' (section 25), 'duty to destroy personal data' (section 27), 'rights of access' (section 37), 'automated individual decision making' (section 38), 'right to object' (section 40) among others. However, section 44 of the Data Protection Act 2017 refers to 'Exceptions and restrictions', which largely can override the rights of data subjects when it deals with matters pertaining to the 'protection of national security, defence or public security', 'the prevention, investigation, detection or prosecution of an offence', 'an objective of general public interest, including an economic or financial interest of the State', and 'the protection of judicial independence and judicial proceedings' (section 44, 1 (a-d)). In fact, the Data Protection Act 2017 goes a step further by vesting power in one person – the prime minister, who can evoke national security, defence, or public security in their widest possible definition to reverse data subjects' protection. Such a clause can have a chilling effect when wrongly used or abused as described in the Mauritian Parliament by the current prime minister, Navin Ramgoolam:

This means that every mobile phone conversation – I am putting stress on 'every' – every landline call, every post on WhatsApp, Facebook, Signal, Tiktok, every e-mail – of everyone, I am saying, not just us, but everyone – was being monitored by the MSM Government. (Government of Mauritius 2025a).

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

Those who have access to and use public space surveillance data are essentially data controllers for the public CCTV cameras and the Mauritius Police Force for the safe city cameras. In the case of CCTV cameras, public organisations must imperatively register with the Data Protection Office as data controllers. For example, both the Mauritius Ports Authority and the Mauritius Airport Terminal Operations have CCTV policy guidelines where they must abide by the Data Protection Act 2017. In both cases, their mandate is driven by 'reinforced security', 'preventing illegal activities' and 'safety reasons'. In the case of the safe city project, the data controller is the deputy commissioner of police and 'ensures that there is no abuse of this surveillance technology and that the processing of personal data is performed in accordance with the Data Protection Act 2017' (Government of Mauritius 2020a). In October 2020, the Data Protection Office issued the 'Code of Practice for the Operation of the

Safe City System(s)'. The code of practice sets out the basic conditions for the use of the safe city systems operated by the Mauritius Police Force. What catches one's attention from the onset is the section dealing with the 'Use of the safe city system(s)' which stipulates the following:

The processing of personal data for the purpose of implementation of the Safe City Project has been exempted through a certificate from the Prime Minister in accordance with section 44(4)(a) of the DPA, from the following sections of the DPA, in the interests of national security, defence and public security: (a) section 23 (2), (3) and (4) – collection of personal data; (b) section 26 – communication of personal data breach to data subject; (c) section 29 but only with respect to personal data pertaining to the commission or alleged commission of an offence by the data subject; (d) section 37 – right of access; (e) section 38 – automated individual decision-making; and (f) section 21, in so far as, it corresponds to the rights provided for in sections 23 (2), (3) and (4), 26, 29 (but only with regard to personal data pertaining to the commission or alleged commission of an offence by the data subject), 37 and 38.
(Government of Mauritius 2020a)

In other words, the Mauritian police have an expansive overview and control of the data emanating from the safe city cameras with the aim of protecting national security, defence, and public security. Under the section, 'Responsibilities and accountabilities', the Mauritius Police Force are required among other things to clearly outline 'responsibility and accountability for all personal data processed under the safe city system(s) including images and information collected, held and used'. As for the section dealing with 'Recorded materials', there is particular attention given to the security, integrity, and storage of the data. Disclosure of recorded material to a third party can only be done if there is a 'lawful requirement under any enactment or court order to disclose the images'.

Test 4. Does the law define mechanisms for citizen redress and accountability?

The Data Protection Act 2017 confers functions and powers to the data commissioner to 'exercise control on all data processing operations, either of his own motion or at the request of a data subject and verify whether the processing of the data is done in accordance with the Act' (section 5). The data commissioner has the right of 'investigation of complaints' (section 6), 'power to request information' (section 7), 'preservation order' (section 8) and 'enforcement notice' (section 9). Data controllers registered with the Data Protection Office need to have a valid 'registration certificate' (section 16 (3)). All data collected by registered data controllers must be 'processed lawfully, fairly and in a transparent manner in relation to any

data subject' (section 21 (a)) and 'collected for explicit, specified and legitimate purposes' (section 21(b)). Data subjects are entitled to request from data controllers information as to whether their personal data has (or is being) processed (section 37 (1a)) and if unsatisfied have the 'right to lodge a complaint with the Data Commissioner' (section 37 (2f)).

On one side there is the Data Protection Office that clearly outlines as per the Data Protection Act 2017 that there is data protection, that data controllers are required to abide, and that data subjects are entitled to privacy and redress. In the case of the Mauritius Police Force – which operates the safe city within the data protection legislation but is exempt when it comes to dealing with national security, defence, and public security – matters are not always straightforward as it is actually difficult to ascertain to what extent the Mauritius Police Force ensures the actual compliance of the Data Protection Act 2017 and of the Code of Practice for the Operation of the Safe City System(s). In fact, the two institutions should work in coordination, but this is not always the case. The Data Protection Office is understaffed and overburdened with overseeing all data protection and privacy cases while the Mauritius Police Force has earned a reputation for having been weaponised during the Pravind Jugnauth regime (2017–2024) and commands very low trust from the Mauritian citizen.

6. Digital rights

Mauritius embarked on the path towards digitalisation in the early 2000s. The move was largely motivated by the need to develop a new pillar of the economy namely the information/knowledge economy and in the process to make the island smart, intelligent, and connected. The constitution of Mauritius guarantees its citizens a number of rights (chapter II) but there is no mention of digital rights (Government of Mauritius 1968). In fact, the digital footprint generated by the average Mauritian citizen is significant as they are avid users of digital platforms. In fact, 80 per cent of the total population have access to the internet, 67.7 per cent use social media platforms, and 168 per cent have SIM cards (DataReportal 2025).

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

The legitimate aim for public space surveillance is ensuring safety and security. National security is often prioritised and the setting up the Mauritius safe city was done 'to provide for an integrated and advanced technological system for ensuring safety and security of the public in general in Mauritius and act as a powerful and effective tool to combat crimes and drug proliferation and also assist in more effective traffic road safety management' (Government of Mauritius 2019). These are often evoked as features relating to national security concerns. The data protection commissioner mentions the need to balance right to privacy and national security interests when developing projects of national security interest:

Although privacy is not an absolute right and can be lawfully constrained for national security purposes, when satisfying necessity and proportionate measures, the right to privacy still remains a fundamental human right. Controllers implementing projects of national security interests must adopt the right approach to avoid any undue harm to any citizen in view of the privacy risks that such projects may entail. (Government of Mauritius 2018c)

In the case of the Mauritius safe city, the protection and privacy of data subjects are partly overridden by section 44 of the Data Protection Act 2017, which deals with 'exceptions and restrictions'. National security, defence, and public security are evoked as necessary and proportionate measures. So how valid is the argument for setting up a comprehensive public space surveillance premised on beating crime and making Mauritius safe? Data released by the National Audit Report (2025) seem to indicate otherwise.

Table 12 indicates the very low cases of crime elucidated using safe city cameras:

Table 12: Number of cases of crime

Financial years	Cases elucidated through safe city cameras	Total number of outstanding cases at the end of financial year	Proportion of elucidated cases to outstanding cases (%)
2021/22	355	11,287	3.1
2022/23	529	13,311	4.0
2023/24	487	12,341	3.9

Source: Author’s own based on data from Mauritius National Audit Report (2025).

Test 6. Does the law control who has access to surveillance data?

The Data Protection Act 2017 guarantees the ‘Rights of Data Subjects’ (part VI) and lays emphasis on a series of rights such as ‘rights of access’ (section 37), ‘automated individual decision making’ (section 38), ‘rectification, erasure or restriction of processing’ (section 39), and ‘right to object’ (section 40). In the case of ‘rights of access’, when personal data of subjects are being processed, the controller has the obligation to mention the purpose for doing so, the type of personal data concerned, to whom the personal data is being disclosed, and the length of time the personal data will be stored (section 37 (2)). Under section 39, inaccurate personal data of a subject will be rectified without undue delay and any personal data no longer necessary in relation to the purpose for which they were collected should be erased (sub sections 1 and 2). Lastly, the ‘right to object’ gives the data subject the right to object in writing to the processing of his/her data unless the controller demonstrates compelling legitimate grounds for its processing (section 40 (1)).

However, in the case of the Mauritius safe city, ‘rights of access’ (section 37) and ‘automated individual decision making’ (section 38) have been exempted through a certificate from the prime minister (Code of Practice, Safe City System(s) 2020). In the case of ‘rights of access’ this debars data subjects from getting access to information from the data controller as to the purpose of the data processing, to whom it has been disclosed, the period for which it will be stored, and the right to lodge a complaint with the data commissioner.

So how safe is the data of subjects when it comes to the Mauritius safe city? In response to a parliamentary question from the then leader of

the opposition, Xavier Duval, concerning storage and access of data from the safe city, the then minister of defence had assured Mauritian Parliament that 'all data will be stored on dedicated Police Storage Servers which have been designed to keep data for that purpose'. 'These servers will be in a Government building and access to these servers will be strictly controlled. Only trained and authorised Police Officers will be allowed access to the stored data' (Government of Mauritius 2019).

However, this assurance concerning the handling and security of safe city footage was questioned in relation to handling of the data of a murder case dubbed by the media as the 'Kistnen case'. On 18 October 2020, the charred body of a political activist from the then ruling party – the *Mouvement Socialiste Militant* – was discovered in a sugar cane field. Initially treated as a suicide case by the Mauritian police the latter turned out to be a murder. The judicial inquiry raised concerns about the handling and manipulation of the safe city footage relating to Kistnen's movements on the day of his death. The case fuelled wide public distrust on the intention and use of the safe city and the possibility of being used as a surveillance tool (Khan 2021; Kanhiya 2023).

Test 7. Does the law create an independent oversight body?

There is no dedicated oversight or regulatory body that oversees public space surveillance. Currently it is only the Data Protection Office and the data protection commissioner that ensure the protection of the data of subjects as per the Data Protection Act 2017. As mentioned earlier, the 'exceptions and restrictions' accorded by section 44 override the duty of protection of data subjects when it comes to matters of national security, defence, and public security. The words of the data protection commissioner ring loud about exercising due caution when it comes to applying national security, defence, and public security in a blanket fashion:

They should be considered on a case-by-case basis for each project after taking into consideration all privacy and data protection implications. In line with the principle of accountability, a controller should be in a position to justify and document the reason(s) for relying on the national security, defence and public security exemption so that it can demonstrate its compliance with the DPA.
(Government of Mauritius 2018c)

The absence of a dedicated regulatory framework was one of the issues that were decried by the then leader of the opposition, Xavier Duval (Government of Mauritius 2019). In fact, Mauritius has a massive public space surveillance system powered by intelligent cameras with facial and number plate recognition that was allowed to come to life with

legislation that rightly focuses on data subjects but has not factored in the operations, procedures, and protocols that would guarantee that public surveillance is done in the public interest and that the Mauritius Police Force (the operator) is transparent and accountable when it comes to the safe city. Having simply a Code of Practice for the Operation of the Safe City System(s) is far from sufficient.

Test 8. Has a human rights impact assessment been carried out?

The Data Protection Act 2017 stipulates that a 'data protection impact assessment' (section 34) must be carried out prior to the processing of data. Further emphasis is laid when it comes to the 'processing and systematic monitoring of publicly accessible area on a large scale' (section 34 (2c)) and the assessment should include 'an assessment of the risks to the rights and freedoms of data subjects' (section 34 (3c)). Discussions in the Mauritian Parliament indicate that the views of the data protection commissioner were sought concerning the implementation of the Mauritius safe city (Government of Mauritius 2019). However, there is no information on whether these views concerned the 'data protection impact assessment' and whether the Mauritius Police Force (the data controller) did carry it out as prescribed by the Data Protection Act 2017.

7. Conclusion

Is the security and safety argument sufficient to provide a legitimate aim for such massive, costly, and intrusive deployment of public space surveillance? The presence of a data protection legislation is a positive and encouraging feature but the latter is diluted or even sidelined when the clause on 'exceptions and restrictions' is applicable to the Mauritius safe city and given that that one person – the prime minister – can override data subjects' right to privacy and protection by evoking national security, defence, and public security.

Concern is voiced on how public space surveillance projects are conducted with a veil of secrecy and lack of open communication giving the impression that there is a hidden agenda. Is it really about security, safety, efficiency, or something more sinister that has to do with surveillance and manipulation? For example, the ownership and control of the data from the safe city is ambiguous and shifting. Is it solely the Mauritius Police Force that is the custodian of the footage or do other parties also have access? Another point of concern is the facial recognition capacity of the safe city cameras. As mentioned in an earlier section, the safe city is equipped with facial and number plate recognition capacity. Debates in the Mauritian Parliament have revealed the possibility of the creation of data banks so that the facial recognition component can be activated (Government of Mauritius 2020a,b). A recent statement in Parliament by the current prime minister, Navin Ramgoolam, confirmed that the safe city cameras 'are equipped with facial recognition technology but they are not being used because it is in contravention with our data protection laws' (Government of Mauritius 2025b). What must be avoided at all costs is for the the decision to activate the facial recognition capacity of the safe city to be taken in a covert manner. To avoid this, digital/data subjects must be active players and not mere recipients as has been the case so far. Understanding and militating for their digital rights is imperative as it will allow public space surveillance to become accountable and transparent in its intended purpose and will protect public interest.

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Smart city surveillance in Mozambique

Dercio Tsandzana

1. Introduction

Mozambique has experienced a rapid expansion of public space surveillance in the last five years. The country's two largest cities, Maputo and Matola, now have extensive 'smart' CCTV camera networks with facial recognition and car number plate recognition capabilities driven by AI systems installed with Chinese funding and equipment alongside smaller contributions from other countries. Although the government justifies the surveillance systems as public safety measures, civil society organisations have expressed concerns about human rights, pointing out that cameras have been deployed in locations where political opposition is concentrated and noting that the surveillance infrastructure was designed and deployed without public consultation in the absence of any adequate legislative framework.

This report presents the most comprehensive documentation to date of digital public space surveillance in Mozambique. It explores the growth of public space surveillance in Mozambican towns, particularly Maputo and Matola. It seeks to comprehend how 'safety' is being reinterpreted and disputed in modern urban Mozambique by using a digital rights lens.

The remaining sections of this report are organised in the following way: the next section provides information on the historical and political history of surveillance in Mozambique as context for this new generation of digital surveillance technologies. A case study description is then provided of the basic features and key dates of the present 'smart' city surveillance project as well as details of the main foreign and domestic actors. The regulatory framework is then discussed before the implications of expanded surveillance are considered from the perspective of human rights. The study concludes with some actionable recommendations for policy and practice.

2. Background

Mozambique's political culture has been characterised by strong state control and centralised governance ever since it gained independence in 1975. However, surveillance dates to colonial times, during which Portuguese security services used it as a tool to suppress voices advocating for self-rule (Araújo n.d.). The ruling party Front for the Liberation of Mozambique (FRELIMO), which has socialist roots, consolidated control in the early post-independence era by nationalising important industries and stifling dissent under the pretence of 'socialist legality'. Citizens were encouraged to participate in politics through strictly regulated organisations, but only through officially approved institutions (Jones 2005). The necessity to safeguard revolutionary victories throughout a protracted civil war (1976–1992) frequently served as justification for the pervasiveness of surveillance and social control in daily life.

The state had a tight hold on politics in spite of decentralisation initiatives, marginalising alternative local governance models and traditional authorities (Nuvunga 2006). Since 1990, Mozambique has operated under the democratic rule of law. But the nation had already experienced a period of one-party dictatorship, having been established in 1977, two years after the Proclamation of National Independence. The political and social environment of the postcolonial era (1975) and independence was characterised by ruling party FRELIMO mistrust due to the dangers of contestation by local and regional pro-democratic movements. This resulted in a strict control and surveillance of the populace (Bussotti 2015).

State control has been embedded in Mozambique's historical trajectory. The colonial authorities routinely surveilled anticolonial activists, especially those associated with FRELIMO. Portuguese security services, such as PIDE and SCCIM, tracked, infiltrated, and suppressed resistance networks in Mozambique and surrounding countries (Araújo n.d.). Working in tandem with the Estado Novo's² larger intelligence

2 The Estado Novo (New State) was the authoritarian regime that ruled Portugal from 1933 to 1974, established by António de Oliveira Salazar. It was characterised by strong centralised power, censorship, repression of political opposition, and a corporatist economic model. Rooted in conservative and nationalist ideology, the Estado Novo rejected liberal democracy and socialism, promoting traditional Catholic values and national unity instead. In the context of Portuguese colonialism, the Estado Novo was fiercely committed to maintaining Portugal's colonial empire, including territories like Mozambique, Angola, Guinea-Bissau, and others. Unlike other European powers that began decolonising after World War II, Portugal under the Estado Novo refused to grant independence, insisting that its colonies were integral parts of the Portuguese nation. This led to prolonged and violent liberation wars throughout the 1960s and early 1970s. The regime ended with the Carnation Revolution in April 1974, a peaceful military coup that ushered in democracy and decolonisation.

network, SCCIM was essential in monitoring nationalist activities and coordinating military intelligence, with a particular emphasis on undermining FRELIMO's organisational capabilities (Araújo n.d.).

From the colonial suppression of nationalist mobilisation to the post-independence state control of political opposition, these narratives highlights significant (dis)continuities in Mozambique's surveillance practices. In fact, the concept of state monitoring persisted after independence in 1975 – it was diverted instead. From the outset, the current ruling party, FRELIMO, used and modified surveillance techniques to keep an eye on internal dissent and, most significantly, to oppose the Mozambican National Resistance (RENAMO) during the civil war (1977–1992).

Under the pretext of national security the FRELIMO government viewed RENAMO – first backed by Rhodesia and then by apartheid South Africa – as a threat to independence. This served as justification for extensive internal surveillance. As Darch (2018) explains, successive administrations in Mozambique have increasingly reinforced centralised control, marked by the party's appropriation of state administrative structures and the systematic delegitimisation of alternative forms of nationalism and political opposition. These dissenting voices were often portrayed – sometimes not without reason – as infiltrators, enemies, *xiconhocas* (a corrupt and opportunistic cartoon figure), or saboteurs.

3. Case study description

In recent years, Mozambique has rapidly integrated digital technology into its urban governance and security architecture, joining a rising number of nations in Africa that have chosen so-called smart or safe city models (Woodhams 2020). In the case of Mozambique, the term 'smart city' is the most appropriate designation and will be applied consistently throughout this report. These programmes, which are frequently marketed as enhancing traffic management, public safety, and urban efficiency, mostly rely on data-driven surveillance technology including digital communications monitoring, CCTV surveillance systems, and facial recognition technology (Nhanale 2021). These tools pose serious concerns regarding transparency, data governance, and civil rights, especially in nations with poor democratic institutions and privacy laws, even while they offer a more modern and safer urban environment.

Maputo, the country's capital, and Matola, its urban twin, have become the main locations for Mozambique's smart surveillance projects. Along with important government infrastructure and institutions, these urban centres are also the sites of growing political opposition and grassroots activism. According to Antonio (2022) asserting governmental presence and authority in crowded, politically charged areas while keeping an eye on crime and traffic is the practical and symbolic goal of installing surveillance infrastructure in major cities. Maputo hosts an extensive CCTV system installed with Chinese support (Passalacqua and Polito 2024). This system, despite being presented as a harmless road safety measure, follows a trend in which development partnerships are used to deploy monitoring infrastructure with little to no public explanation or consultation (Nielsen 2011; Mazzolini 2016).

The government of Mozambique has several key partners in these smart city ventures as illustrated in Table 13. below. Mozambique was first introduced to the concept of smart cities and the installation of surveillance cameras in 2016 (@Verdade 2016). As of 2025, it is estimated that more than 450 cameras are in operation across the capital, Maputo, and the neighbouring city of Matola. As a strategic development partner, Huawei frequently provides technical know-how and concessional funding. However, civil society groups have expressed concerns about data sovereignty, surveillance issues, and the reliance on Chinese technology in key infrastructure (Club of Mozambique 2017). Huawei is able to operate in a privileged position within the nation's digital transformation programme because of its close

political links with Mozambican leaders and the high-level diplomatic ties that exist between China and Mozambique (Nhanale 2021).

Similar to Huawei, ZTE functions under a framework of Sino-Mozambican collaboration that encompasses infrastructure development, technology transfer, and strategic alignment with governmental interests. Its engagement in surveillance and security technologies – like smart city projects – places it in politically delicate regions, posing questions about accountability, transparency, and the possibility of authoritarian tech adoption (O Pais 2024). Another key partner for smart/safe cities in Mozambique is the Korea International Cooperation Agency. In fact, during a June 2024 visit to Korea, the former president of Mozambique, Filipe Nyusi, expressed his belief that South Korea is willing to share its growth experience with Mozambique (Club of Mozambique 2023).

In recent years, digital surveillance has been extended to include public space surveillance as a key component of Mozambique's urban security strategy, particularly through China-backed initiatives under the safe city programme. According to Canal de Moçambique, one of the country's most credible newspapers, 450 surveillance cameras were installed in Maputo and Matola in 2016 as part of the National Information Interception Command project, which allegedly includes a 'phone tapping' scheme aimed at spying on citizens (Global Voices 2016). The mentioned project aims to provide live information related to urban-public crimes in the capital Maputo. Although the value of the safe city surveillance camera contract has not been made public, we do know that CCTV surveillance is a growth business sector in Mozambique. The project is estimated at 140 million, according to @Verdade (2016).

According to Statista (2025), the smart security cameras market³ in Mozambique is projected to generate revenue of US\$7.2 million in 2025, with an expected annual growth rate of 13 per cent between 2025 and 2029, reaching a market volume of US\$11.8 million by 2029. Statista (2025) explains that the projections above are based on the sales of smart home products, excluding taxes.

The Ministry of Interior awarded the CCTV installation contract to Msumbiji Investment Limited without considering alternative bids (Tsandzana 2016). Notably, this company is owned by the son of former president Armando

3 Statista defines smart security cameras as advanced surveillance cameras equipped with features such as motion detection, subject recognition, and remote access via the internet or mobile applications. These devices connect to Wi-Fi networks and can be integrated into broader home automation systems, enabling real-time monitoring, automatic alerts, and cloud storage for recorded footage. Smart security cameras are equipped with built-in intelligence and communication capabilities, allowing them to interface seamlessly with automation equipment. This segment focuses specifically on the cameras themselves and excludes additional components like storage devices or monitoring.

Guebuza, who subcontracted the Chinese firm ZTE (Club of Mozambique 2023). Many officials, including city mayors, were not aware of who put the cameras in their towns or their intended purpose, according to enquiries by journalists and activists (MISA 2022). Table 13 charts elements of the safe city programme.

Table 13: Chronology of the projects

Project	Country involved	Amount (US\$)	Year	City(ies)	Who is involved?
Safe city programme	China	Not public	2016	Maputo and Matola	Ministry of Interior
National information interception command (data centre)	China and Mozambique	140 million	2016	Maputo	Ministry of Interior (Defence and Security Forces)
Road safety improvement and institutional capacity building project	Korea	7 million	2023/2024	Maputo	Ministry of Transports and Communication
Road safety and public transport system	Not known	Not known	2025	Matola	Municipality of Matola

Source: Author's own.

The degree of opacity about the safe city programme heightens worries that surveillance infrastructure is just one component of a larger social and political control agenda rather than just a public safety instrument (Tsandzana 2016). China are not Mozambique’s only international partners in expanding public space surveillance. In August 2023, the Korea International Cooperation Agency (KOICA) announced a project to improve road safety in Maputo, which includes the installation of cameras for speed control and traffic offence detection – it is not clear whether the technology includes number plate recognition and/or facial recognition (Integrity 2024). Apparently, the system is operated by the Ministry of Interior. The funding of around US\$7 million (€6.6 million), as a donation, was awarded to the government of Mozambique through the KOICA for the implementation of the ‘Road safety improvement and institutional capacity building project’.

Civil society groups have raised concerns about the potential misuse of surveillance technology, particularly for monitoring political opposition and activists (DW 2024). In both cases – 2016 (surveillance cameras in

Maputo/Matola) and 2023 (road safety in Maputo) – the Mozambican government has never officially disclosed the intended purpose of the surveillance equipment. There is no public information about expansion of the camaras since 2016. Even the police and municipal authorities claimed they were unaware of the cameras' function or the entity responsible for their installation (O Pais 2024). This opacity suggests either covert surveillance practices or a broader lack of digital transparency, which is particularly concerning given that Mozambique has no data protection law.

4. Key actors

In August 2025, the Municipal Company for Mobility and Parking of Matola (EMMOBE) in the Municipality of Matola launched the Road Safety and Public Transport System 'to address irregular practices by transport operators, such as shortening routes' (A Carta 2025).

We need to make use of electronic tools, invest in videos and inspection cameras – in other words, a kind of fiscal counter-intelligence – so that offenders can be identified. They won't know who is monitoring them at the stops, and they will be summoned using their licence plate numbers, along with proof, so they can be penalised.

(Francisco Lichucha, Chairperson of EMMOBE, July 2025)

Despite the initiative's apparent potential, very little information on its funding, project start date, coverage areas, and populations it is meant to assist is publicly available. The planning and execution process, as well as EMMOBE's responsibility and openness in overseeing a project that directly impacts everyday commuters and the general public, are all seriously called into doubt by this opacity. It is challenging to evaluate the system's possible influence or efficacy in the absence of these crucial details.

Without transparent reporting on operational logistics, monitoring systems, or funding allocation, there is a chance that the project won't meet its objectives and might even exacerbate already-existing disparities in access to public transportation. Furthermore, the absence of public consultation or involvement with impacted communities erodes public confidence and restricts accountability prospects. The current opacity draws attention to a larger governance issue in municipal programmes that are supposed to improve public services but offer little insight into how they are actually carried out or how they affect society.

Political and business interests stand out as the primary themes in Table 13. This can be explained by the fact that many of the organisations engaged have strong relationships with government institutions, either directly or through elite networks or strategic investments in areas that the state has designated as priority. These connections frequently make it difficult to distinguish between the provision of public services and private benefit, giving some players an unfair edge whether it comes to state contracts, access to infrastructure projects, or regulatory benefits (Lysova 2022).

It appears that state security agencies, private technology firms, and political elites – especially those with close connections to government

actors – benefit from Mozambique’s rapid growth of public space surveillance. The implementation of surveillance infrastructure, such as CCTV networks and smart city systems, frequently occurs in many countries (Lyon 2001) without sufficient transparency or public consultation, which begs the question of whose interests are being served. According to Gray (2002) these technologies are being used as means of political control, allowing the surveillance of dissent, the administration of public space, and the strengthening of state authority, even though they are presented as tools for modernising public management and enhancing urban safety. Thus, a deeper logic of centralised control and elite consolidation is concealed by the language of public security (Gray 2002). In the case of Mozambique, the government did not provide information about the need to use the cameras.

Due to opaque procurement procedures that reward those with political clout, private companies – both domestic and foreign – benefit greatly from the expansion of public space surveillance. Examples of beneficiary corporations include Huawei and ZTE. A domestic beneficiary is Msumbiji Investment Limited, a business of which not much is known to the general public, but which is thought to be connected to the son of the former president of Mozambique. It appears that public funding and digital infrastructure are being used for private enrichment and influence building, as evidenced by the possible involvement of politically connected companies in the surveillance industry. Under such circumstances, the purpose of surveillance technologies shifts from safeguarding the private enrichment rather than public safety. Table 14 records the key actors shaping the expansion of public space surveillance in Mozambique.

Table 14: Key actors in Mozambique’s safe city programme

Project	Actor(s)	Interest
Safe city programme	Huawei	Huawei: Working collaboratively with government agencies to create telecommunications networks, mobile broadband, and digital public services, Huawei has been instrumental in the growth of Mozambique’s ICT infrastructure. The company has collaborated with the Ministry of Science and Technology on training initiatives, e-government platforms, and the rollout of fibre-optic infrastructure, among other significant national projects.
National information interception command	ZTE and Msumbiji Investment Limited	ZTE: Despite frequently having a lesser profile than Huawei, ZTE, another significant Chinese ICT business, also has a significant presence in Mozambique. ZTE has supported government initiatives to upgrade communication networks and supplied state-owned businesses with equipment and services. Msumbiji Investment Limited: There is no public information available, but it is believed to be part of the former president’s son’s company.
Road safety improvement and institutional capacity building project	Korea International Cooperation Agency	KOICA: Since the early 2000s, the Korea International Cooperation Agency (KOICA) has operated in Mozambique, concentrating on development cooperation in fields such agriculture, water and sanitation, education, health, and institutional capacity building. KOICA has provided funding for projects in Mozambique, frequently in conjunction with government ministries, such as the establishment of ICT infrastructure, rural health programmes, and vocational training centres.
Road safety and public transport system	EMMOBE	EMMOBE: It is a newly established company, created in July 2025. According to its chairperson, its goal is to put an end to persistent misconduct often committed by public service operators to the detriment of passengers – the introduction of this mechanism will not require deploying additional municipal police officers, as the aim is instead to enhance the performance of the existing municipal police.

Source: Author’s own.

This relationship between surveillance, business, and elite networks is indicative of a larger trend in Mozambique’s political economy, where clientelism and unofficial power structures influence governmental initiatives. Instead of promoting safety, accountability, or trust, the growth of surveillance infrastructure runs the risk of escalating inequality, bolstering authoritarian tendencies, and alienating the very people it is meant to protect.

As far as the public is aware, the Mozambican government did not perform any kind of risk analysis or human rights effect assessment prior to enforcing citizen surveillance in public areas. Without any public participation, parliamentary discussion, or disclosure of measures to protect residents’

rights and privacy, monitoring technologies – particularly CCTV systems – were implemented in large cities like Maputo. Given Mozambique's history of political repression and limited institutional accountability, this lack of transparency raises fundamental questions regarding the necessity, appropriateness, and constitutionality of such tactics in a democratic setting.

5. Regulatory landscape

The right to privacy is established by article 41 of the Mozambican constitution (1990). It states that 'everyone has the right to honour, a good name, a reputation, the protection of their public image and the privacy of their private life'. However, the same constitution is silent about the digital surveillance. In response to the difficulties brought about by the growth of the digital technology industry, Mozambique has been enacting a number of legislative revisions and legal instruments that either directly or indirectly address the problem of electronic surveillance (MISA 2022). Several legal sources provide general guidelines and principles for data protection. These include the Electronic Transactions Law (3/2017), the Labour Code (Law 6/2019), and the Civil Code (decree law 47344). Notably, the Electronic Transactions Law and the Labour Code have sector-specific regulations.

Additionally, the National Institute for Information and Communication Technologies (INTIC), the primary administrative entity in Mozambique overseeing the information and communications sector, has been engaged in the drafting process for a national data protection law. As part of this process, Mozambique ratified the African Union Convention on Cybersecurity and the Protection of Personal Data in 2019 with resolution 5/2019. Significant changes were also made to the criminal code framework, which was authorised by law 24/2019. These changes were made to address issues related to cybercrime and to expand the criminal framework to include crimes that may arise from the use of new technologies, data forgery, improper use of information devices, and highly pertinent wiretapping issues.

The Law on Electronic Transactions (law 3/2017), which was enacted in 2017, governs e-government and e-commerce matters as well as the security of ICT users and suppliers. It is under this law that crucial data protection talks take place. Law 2/2017, which simultaneously introduces several pertinent elements of access to personal data for criminal investigation purposes, particularly in its article 18, replaced the Criminal Investigation Police in 2017 with the National Criminal Investigation Service (SERNIC). However, the Electronic Transactions Law includes particular clauses that control third-party access to or exchange of data, controlling data transfers to foreign organisations or services. Unless required by another statute or a court order, these laws limit such transfers. As a result, these clauses limit transfers between parties inside Mozambique and have an effect on dealings with foreign third parties.

The Council of Ministers introduced decree 44/2019 on Telecommunications Service Consumer Protection in 2019. Article 10 of the decree outlines the consumer's right to privacy and protection from unauthorised use of their personal information, including their addresses, communications, and personal data. In keeping with this spirit, article 7 of decree 66/2019, which deals with the 'regulation of the security of telecommunications networks', emphasises the importance of consumer privacy and data protection, as well as the necessity of obtaining consent before sharing consumer data with third parties.

Since telecommunications network security directly safeguards customer privacy and personal information, it is essential for combating surveillance. Such regulations reduce the possibility of illegal monitoring and misuse of sensitive data by laying out explicit guidelines for data collection, storage, and sharing, such as the need to get consent before disclosing information to other parties. Strong telecommunications security and data protection frameworks are crucial safeguards in situations where state or private actors use surveillance technologies in an opaque manner.

At the time of writing this report, Mozambique lacks a data protection legal framework. Mozambique has not yet passed any specific data protection laws, even though it is a signatory to the African Union Convention on Cyber Security and Personal Data Protection, which was codified through resolution 5/2019. There are no legally obligatory guidelines on the collection, storage, sharing, or deletion of data in the absence of defined legal criteria. Because they lack institutional channels to seek redress in the event that their data is misused, citizens are left in a state of ambiguity and have no awareness or control over how their personal information is handled. This absence erodes confidence in organisations in charge of handling sensitive data in the face of increasing digitisation and public space surveillance.

In addition, there is no extensive legal structure in Mozambique that governs digital monitoring, data protection, or the use of AI in public. Although the right to privacy and the inviolability of communications are recognised by the constitution (article 71), there are no strong enforcement measures to support these rights. The current legislative requirements pertaining to public safety and telecommunications interception are ambiguous, permitting state actors to apply them however they see fit. The country's current laws pertaining to interception and monitoring are either unclear or not enforced (Tsandzana 2025).

When illegal surveillance or misuse of data occurs, citizens have no remedy because there is no data protection authority or privacy oversight organisation. Additionally, public officials, such as local governments and the police, frequently assert that they are unaware of the particulars of

the surveillance infrastructure in their areas (Nhanale 2021). A culture of impunity and clandestine operations by the current administration, since the ruling party came to power, has been enabled by this legal void, as noted by Global Voices (2016) and the Club of Mozambique (2017). To ensure accountability, define roles, and bring Mozambique's surveillance methods into compliance with international human rights standards, legal reform is required. The restrictive environment that predominates in Mozambique may be the reason for the lack of study or reporting on these topics, rather than a lack of interest.

In summary, this section has shown that Mozambique lacks a legal framework governing public space surveillance. As a result, the legitimate objectives remain undefined, the designation of actors authorised to conduct surveillance is ambiguous, and mechanisms for redress, proportionality, necessity, and independent oversight are absent. In other words, Mozambique faces a situation in which public space surveillance can be deployed without accountability measures.

6. Digital rights

The aforementioned cameras' installation has sparked serious worries about digital rights. The lack of transparency in the decision-making process, the absence of explicit rules for the gathering of, storing of, and access to CCTV surveillance data, and the possible abuse of surveillance have all drawn criticism from citizens and activists (Tsandzana 2016; DW 2024). The fundamentals of accountability and supervision, as well as the rights to information and privacy, are the main areas of concern. Instead of improving public safety, these cameras might be used to violate human rights by surveillance of people's whereabouts, keep tabs on political activity, or silence critics if there is no independent oversight or legal protection.

In Mozambique, journalists, civil society organisations, and proponents of digital rights have continuously expressed concern about the growth of surveillance without public participation or legal protections (Bussotti and Nhaueleque 2023). The possible targeting of opposition voices, data misuse, and the suppression of free speech and protest are among the issues. Public mistrust has been exacerbated by the fact that the surveillance systems were put in place through ambiguous contracts, frequently involving middlemen with political connections. In order to expose these networks of influence and hold officials responsible, investigative journalism like Canal de Moçambique and others has been essential in denouncing projects related to surveillance in Mozambique (Nhanale 2021). However, evidence of smart/safe city surveillance in the case of Mozambique is still limited, mainly because the access to public information related to the projects is still unaccusable.

The conflict between democratic principles and official secrecy has also been brought to light by activists. Despite the government's promotion of surveillance as a weapon for the public good, concerns about authoritarian drift are raised by its unwillingness to reveal fundamental details regarding ownership, operation, or oversight. Civil society has demanded rapid legal reforms and public discussion on digital rights in the absence of a data protection law (Nhanale and Verde 2023). Although these demands have not yet been addressed, the topic is expected to continue to be on the public agenda due to the increased worldwide attention and the tenacity of investigative media.

The Defence and Security Forces are in charge of operating and managing the security cameras positioned across Maputo City and Province, according to a February 2024 report by the media site O Pais (2024). Initially, it was anticipated that these gadgets would aid in lowering crime rates and, more

specifically, expedite the conclusion of kidnapping cases, a problem that has been documented by successive governments since 2011. Soico Television did point out that it's still unclear what mechanism controls how these cameras work. In a situation where monitoring is growing without any kind of independent supervision or oversight, this lack of transparency emphasises the necessity for unambiguous justifications to protect citizens' rights.

There are significant issues regarding digital rights with the police's refusal to reveal the usage of security camera footage, citing state secrecy as the reason. In actuality, this kind of opacity erodes public confidence and fosters the possibility of surveillance technology abuse, undermining fundamental rights to accountability, privacy, and the rule of law. Social activist Mirna Chitsungo reinforces the idea that the project may have served political and personal interests rather than public ones:

The process of installing these surveillance cameras, combined with the company involved in setting up this equipment, brings to mind corruption cases that those connected to the Mozambican government have made us accustomed to. There is no explanation as to how a member of the former President's family was involved in this business.
(Activist Mirna Chitsungo, Interview DW 2024)

Mirna Chitsungo also speaks about the violation of the right to information: 'By installing these cameras without informing Mozambicans, the government violated a fundamental right – the right to information', she argues. Privacy concerns are also a central point in this debate. With cameras scattered across the city, many citizens fear that their lives are being monitored not for their protection, but to serve obscure interests. In September 2024, DW questioned journalists about the surveillance cameras in Maputo. According to journalist Rafael Machalela, these cameras were installed by a closed government that communicates little – or not at all – with the people.

The government made the decision on its own to install the cameras without informing Mozambicans through Parliament. That is unfortunate, and in this case, there was a violation of the right to participate in decisions that affect the country.
(Journalist Rafael Machalela, Interview DW 2024)

It is also claimed that the installation of the cameras has not reduced public crime (DW 2024). On the contrary, cases of abduction have increased, and the police have been unable to provide concrete information on how many crimes have actually been solved with the help of the cameras. 'We don't know how many kidnapping cases have been resolved thanks to the

cameras', said Hilário Lole, spokesperson for the Criminal Investigation Police, in September 2024.

There is considerable doubt about the police's assertions about the surveillance cameras' efficacy. Although they claim that the cameras are meant to improve public safety, there is evidence that crime rates – especially those related to kidnappings – have increased since they were installed. Furthermore, the authorities have not produced any credible statistics showing the number of instances that have been resolved using this technique. The fact that they do not know how many kidnapping cases have been resolved emphasises the project's lack of transparency and accountability. Concerns are raised that the cameras might be more symbolic than useful in genuinely lowering crime because the police's statements are unsupported by precise statistics or an impartial assessment.

7. Conclusion

Given the absence of a thorough data protection framework, this report shows that the installation of surveillance cameras across Maputo and Matola presents serious obstacles to citizens' digital rights in the country. Despite the fact that these cameras are marketed as instruments to increase public safety and fight organised crime, the lack of legal protections exposes personal information to possible abuse. People lack adequate channels to seek redress in the event that their rights are violated, and they have little to no clarification on how data is gathered, stored, or accessed. In addition to undermining confidence in governmental institutions, this regulatory gap increases the possibility that monitoring may be utilised for private or political gain rather than for the benefit of the general public. The lack of a formal data protection law in Mozambique is especially troubling in the context of government-led surveillance initiatives.

The intricate conflict between digital rights and public security in Mozambique is brought to light by the installation of surveillance cameras throughout Maputo and Matola. Although lowering crime and more effectively resolving crimes like kidnapping are the declared goals, the process has revealed serious flaws in accountability, transparency, and legal supervision. Concerns over the possible abuse of surveillance, whether for social control, political reasons, or personal benefit, are raised by the absence of established procedures for regulating and overseeing the use of these cameras. People are left in a state of ambiguity, unable to determine how and when they are being watched, and without recourse in the event that their rights are infringed.

This circumstance further emphasises how urgently the government must implement a rights-based strategy for digital surveillance. In addition to legal protections, open processes and impartial scrutiny are necessary to preserve citizens' privacy, guarantee information access, and foster confidence in public institutions. In the absence of these safeguards, surveillance technologies run the risk of undermining core democratic values and solidifying a climate of distrust and secrecy. To make sure that security activities advance rather than impede the public interest, it is imperative to strengthen governance structures, interact with civil society, and educate the public about digital rights.

To address these challenges and safeguard digital rights while pursuing public safety, the government should take the following actions:

- Introduce comprehensive data protection laws that set down precise guidelines for the gathering, storing, and use of personal data, as well as independent oversight and public accountability procedures.
- Establish a transparent regulatory framework for all surveillance technologies, including clear rules on data collection, storage, access, and deletion.
- Implement independent oversight mechanisms to monitor and evaluate the use of surveillance cameras, ensuring accountability.
- Ensure public awareness and consent, informing citizens about camera locations, data usage, and their rights regarding surveillance.
- Guarantee data protection and privacy safeguards, including legal remedies to prevent or address misuse of collected information.
- Conduct regular audits and public reporting to maintain transparency, build public trust, and demonstrate that surveillance serves the public interest.

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Smart city surveillance in Nigeria

Jake Okechukwu Effoduh and Odeh Friday

1. Introduction

In recent years, Nigeria has seen the government increasingly adopt the safe city initiative to boost urbanisation and security of cities in the country. While there are other intricate components to these initiatives, the claim by the government is that at the centre of the safe city initiative is the use of surveillance technologies to monitor public spaces, combat nefarious activities, and ensure safety in cities. Although other states and cities in Nigeria are increasingly embracing these initiatives to foster the safety of lives and property within their domains, Lagos and Abuja stand out as notable testing grounds for these technologies (Oluwadiya 2012). This is because these states are the most technologically inclined states in the country, in addition to the fact that both cities are renowned for their partnerships with foreign companies that provide surveillance technologies for the cities. Public observers note that there is the question of whether these initiatives have been diligently implemented in these cities since their launch by respective governments. However, commentators like Oluwadiya (2012) hold the view that there is no denying its benefits. This report will assess evidence on the benefits and risks claimed before reaching conclusions.

However, although the surveillance technologies and the initiatives behind them are important additions to established security mechanisms for the protection of lives and properties in Nigeria, the legal implications of such technologies on the personal lives and privacy of Nigerians remain a matter for debate. Hence, this report examines the current condition of public space surveillance in Nigeria. It also provides a detailed timeline of the safe city projects in Nigeria, with the actors involved in these projects. Additionally, it explores the regulatory landscape behind the use of surveillance technologies to protect lives and properties in Nigeria. Altogether, this report seeks to address the following questions:

- To what extent have different levels of government in Nigeria progressed their safe city projects?
- To what extent is Nigeria's initiative of safe city public space surveillance legal, necessary, and proportionate?
- How does this initiative affect the practice of digital rights and justice?

2. Background

Nigeria's history with surveillance dates back to the colonial era when a Special Branch was created by the colonial authorities to spy on local dissidents who sought to challenge its control in the country. While the colonial masters left the political shores of Nigeria in 1960, the mechanisms for government-backed surveillance remained and became institutionalised into the security architecture of the newly independent state. The institutional surveillance systems took the front burner again as the military took over political and economic control of Nigeria in 1967. As part of its strategy to keep the people in check and obliterate criticisms of its high-handed approach to governance, the military, especially the General Olusegun Obasanjo-led regime (1976–9), utilised state surveillance systems to violate the constitutional right to privacy enjoyed by Nigerians. This right was repeatedly violated by the General Obasanjo-led regime to suppress the voice of critics and dissenters to military rule (Carver, n.d.). However, while Nigeria has enjoyed an uninterrupted democratic process for over two decades from the year 2000 to date, the fear of the violation of the right to privacy still weighs heavily in public circles. Therefore, the introduction of surveillance technologies as part of the safe city initiative further exacerbates this fear and scepticism. Worthy to note is that despite democratisation, the surveillance architecture has been further expanded by the emergence of digitalisation and its rapid immersive growth into economies (Carver n.d.).

It has been suggested that the deployment of surveillance technologies in the context of the safe city initiative is not aimed at political repression as was the case during the military era, but as a strategy to enhance the safety of lives and properties in cities where they are deployed (Feldstein n.d.). Importantly, the Nigerian government across federal and state levels has maintained that the deployment of surveillance technologies and introduction of the safe city plan is targeted at ensuring the protection of national security, prevention of crime, enhancement of the dynamics of crime investigation, improvement of economic gains and stability, enhancement of public safety, and a commitment to international agreements with national security as the basis (Laibuta *et al.* 2024). These objectives are important as Nigeria struggles to contain the scourge of domestic insecurity, such as the Boko Haram insurgency, the kidnappings by herdsmen, the unknown gunmen saga, that has engulfed the country since the advent of its current democratic dispensation. However, while the integration of surveillance into the security architecture of the nation aims at strengthening its capacity to pre-empt and quickly respond to

security threats, there are still associated risks which are bound to arise if left unchecked, including the risk of misuse by overzealous actors, violation of the rights to privacy by state actors, among others (Ali 2025).

It is against this background and the need to stem this wave of insecurity that the federal and state governments have made a move to onboard safe city surveillance systems across cities to help monitor and prevent crime. This report seeks to highlight and discuss important steps and milestones taken by governments across Nigeria to integrate surveillance systems in their domains. It highlights key dates, main actors, contracts for these surveillance systems, and funding mechanisms, among others.

3. Case study description

3.1 Nigeria National Public Security Communication System Project (Lagos and Abuja)

As part of its safe city initiative drive, the Nigerian government invested in the procurement and deployment of CCTV cameras in Lagos and Abuja through the Nigeria National Public Security Communication System Project in 2010. In furtherance of this project, the government entered into a US\$470 million contract with ZTE, a Chinese company, to deploy 2,000 CCTV cameras across Lagos and Abuja (Laibuta *et al.* 2024). Further, the federal government of Nigeria signed a US\$399.5 million preferential buyer's credit (PBC) agreement with China Eximbank to finance 85 per cent of the CCTV procurement contract the government signed with ZTE (Laibuta *et al.* 2024). The remaining 15 per cent of the contract was agreed to be financed by the Nigerian government. The contract featured the deployment of 2,000 CCTV cameras across Lagos and Abuja, video conferencing subsystems, and e-policing (Aiddata n.d.). The CCTV surveillance system was primarily set up to facilitate real-time communication between security agencies in Lagos and Abuja using data captured by the CCTV cameras (Igbintade 2022).

However, evidence suggests that this project failed due to the commitment of the government officials, not to the success of the project but to the enrichment of their pockets (Business Day 2022a). Queries as to the level of implementation of the project arose due to instances like the multiple bomb blasts in Abuja that affected the United Nations Building and operational headquarters of the Nigeria Police Force (NPF), the Christmas Day bombing of a Catholic Church among others between 2011 and 2014 without the perpetrators being captured by any surveillance system in the city (Effoduh 2021).

Essentially, while this project started in 2011 and was slated to be completed in 2012, its level of completion has been marred by controversies and allegations of corruption. Monitoring the extent to which the project had been undertaken in the states of Lagos and Abuja, an investigative report in 2016 found that only 40 CCTV cameras were installed in Abuja out of a possible 1,000, with no traces of the cameras installed in Lagos (*This Day* 2023). Also, there has been no evidence of data from the CCTV cameras used for any purpose, such as evidence in court cases or data to track perpetrators of crime. Furthermore, the report found that the 40 CCTV cameras installed in Abuja were previously rejected in Ghana and then shipped down to Nigeria, with further investigation revealing that ZTE

was underpaid for the project, as a significant part of the borrowed funds mapped out for this project ended in the pockets of private individuals (Business Day 2022a). Reports reveal that the 40 CCTV cameras installed in Abuja went offline in June 2013 (Culture Custodian 2019).

Figure 3: Timeline of the Nigeria national public security communication system project (Lagos and Abuja)

Year	Key milestones
2010	Announcement of the project and extension of loan facility for the project by China Eximbank.
2012	Slated year for completion of the project.
2013	Project abandoned and equipments left to rot.

Source: Author's own based on data from Igbintade (2022)

3.2 Lagos state safe city initiative

In 2016, the Lagos state government launched a safe city project that was distinct from and unaffiliated with the Nigerian Public Service Communication System project discussed above. The state commenced the installation of 13,000 CCTV cameras to curb the incidence of crime in the state (Omisakin 2016). In addition to the CCTV cameras, a command and control centre was commissioned as the monitoring station for the project. While the Lagos state government entered into a partnership with the city of Dubai to help actualise the Lagos safe city project when it conceived the idea in 2016, there are currently no details of other partners or external funders of the current configuration of the safe city project. It is also important to note that the city of Dubai merely provided expertise for the project, with no record of it providing funding. Based on the available information, the state government continues to pursue the safe city mandate with plans to complete the safe city project by 2030 (NAN 2022).

Furthermore, in 2019, the government of the state ramped up investment in surveillance technologies by deploying an additional 10,000 high-definition surveillance cameras across the state with facial recognition capabilities in order to deepen tech-enabled security (Laibuta *et al.* 2024). This is in addition to the recent roll-out of wearable surveillance cameras distributed to security personnel in Lagos state (Business Day 2022b). More recently, the state has been deploying ANPR to

manage traffic and enforce traffic laws and regulations (Aduloju 2023). However, funding details for this project are not publicly available.

These tech-enabled safeguards herald the state's ambition to build a safe city that will be a model for future safe cities across the country. Conceived in 2016, the Lagos safe city initiative was led by the government of the state, with investments from both the state and private actors (Proshare 2016). At the time of writing this report, there remains no report of foreign financiers funding this project. Interestingly, the government of Lagos state has already earmarked US\$161,000 as its investment in the project, with more investments expected from the private sector (Borgen Project 2021).

An integral part of the plan in this safe city is the installation of 3,000 kilometres of fibre metro network cables and broadband infrastructure aimed at boosting internet connectivity across buildings, entities, and agencies in the city by 2030 (Borgen Project 2021). However, the project is still ongoing, with the first phase of the 3,000 kilometres of fibre cables having been laid (Borgen Project 2021).

Figure 4: Timeline of Lagos safe city project

Year	Key milestones
2016	Project initiated with the installation of 13,000 CCTV cameras across Lagos
2019	Installation of additional 10,000 CCTV cameras across Lagos
2021	The Lagos state governmetn invests \$161,000 as seed capital for specially designated Lagos smarrt city project
2030	Project slated to be completed by 2030.

Source: Author’s own based on data from Borgen Project 2021

3.3 Oyo safe city project

In collaboration with the Management Information Centre at the Oyo State Governor’s Office and the Office of the Special Adviser on Security, the Oyo state government launched the safe city pilot project in 2018 to improve the security of lives and property in the state. The project sought to use artificial intelligence to identify abnormalities and discrepancies in crowd behaviour (Olatayo 2018). A fundamental part of the project was the mounting of 46 high-definition CCTV cameras across the state capital, Ibadan (Olatayo 2018). The state government entered into technical partnerships with Huawei Technologies Nigeria Limited and MTN Nigeria Limited to execute and sustain the project, with the former acting as the government’s technical

vendor partner for supplying and mounting the CCTV cameras while the latter provided internet connectivity and data transfer using fibre optics and microwave technology (Olatayo 2018). While the project became operational in 2018, the number of installed cameras and the monetary value of the project are not available to the public, as research on the state’s budget for the year does not show any specific expenditure on the project.

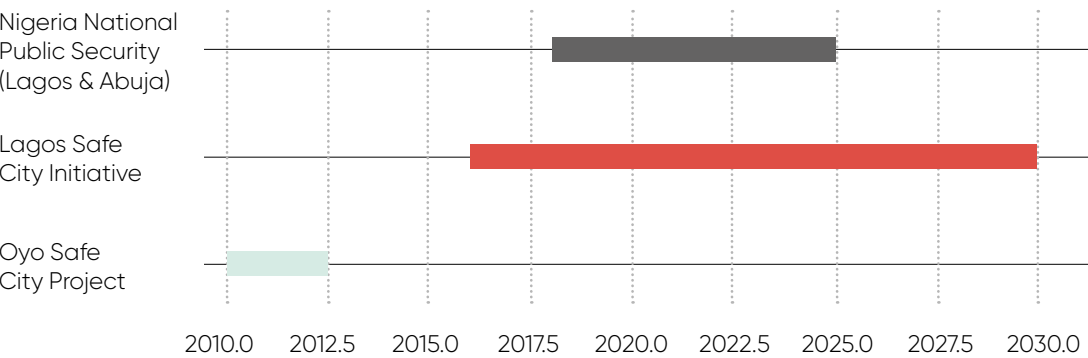
The state government planned to sustain the project through a public–private partnership with levies sought to be imposed on private businesses and religious centres in the state, with an annual target of ₦2 billion (Sadeeq 2018). The current nature and configuration of this project are also not known as of the time of writing this report.

Figure 5: Timeline of the Oyo safe city project

Year	Key milestones
2018	Conception and launch of project
2019	Partnership with technical partners for the procurement of 46 high-definition CCTV cameras in Ibadan Public–private partnership for the maintenance of the project

Source: Author’s own based on data from Olayo (2018).

Figure 6: Graphical representation of safe city projects in Nigeria



Source: Author’s own.

4. Key actors

Table 15 shows the safe city projects that have successfully kicked off across Nigeria and the key players associated with each of these projects.

Table 15: Summary of safe city projects in Nigeria

S/N	Project	Actor	Role	Detail
1	National Public Security Communication System Project (Lagos and Abuja)	Federal Government of Nigeria	Initiator and financier	Provided 15% of US\$470 million and secured an 85% loan from China Eximbank
	China Eximbank	Lender	Advanced US\$399 million loan to the Nigerian government	
	ZTE (China) and Hikvision	Contractor	Responsible for the full deployment of the project infrastructure	
	Nigeria Police Force (NPF)	Key user/operator	Intended primary users of the surveillance and video conferencing system	
	Ministry of Communications, Innovation and Digital Economy	ICT oversight	Oversight on telecommunications and GoTA (Global Open Trunking Architecture) system integration	
2	Lagos state safe city initiative	Lagos state government	Initiator and financier	Funded installation of 13,000 CCTV cameras (2016), 10,000 more (2019), ₦250 million seed capital
	Contractors/city of Dubai	Technical partner/contractor	Supported early-stage design and installation; details of current partners undisclosed	

S/N	Project	Actor	Role	Detail
	Lagos state Ministry of Science and Technology	Implementation agency	Drives digital infrastructure and safe city deployment	
	Lagos State Police Command	User/operator	Uses CCTV and surveillance data for policing	
	Lagos State Traffic Management Authority/traffic management agencies	Users	Use safe city surveillance for traffic enforcement and automated plate recognition	
3	Oyo safe city project	Oyo state government	Initiator and financier	Funded and structured the project, engaged technical partners
	Huawei Technologies Nigeria Ltd (China) and Hikvision	Technical vendor	Supplied and mounted high-definition CCTV cameras	
	MTN Nigeria Limited	Internet provider	Provided internet and data transfer via fibre/microwave	
	Oyo state Ministry of Information and Technology	Supervising agency	Oversees project execution and digital infrastructure	
	Oyo State Police Command	User/operator	Uses surveillance data for public safety	
	Office of the Special Adviser on Security	Strategic lead	Coordinates state-level security planning and integration	
	Private sector and religious centres	Funding partners	Provide ₦2 billion annually for sustainability	

Source: Author's own.

5. Regulatory landscape

Currently, there are no safe cities-specific laws or regulations in Nigeria. The implication of this is that there is no one law that (1) addresses concerns of surveillance under the safe city project; (2) defines and sets out the scope of application and proportionality of these surveillance measures; (3) defines the term 'legitimate aim' for public space surveillance; and (4) sets out a specific regulatory framework or body for the operation and regulation of the safe cities project.

Practically, the fragmentation of applicable rules can be problematic as these laws historically and ordinarily would not have contemplated the challenges posed by safe cities. This gives credence to the call for a comprehensive and fit-for-purpose body of rules that apply specifically to safe cities.

As a guide, in this segment we consider the following issues:

- Necessity and proportionality: whether privacy violations arise from the surveillance occasioned by the safe city projects and to the extent that violations arise, whether these violations are necessary and proportionate to achieve any legitimate aim?
- Access control: are there protections in place to limit access to the digital trail of personally identifiable information created through the use of surveillance?
- Oversight mechanism: does the law provide an adequately funded and fully independent oversight body to ensure public space surveillance is operated in accordance with law and consistent with national commitments to human rights?
- Risk assessment: was a human rights risk assessment completed for any of the safe city projects?

Accordingly, in the following paragraphs, we consider the varied laws and regulations that tangentially touch on aspects of human living that would typically fall within the operationalisation of safe cities.

5.1 Laws on public surveillance

The prohibition of private surveillance is well entrenched in Nigerian law. The primary point of reference is sections 37 and 39 of the 1999 Constitution of the Federal Republic of Nigeria (the Nigerian constitution),

which provides that the privacy of citizens, their homes, correspondence, telephone conversations and telegraphic communications, is protected. Furthermore, under Cybercrimes (Prohibitions, Prevention, etc) Act 2015 (the Cybercrimes Act), section 12 provides that it is an offence for any person to intercept, by technical means, **non-public** transmissions of computer data, content, or traffic data.

However, these data surveillance rules are not absolute and contain loopholes that can be exploited to (potentially) infringe upon the rights of citizens. For instance, while the Nigerian constitution expressly provides for a citizen's right to privacy, this right can be derogated from, pursuant to section 45, in the limited instances of the interest of defence, public safety, public order, public morality, or public health, or for the purpose of protecting the rights and freedom of other persons. Furthermore, beyond the provisions of the Nigerian constitution, broad powers of 'lawful' surveillance are likewise contained in a number of laws and regulations. For instance, interception of communications is a form of surveillance that has been increasingly adopted by governments, including the Nigerian government, typically under the guise of crime prevention.

Section 68 of the Terrorism (Prevention and Prohibition) Act 2022 empowers a relevant agency to apply to the court for an *ex-parte* interception of communication order for the purpose of prevention, detection and prosecution of offences under the Act, albeit subject to the approval of the National Security Adviser. To this application, the court may make an order (1) requiring a communication service provider to intercept and retain a specified communication, including the call records or meta data of a data subject; and (2) authorising a relevant agency to enter any premises and to install a device for the interception and retention of communications and to remove the same for the purpose of intelligence gathering. The order will specify the duration for retention. The thought is that perhaps the rationale for the requirement of a court order is to have the court determine whether surveillance is necessary and proportional in the instance in which the interception is sought.

In the same vein, Nigerian law permits the Nigerian Communications Commission (NCC), in sections 147 and 148 of the NCC Act 2002, to allow authorised interception of communications on the system of third parties which it had licensed during a public emergency or in the interest of public safety. Similarly, the Cybercrimes Act provides room for a judge to order a service provider or a law enforcement agent to collect, record, permit, or assist competent authorities with the collection or recording of content data and/or traffic data associated with specified communications transmitted by means of a computer system.

The exception granted on the grounds of public safety or combating of crimes is also contained in the Code of Practice for Interactive Computer Service Platforms/Internet Intermediaries and Conditions for Operating in Nigeria (the NITDA Code), which requires internet intermediaries and interactive computer service platforms to provide information, including personal data, under their domain or any assistance to any authorised government agencies regarding content put on a platform, to carry out an investigation, combat cybercrimes, or prosecute an offence.

Notably, no law permits public surveillance in furtherance of the activities/operation of safe cities. However, it may well be possible that, given the broad language in some of these exculpatory laws, actors may be able to find exclusionary provisions that permit them to carry out surveillance or surveillance-like activities while operating safe cities.

5.2 Necessity and proportionality

This aims to address two questions: (1) whether privacy violations arise from the surveillance caused by the safe city projects; and (2) to the extent that violations arise, whether these violations are necessary and proportionate to achieve any legitimate aim?

To the first question, the answer is yes. Presently, there is no law that categorically excuses public surveillance for any purpose as an exemption to the privacy rights of citizens under the Nigerian constitution and international treaties signed and ratified by the nation. Following the analysis in section 5.1 of this segment above, it is deducible that there are some laws in Nigeria which tangentially touch on the issue of surveillance but none which directly point to the legality of public space surveillance. Accordingly, by the fact of the surveillance itself, there is violation of the rights to privacy of citizens.

Notably the Nigerian constitution in section 45 provides an exception to this right to privacy, which can be conceived as a metric for determining the extent to which the violations arising from the public surveillance may be necessary or proportionate. Bearing in mind that no law in Nigeria defines 'legitimate aim' within the context of public surveillance, there is presently no threshold against which this report can determine the extent to which the violations are necessary and proportionate to achieve a legitimate aim.

However, since the claim by the Nigerian government is that the aim of the safe city projects is the prevention of crime, it can be concluded that this is perhaps what the 'legitimate aim' would be. Against this claim, have the safe city projects and its attending surveillance measures led to crime prevention or a reduction in crime rate? No. Instances of increasing crime rate include the multiple bomb blasts in Abuja such as those affecting the

United Nations Building and the operational headquarters of the Nigeria Police Force (NPF), and the Christmas Day bombing of a Catholic Church, among others between 2011 and 2014 without the perpetrators being captured by any surveillance system in the city, as well as the kidnapping of 55 students of the University of Nigeria along the Lagos-Ibadan Expressway while on their way back to school (Effoduh and Odeh 2023). Recently, in March 2025 alone, nearly 400 people were reportedly kidnapped in Nigeria including 287 school children (Human Rights Watch 2025). None of these were captured by any CCTV in the country or at least in the regions where the safe city projects have been purportedly commenced.

5.3 Access control

Are there measures and protections in place to limit access to the digital trail of personally identifiable information created through the use of surveillance? This requires a consideration of the legal framework for data protection in Nigeria considering that there is no safe city-specific law in Nigeria. Further, a consideration of the data protection framework is important to this report because it contributes to the understanding of how surveillance is treated in the country and the existence or otherwise of any measures for the preservation of the data rights of Nigerian citizens in the wake of the public surveillance.

The enactment of the Nigeria Data Protection Act (NDPA) in 2023 heralded a new era for data protection in Nigeria. As a starting point we note that the NDPA does not make any reference to video surveillance, CCTV footage, or public space surveillance. However, because from the read of the Act, its provisions are arguably not sector-specific and could be interpreted to apply to all instances where concerns about the data rights of citizens arise, its provisions become relevant to this report.

The NDPA primarily provides that the processing of personal data of a data subject by data controllers or data processors must be strictly for lawful purposes. In addition to the requirement for lawful processing of personal data, the NDPA likewise contains several guiding principles that inform the nature, type, and procedure for processing personal data.

Apart from the NDPA, data protection rules are found in various other laws in Nigeria. For example, the NCC Regulations require all licensees to take reasonable steps to protect customer information against improper or accidental disclosure and ensure that such information is securely stored and not kept longer than necessary. The Freedom of Information (FOI) Act 2013, in section 14, provides that a public institution is obliged to deny an application for information that contains personal information unless the individual

involved consents to the disclosure, or where such information is publicly available.

While there is currently no case law by the Nigerian courts which has established that video surveillance is captured under the NDPA, it is argued that being that the NDPA and other data protection laws are sector-agnostic, it means that the legal requirements espoused therein (in relation to the processing of personal data) should apply even in the context of safe cities. Hence, the processing of personal data in the context and for the purpose of carrying out safe city-related activities must be in accordance with the provisions of the NDPA and other rules.

5.4 Oversight mechanism

Does the law provide an adequately funded and fully independent oversight body to ensure public space surveillance is operated in accordance with law and consistent with national commitments to human rights?

First, further to the understanding that there is no safe cities-specific law, there is also no body specifically charged with the operationalisation of safe cities projects in the country and more so, the oversight of public surveillance mechanisms. The NDPA however, establishes a National Data Protection Commission (NDPC) in section 4(1). The NDPC is charged with the powers, among other things, to receive complaints relating to the violations of the NDPA or subsidiary legislation and ensure compliance with national and international data protection obligations and best practice.

Presently, this is the only body that is charged with obligations similar to what an oversight body would do in relation to the regulation of public surveillance. This lack of a specifically charged oversight body would be addressed if a safe city law is enacted.

5.5 Risk assessment

Was a human rights risk assessment completed for any of the safe city projects?

At the time of this report, there is no record of any form of human rights risk assessment conducted prior to the launch of any of the discussed safe city projects in Nigeria.

6. Digital rights

Perhaps, the greatest challenge posed by the growing proliferation of sensors, data collection mechanisms, and public surveillance mechanisms is the possible erosion of human rights. For example, some of the data gathered in a safe city context consists of sensitive and personal information such as current location information, habits, and personal preferences. This allows the owner of the data to create rich and in part personally identifiable knowledge about consumers, patients, clients, customers, and products (Leif and Hofmann 2020).

The implementation of safe city projects and public space surveillance in Nigeria has significant implications for the practice of digital rights and justice. Digital rights, understood as the extension of human rights into digital contexts, encompass entitlements such as privacy, freedom of expression, association, and protection from discrimination (Oloyede *et al.* 2021). While these initiatives are framed by government agencies as essential to combating insecurity and improving urban management, their design and deployment raise questions about proportionality, accountability, and equity in the exercise of rights, most of these concerns having been supported by evidence in this report.

To the extent that digital rights are essentially the human rights of persons exercisable when they are in digital spaces, and Nigeria being a United Nations member state, the implication of this is that the digital rights of Nigerians are the rights recognised under the Nigerian constitution and other human rights frameworks operational in the country. To this end, the general human rights challenges faced in the implementation of rights in Nigeria also arise in the implementation of digital rights in the wake of safe cities and public surveillance in the country. (Oloyede *et al.* 2021)

Notably, there is a Digital Rights and Freedom Bill currently before the Nigerian National Assembly. The objective of the Bill is to, among others, promote freedoms of expression, assembly, and association online, and guarantee the fundamental privacy rights of citizens in the use, development, and deployment of security-related technologies.

There is no denying that the passage of the Bill will be very important to address the challenges posed by the Internet of Things and advancement in technology generally, as well as the changes to the landscape that will be heralded by safe cities. It is, however, noted that the Bill, as is, does not address safe cities' specific questions and, as such, cannot replace the need for tailored legislation for that purpose.

7. Conclusion

This report sets out the factual and regulatory landscape of safe city surveillance in Nigeria, spanning efforts at the national level as well as state-level projects. Considerable investments have been made, and partnerships have been entered into to advance safe city objectives at the federal and state levels. Notably, some positive outcomes have been recorded at state levels; however, the pace and results of implementation of most of these projects have remained marred by inefficiencies, opacity, and allegations of misappropriation.

In this report, available evidence suggests the conclusion that not much progress has been made in pursuit of the safe city projects at different levels of government in Nigeria considering that none of the projects have been successfully completed and launched and no positives have been recorded in furtherance of the purported goal of crime prevention. Furthermore, the report also lends credence to the conclusion that there is no safe city-specific legislation in Nigeria, and as such its legality is inferred through fragmented laws. The necessity and proportionality of the projects and attending public space surveillance is questionable. There is no oversight body regulating the implementation of the projects and public space surveillance in the country nor are they comprehensive measures for safeguarding the rights – including digital rights – of persons in the wake of the projects.

Accordingly, against the foregoing, the following recommendations are proposed:

- Enactment of a dedicated Safe Cities and Public Surveillance Law specifically tailored towards the nuances and necessary safeguards for the operation of safe city projects.
- Establishment of a multistakeholder regulatory body charged with the responsibility to monitor development, deployment, and compliance with relevant laws in the operationalisation of safe cities.
- Requirement of independent impact assessments before and after the deployment of safe city technologies. These assessments would be submitted to the regulatory body.
- For financial viability and better outcomes, the government at the federal and state levels should adopt sustainable public-private

partnership models with clear contractual safeguards and accountability benchmarks.

Table 16: Summary responses to guiding research queries

S/N	Query	Response
1	Date of first CCTV cameras deployed	2010 – under the Nigeria National Public Security Communication System project (NNPSCS project).
2	Date safe city project announced	NNPSCS project – 2010 Lagos safe city initiative – 2016 Oyo safe city project – 2018
3	Date of first smart CCTV cameras deployed	2019 – Lagos state deployed 10,000 high-definition CCTV cameras with facial recognition capabilities.
4	Total number of smart cameras by 2025	This data is not readily accessible, publicly.
5	Date the Data Protection Law was passed	2023 – The Nigeria Data Protection Act (NDPA) was enacted.
6	Was a human rights risk assessment completed on the safe city projects?	No evidence or record of a human rights risk assessment being conducted for any project.
7	Is there a validly enacted law on safe cities covering public space surveillance?	No – there is no dedicated safe city law in Nigeria. Existing legal frameworks only provide partial and tangential coverage.
8	Is the legitimate aim of public space surveillance defined/limited in law?	No – there is no legal definition or statutory limitation specifically tied to safe city surveillance initiatives.
9	Does the law provide citizens with redress mechanisms?	Partially – general laws like the NDPA, FOI Act, and constitution provide redress, but no safe city-specific framework or mechanism exists.
10	Is safe public space surveillance legal, necessary, and proportionate?	Partially – legality is inferred through fragmented laws; necessity and proportionality are not explicitly addressed or evaluated through formal legal thresholds. However, evidence suggests a lack of proportionality.
11	Are controls in place to restrict access to data?	Yes – NDPA and NCC Regulations provide some control measures, but enforcement and adequacy are unclear in the safe city context.
12	Does an independent oversight body exist?	Partially – NDPC oversees data protection under the NDPA; however, there is no specific oversight body for safe city surveillance or public space surveillance broadly.

Source: Author's own

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Smart city surveillance in Rwanda

Joseph Antwi-Boasiako

1. Introduction

The report focuses on surveillance of public space in Rwanda, with the research question: How are smart city technologies used to surveil public spaces in Rwanda and for what reasons are these technologies used? Rwanda's rapid deployment of comprehensive surveillance infrastructure across public spaces represents a critical case study for understanding the intersection of digital transformation, governance, and civil liberties in contemporary Africa. As one of the continent's most technologically ambitious nations, Rwanda has emerged as a test case for smart city initiatives that integrate advanced surveillance capabilities with broader development objectives. There are, however, reports of how these technologies have been used to monitor dissents in Rwanda. This report addresses a significant gap in scholarly and policy analysis regarding the implementation of public space surveillance in African contexts. While global attention has focused on surveillance developments in China and Western democracies, Rwanda's surveillance expansion offers unique insights into how post-conflict states leverage digital technologies to achieve multiple governance, security, and economic objectives simultaneously. The need for this analysis is particularly urgent given Rwanda's influential position within African digital transformation discourse and its role as a model for other developing nations. Understanding the institutional frameworks, international partnerships, funding mechanisms, and underlying political economy of Rwanda's surveillance infrastructure provides essential insights into broader patterns of technology-enabled governance across Africa.

The report is structured in seven sections: this introduction provides a general justification for the report; section 2 presents the background, which situates surveillance practices in Rwanda in a historical context; section 3 documents key dates and features of current smart city surveillance of public space in Rwanda; section 4 identifies the key actors involved in the surveillance of public space; section 5 and section 6 present the regulatory landscape and digital rights; and section 7 concludes the report.

2. Background

Surveillance in the context of Rwanda is not a recent occurrence, as it long precedes its present-day digital forms. Over the years, there have been surveillance practices in Rwanda under precolonial, colonial, postcolonial, and recent government administrations. Understanding this historical context illustrates continuities with the social monitoring systems of the past.

2.1 Precolonial context

Social monitoring and surveillance in Rwanda have deep historical roots in precolonial administrative systems (Purdeková and Mwambari 2021). The Nyiginya kingdom, which formed the foundation of modern Rwanda, developed sophisticated mechanisms for overseeing local life through what Purdeková (2016) describes as 'mundane sights of power'; the everyday forms of state presence and monitoring enabled systematic social surveillance. The kingdom's centralised administrative structure, operating from approximately 1650 to colonial intervention, relied on hierarchical networks of chiefs and local officials to exercise authority through labour management, tax collection, and maintenance of social cohesion (Vansina 2005). These precolonial surveillance practices established enduring patterns of state monitoring that would later be transformed and intensified under colonial administration, demonstrating remarkable continuities in 'interactive surveillance' across centuries of political transformation (Purdeková 2016). According to Purdeková, 'interactive surveillance is one that fundamentally depends not only on "presencing" of agents of state (or state-aligned or state associated actors) but "presenting" of tokens by those being surveilled, either in forms of cards and permits, information for censuses, taxation, displays of loyalty or a friendly roster, or attendance at meetings' (2016: 4).

2.2 Colonial period (1897–1962)

Colonial surveillance in Rwanda evolved through two distinct phases of European administration, beginning with German rule from 1897 to 1916, followed by Belgian administration from 1916 to 1962 (Des Forges 2011). From 1894 to 1918, Rwanda, along with Burundi, was part of German East Africa, and after Belgium became the administering authority under the mandates system of the League of Nations, Rwanda and Burundi formed a separate administrative entity. Like many other colonies, the German colonial government implemented a policy of indirect rule that deliberately strengthened existing hierarchical structures, working through the Rwandan monarchy and traditional chiefs to exercise administrative

control (Newbury 1988). The German colonial government pursued a policy of indirect rule that strengthened the hegemony of the Tutsi ruling class and the absolutism of the Tutsi monarchy, led by the mwami, and that approach continued under Belgian administration in Rwanda. This system of indirect governance included systematic monitoring of population movements and political activities through established traditional networks (Longman 2017). Most significantly, first German and then Belgian colonial powers reified and institutionalised racialised identification, creating documentation and classification systems that would later facilitate ethnic targeting during the 1994 genocide (Purdeková and Mwambari 2021).

2.3 Post-colonial period

Following independence in 1962, Rwanda's surveillance mechanisms underwent a fundamental transformation towards systematic ethnic monitoring and exclusion of Tutsi populations. The transition to Hutu-dominated governance emerged from the violent 'Hutu Revolution' between 1959 and 1961, which established new power structures that institutionalised anti-Tutsi discrimination (Purdeková and Mwambari 2021). Under President Grégoire Kayibanda's administration (1962–73), the new republic implemented a systematic form of social surveillance that institutionalised ethnic discrimination, which excluded Tutsis from prominent careers, higher education, and government positions (Des Forges 2011). The social surveillance functioned through bureaucratic mechanisms that identified, monitored, tracked, and controlled Tutsi participation in the society. The surveillance apparatus was further consolidated under Juvénal Habyarimana's regime (1973–94), whose 21-year rule, along with the informal network based around his family known as the Akazu, established penetrating control mechanisms throughout Rwandan society (Wallis 2020). These systems relied extensively on local administrators, militia groups, and community informants to monitor suspected 'enemies' and enforce discriminatory policies against Tutsi populations and political dissidents (Longman 2017). The surveillance infrastructure developed during this period proved instrumental in facilitating the systematic identification and targeting of Tutsis during the 1994 genocide, demonstrating how state monitoring systems can be weaponised to enable mass atrocities (Straus 2006). The surveillance infrastructure helped in identifying, tracking, and targeting Tutsis who suffered atrocities during the genocide.

2.4 Contemporary Kagame administration (1994–present)

On 6 April 1994, after President Habyarimana's plane was shot down, Hutu extremists went on a rampage targeting the Tutsi minority leading to the death of around 800,000 people within 100 days (Voice of Africa 2024). The bloodletting only stopped when President Kagame's Rwandan Patriotic Front militia captured Kigali, the capital of Rwanda, which led to him becoming the shattered country's *de facto* leader (Voice of Africa 2024). The current administration under Paul Kagame has developed sophisticated surveillance capabilities that extend both domestically and internationally. The government has faced a barrage of allegations of kidnappings, assassinations, torture, and extreme levels of surveillance of the opposition (Council on Foreign Relations 2024). The Voice of Africa (2024) reports that the administration under Kagame tracked and pursued anyone seen as the enemy of the state prominently including the 'Hotel Rwanda' hero Paul Rusesabagina who called Rwandans 'prisoners in their own country'. Contemporary surveillance in Rwanda encompasses multiple dimensions: digital monitoring, physical surveillance of dissidents, and transnational repression of critics even abroad. Digital surveillance has become particularly prominent, with civil society and human rights organisations specifically challenging the patterns of technological social governance: mass surveillance and use of spyware against political opponents, online censorship, and the blocking of social media content (Media Defence 2023).

The government has invested heavily in digital infrastructure that enables comprehensive monitoring of online activities, social media engagement, and digital communications. The surveillance system is also allegedly operating internationally, targeting Rwandan dissidents and critics living abroad. Digital transnational repression refers to the use and exploitation of digital technologies to target people abroad, including surveillance and online harassment (Human Rights Watch 2024). This has included accusations of using sophisticated spyware, harassment campaigns, and even physical intimidation of critics in foreign countries. There are many examples of how the state has allegedly deployed technology to surveil and monitor dissidents and opposition members, as evidenced in several court cases where the prosecution presented private communications sourced from mobile chat apps (Gitinywa 2020). In a prominent case of the State versus Kizito Mihigo, according to The New Times report, the prosecution detailed that 'in WhatsApp and Skype (social media) conversations between Mihigo and Callixte Nsabimana, an RNC (Rwanda National Congress) member, the two discussed plans of overthrowing the government by use of

violence’ (Musoni 2014). One wonders how the prosecution knew about these plans, especially from personal applications like WhatsApp and Skype.

Table 17: Key surveillance methods by period

Period	Primary methods	Key targets	Geographic scope
Pre-colonial	Administrative oversight, traditional networks	General population	Local/regional
Colonial	Indirect rule, identity documentation	Political activities, movements	National
Post-colonial	Ethnic monitoring, community informants	Tutsi populations, dissidents	National
Contemporary	Digital surveillance, spyware, transnational repression	Political opponents, critics	National/international

Source: Author’s own.

3. Case study description

Rwanda has emerged as a continental frontrunner in digital transformation and smart city development, guided by a national vision to become Africa's information and communications technology (ICT) hub (UN-Habitat 2017). Central to this agenda is the integration of public space surveillance as a key pillar of urban modernisation, officially aimed at enhancing safety, service delivery, and governance efficiency (Ministry of Youth and ICT 2020). Rwanda's digitalisation approach is considered one of the most comprehensive in sub-Saharan Africa, reflecting a broader strategy of embedding digital infrastructure into governance frameworks to improve urban resilience and competitiveness (The Atlas of Urban Tech 2017).

Buchanan (2016) reports that the potential value of public surveillance in Rwanda took on a new meaning in 2011 during Diana Tukundane's case of stealing US\$400 belonging to another passenger at the Kigali International airport. According to reports, the woman denied stealing the money until the Rwanda police used 'plain' CCTV to pin her down with evidence (The New Times 2011). It was only her identification through these devices that completely rested the case (Buchanan 2016). Following this viral incident, the Rwanda National Police has also since relied extensively on public video surveillance as a primary tool to prevent crime, fight theft, monitor car and motor traffic, prevent terrorism, and also monitor people's movement in some public areas (Buchanan 2016).

In 2017, Rwanda unveiled its smart cities blueprint during the Transform Africa Summit, a framework that aimed to serve as a catalyst to fast-track ICT-driven initiatives in cities (MINICT 2017). The blueprint provides an overview of smart and sustainable cities programmes meant to integrate available technologies at every level of urban management. It emphasises the role that technology plays in enhancing the delivery of services both in the public and private sectors.

In May 2018, The New Times reported that, according to the budget estimates proposed by the Rwanda Information Society Authority (RISA), which was discussed in the Parliamentary Standing Committee on National Budget and Patrimony, the government had planned to spend RWF5 billion (about US\$5.3 million) to purchase a smart CCTV security system to be installed in different public spaces in the city of Kigali to enhance security (Kwibuka 2018). According to the report, the chief innovation officer at RISA has said in an interview that the CCTV cameras were to be installed in different public spaces in Kigali with a focus on areas where

many people meet and that each site in the city was to be outfitted with three to four cameras, making a total of 349 cameras throughout the city.

Following the plan from May 2018, Rwanda has implemented automated speed enforcement systems as part of its comprehensive road safety strategy, with the Rwanda National Police introducing speed enforcement cameras utilising AI technology to monitor traffic violations (Rwanda National Police 2024). The first speed enforcement camera was erected in July 2019 in Kanzenze, Bugesera district, followed by installations in Nyagasambu, Kamonyi, Ryabega, and Kanyinya during the pilot phase (Rwanda National Police 2024).

The most significant expansion of Rwanda's public surveillance infrastructure began in May 2020, marking a coordinated deployment across Kigali and major highway networks. Muvunyi and Gitinywa (2021) document this critical implementation phase, noting that starting from May 2020 'the government of Rwanda has embarked on putting up the Closed-Circuit Television (CCTV) cameras all around the main roads of the city of Kigali'. At the time of the significant nationwide expansion of the surveillance and safe city project in May 2020, the police established a centralised command system, with CCTV data collected through a dedicated private network, which cannot be accessed over the internet. The surveillance network that expanded beyond the initial pilot utilised both fixed and mobile camera systems equipped with light detection and ranging (LiDAR) measuring technology capable of monitoring traffic from both directions under various weather and traffic conditions (Rwanda National Police 2024). The Rwanda Development Board (2020) reports that the technology employed for the nationwide expansion in 2020 was from an agreement signed between the Rwanda National Police and Eastern Ventures, a company from the United Arab Emirates, to procure and install 500 cameras. De La Croix Tabaro (2023) reports that the supplies included fixed, mobile, and movable cameras, with speed cameras constituting the lion's share with 400 units. Despite the unavailability of current figures on cameras installed in Rwanda, Delavary *et al.* (2024) revealed that between 30 July 2019 and 30 December 2022, 336 automated speed enforcement cameras were installed throughout Rwanda. In the City of Kigali alone, 93 automated speed enforcement cameras had been mounted during this same period.

The fixed speed enforcement camera is popularly known under the name 'Sophia', named after a popular and intelligent Robot-Sophia, which attended Transform Africa in Kigali (De La Croix Tabaro 2023). Apart from the Sophia, there are also mobile cameras to monitor drivers. The inspector general of police, Dan Munyuza has emphasised that the cameras are capable of monitoring speed violations, recognising number plates and

transmitting data directly to a centralised command centre at the police headquarters (Muvunyi and Gitinywa 2021). Once the camera captures the number plates of an speeding car, a short message is directly sent to the registered mobile phone, including the amount of fine and when it is due to be paid via the government payment platform called the 'Irembo' (De La Croix Tabaro 2023). If the car owner refuses to pay by the due date, the fine incurs an extra RWF10,000 per each new due date.

Rwanda's surveillance infrastructure extends significantly beyond traffic-related monitoring to encompass comprehensive public space coverage throughout major urban areas. Over the past three years, the state has broadened its local mass surveillance capacities with the recent implementation of smart CCTV networks throughout neighbourhoods in Kigali, the capital (Gitinywa 2020). The government's CCTV deployment focuses on strategic public locations where large gatherings occur. According to RISA chief innovation officer Antoine Sebera, the CCTV cameras will be installed in different public spaces in Kigali with a focus on areas where many people meet (Kwibuka 2018). This includes commercial districts, public squares, and transportation hubs beyond traditional traffic intersections. For instance, in places like Kigali, car-free zones are equipped with surveillance cameras for security purposes, even though specific details about the number and type of cameras are not publicly available.

The Rwandan government has long emphasised the strategic deployment of surveillance infrastructure across major urban centres as part of its broader smart city and public safety agenda since the establishment of the smart city blueprint in May 2017 (Ministry of Youth and ICT 2020). The inspector general of police has publicly affirmed that the nationwide deployment of surveillance cameras is part of a broader presidential directive aimed at enhancing public safety (Muvunyi and Gitinywa 2021). Research conducted between 2022 and 2023 demonstrated a significant effectiveness of these automated speed enforcement systems, with case-control studies showing a 69 per cent decrease in speed violations at deployment sites compared to control locations without enforcement (US Department of Transportation 2024).

Figure 7: Rwanda’s surveillance timeline

Year	Key milestaone
2011	The breakthrough CCTV case at Kigali Airport that demonstrated surveillance effectiveness
2017	Establishment of Rwanda's smart city vision and JCT hub strategy
2019	Installation of the first speed camera 'Sophia' in Kanzenze
2020	Major CCTV deployment across Kigali with 500 camera agreement
2021	Red Light Camt-ra system implementation
2025	Introduction of drone-based traffic monitoring

Source: Author’s own

4. Key actors

The implementation of a safe city and surveillance infrastructure involves multiple government agencies working in institutional coordination. While the Ministry of Information, Communication Technology and Innovation provides the overall policy direction for all ICT related programmes, Muvunyi and Gitinywa (2021) identify RISA and the Rwanda National Police as key institutions in charge of implementing this policy. These two agencies serve as the primary implementing bodies for the deployment and operation of surveillance infrastructure. Since 18 November 2019, the Rwanda National Police has moved from being under the Ministry of Justice and Attorney General to the Ministry of Internal Security, making the Ministry of Internal Security a key policy stakeholder in the process (MINIJUST 2025). The Presidency of Rwanda is also a key stakeholder, especially as presidential orders are critical to policymaking in Rwanda. Below is an overview of how these stakeholders contribute to safe city management.

The Ministry of Information, Communication Technology, and Innovation (MINICT) is the key policy institution responsible for overseeing the implementation and development of information and communication technology policies, strategies, and programmes (MINICT 2025). This positions the ministry as the central policy architect for all smart city initiatives, include the public space surveillance systems. The ministry operates through two key departments: Digital Government Transformation, and Innovation and Business Development, with the vision of fostering the development and diffusion of information and communication technology throughout the economy and society in Rwanda. The ministry, through its Department of Digital Government Transformation, has been leading the implementation of the Smart Rwanda Master Plan, which was approved in November 2015 (Shanbhough 2017). Apart from that, the MINICT serves as the overseer of the two key institutions, the Rwanda Information Society Authority and the Rwanda National Police, in charge of implementing the smart city public surveillance systems. As such, these make the ministry serve as the overseer of implementation for Rwanda's smart and safe city transformation.

The Rwanda Information Society Authority (RISA) serves as the central implementation agency for Rwanda's smart and safe city transformation, positioned at the forefront of all ICT projects' implementation, research, infrastructure development, and innovation within the ICT sector (Shanbhough 2017). Established as a public institution in 2017 under the Ministry of ICT and Innovation and governed by presidential order 077/01 of 09/12/2022, RISA has the mission of digitising Rwandan society through increased usage of

information and communication technologies and innovation technology as a cross-cutting enabler for development (RISA 2025). RISA derives from the former IT department of the Rwanda Development Board and it is currently affiliated to the MINICT (Cyber Security Intelligence 2025). RISA serves as the key implementing body for the Smart Rwanda Master Plan and has the mandate of planning and coordinating the implementation of the national ICT for Development Agenda. Regarding surveillance systems, RISA works in partnership with the Rwanda National Police to implement surveillance policies, claiming that the use of CCTV cameras will significantly enhance security and make the city and country as a whole safer by establishing a robust mechanism and deterring, preventing, and detecting crime (Byishimo 2021; Muvunyi and Gitinywa 2021; Rwanda National Police 2025).

The Ministry of Internal Security (MININTER) plays a crucial oversight role in Rwanda's smart and safe city initiatives through its administrative authority over the Rwanda National Police. MININTER provides high-level oversight for surveillance operations. On 18 November 2019, the minister of justice and attorney general (MINIJUST) handed over Rwanda National Police responsibilities to General Patrick Nyamvumba, the minister of internal security, positioning MININTER as the primary ministry responsible for coordinating law enforcement activities, including surveillance operations (MINIJUST 2025). This administrative restructuring placed the implementation of smart city surveillance systems under the Ministry of Internal Security's purview, as the Rwanda National Police serves as the operational arm for CCTV monitoring and public safety enforcement. During the Covid-19 pandemic, the Rwanda National Police under MININTER's oversight utilised advanced technologies including drones to spread awareness messages through remote and densely populated neighbourhoods and implemented bracelet-enabled trackers for surveillance teams to enforce travel restrictions demonstrating the ministry's role in coordinating technology-enabled security measures that extend beyond traditional policing into comprehensive urban surveillance and public safety management (World Health Organization Rwanda 2020). The ministry, during the Covid-19 pandemic, implemented a policy that attached a watch-like tracker device to the wrist of travellers to monitor their movement (Tasamba 2021). Rwanda's comprehensive urban surveillance represents a systematic approach to monitoring public spaces across multiple cities, which the state has been focusing on over the past three years.

Rwanda National Police operates as the primary implementation partner, working directly with RISA in safe city and surveillance deployment. The Rwanda National Police manages day-to-day surveillance operations and data processing. According to Muvunyi and Gitinywa (2021), the inspector general of police, Dan Munyuza, emphasised that the police are using LiDAR

technology that could run number plate recognition of vehicles, and that they can see them from the commanding post at the police headquarters.

The head of the Information Technology Department, assistant commissioner of police, Oscar Sakindi has stated that the Rwanda National Police 'can't have an officer on every street, but we do have cameras and drones monitoring key zones', especially in Kigali and other hard-to-reach areas to help detect crime in real time (IGIHE 2025; Rwanda Dispatch 2025; Rwanda National Police 2025). Rwanda National Police (2025) claim that as well as crime detection and prevention, they have made strides in digitising other operations, such as road safety management, motor vehicle inspections, driving and licensing services, among others.

4.1 Presidential authority and strategic direction

In the Rwanda case, another important institution in smart city conversation is the Presidency. The smart and safe city initiative operates under direct presidential mandate, with President Paul Kagame providing high-level political backing that positions smart city surveillance as a strategic priority within Rwanda's development agenda (Muvunyi and Gitinywa 2021). This demonstrates the presidential administration's emphasis on centralised security control to ensure the safety of the citizens. President Paul Kagame has personally championed smart city initiatives, convening events such as the Transform Africa Summit while positioning surveillance technology within broader economic development objectives. As the chair of the Smart Africa Alliance, President Paul Kagame is personally driving the smart/safe city agenda in Rwanda and even Africa as a whole. 'The Inspector General of Police Dan Munyuza indicates that the enforcement of cameras networks across the country, which is anchored into the special presidential directives, is in line with ensuring the general public security' (Muvunyi and Gitinywa 2021).

4.2 Technology suppliers

The African surveillance technology market represents substantial investment, with African governments spending over US\$1 billion per year on digital surveillance technologies, which are being used without adequate legal protections in ways that regularly violate citizens' fundamental human rights (Roberts *et al.* 2023). Research indicates that governments buy packages of safe city surveillance technologies from companies, including China-based Huawei and ZTE, often including the installation of thousands of CCTV cameras and command and control room data centres to monitor citizens (Roberts *et al.* 2023). Analysts warn that their tools provide

authoritarian governments with the means to spy on citizens and promote repression (The Africa Report 2024).

Rwanda's surveillance expansion is dominated by Chinese technology firms, reflecting broader patterns across Africa. Chinese technology companies, such as Huawei, are contributing to Rwanda's smart city development. Close to two years after the Smart Rwanda Master Plan was approved in cabinet, Huawei signed a memorandum of understanding with the government of Rwanda to open all-round cooperation in the ICT industry (Huawei 2017). In the memorandum of understanding, Huawei was supposed to support Rwanda in building national broadband, regional data centres, and other smart infrastructures to promote Rwanda to become Africa's ICT Hub. The Chinese model of smart city development appears to have influenced Rwanda's approach, with firms like Huawei employed by the state to support the establishment and application of smart city systems and auxiliary technologies like cloud computing and AI surveillance. This suggests that commercial interests from China are directly shaping the technological architecture of Rwanda's surveillance systems. One key international technology supplier in Rwanda has also been Korea Telecom. Korea Telecom Rwanda Networks (KTRN), a joint venture between the government of Rwanda and Korea Telecom, has been foundational in providing the 4G long-term evolution (LTE) infrastructure that supports smart city applications, which served as the basis for most of the deployed surveillance systems (UN-Habitat 2017). KTRN's infrastructure underpins many digital services deployed in Kigali and other cities. The government of Rwanda in 2013 signed a 25-year deal with Korean Telecom to launch 4G wholesale in the country with a cash injection of about US\$140 million (Sehloho 2023). The partnership also meant that the Rwandan government would own 49 per cent of KTRN and Korean Telecom the other 51 per cent (Sehloho 2023).

In 2017, Inmarsat partnered with the Rwandan government to support Kigali's smart city transformation initiative (Inmarsat 2017). The partnership involved deploying a low-power wide-area network (LoRaWAN) infrastructure across the city of Kigali, developed in collaboration with Actility, the industry leader in LoRaWANs for industrial Internet of Things (Inmarsat 2017). This solution provides city-wide coverage that enables a variety of organisations to develop and deploy IoT applications on a large scale (Barton 2017).

Specifically, with regard to the surveillance of public spaces, the government of Rwanda, in February 2020, signed an agreement with Eastern Ventures, a local special-purpose vehicle, in partnership with Vitronic, a firm based in the UAE (Rwanda Development Board 2020; Gahigi 2020). The deal agreement was signed on behalf of the government of Rwanda by the Rwanda National Police. Gahigi (2020) asserts that even though the

agreement is termed as a multimillion-dollar deal, how much the Rwandan government is paying towards this agreement is unknown. The deal agreement was to see the installation and operation of over 500 German high-tech mobile, fixed cameras, and trailer light bars used to monitor speed and traffic regulation (Rwanda Development Board 2020; Gahigi 2020).

4.3 Local commercial ecosystem

The surveillance expansion has also stimulated local commercial activity, with various companies offering CCTV installation and maintenance services. However, these appear to be primarily implementation partners rather than primary technology manufacturers, suggesting that the core technological infrastructure remains dependent on international suppliers.

Table 18: Key actors in Rwanda’s public surveillance system

Key actors	Roles/responsibility
Ministry of Information, Communication Technology and Innovation (MINICT)	Is the key policy institution responsible for overseeing the implementation and development of information and communication technology policies, strategies, and programmes.
Rwanda Information Society Authority (RISA)	Is the central implementation agency for Rwanda’s smart and safe city transformation, positioned at the forefront of all ICT projects’ implementation, research, infrastructure development, and innovation within the ICT sector.
Rwanda National Police	Manages day-to-day surveillance operations and data processing and works directly with RISA in safe city and surveillance deployment.
Ministry of Internal Security (MININTER)	Provides high-level oversight for surveillance operations through its administrative authority over the Rwanda National Police.
Presidency of Rwanda	Provides direct presidential mandate for smart and safe city initiative; offers high-level political backing positioning smart city surveillance as a strategic priority.

Source: Author’s own.

5. Regulatory landscape

Test 1. Are there validly enacted laws governing public space surveillance?

Rwanda's legal framework for managing smart city surveillance and data protection has evolved significantly in recent years, but critical gaps remain in specifically addressing mass surveillance of public spaces. The country's regulatory environment presents a mixed picture regarding human rights protection. First, article 23 of the constitution protects the right to privacy by establishing a constitutional foundation for privacy rights. Additionally, Rwanda ratified the International Covenant on Civil and Political Rights, and is therefore bound by article 17, which states: 'No one shall be subjected to arbitrary or unlawful interference with his privacy, family, home or correspondence, nor to unlawful attacks on his honour and reputation'. This international commitment theoretically provides strong privacy protections.

Rwanda's surveillance initiatives are grounded in its broader ICT legal framework, particularly Law 60/2018 on the Prevention and Punishment of Cybercrimes, which outlines the legal basis for digital monitoring and data collection practices (Government of Rwanda 2018; Laws.Africa Legislation Commons 2018). This law complements the 2016 Law Governing Information and Communication Technologies, which grants the government extensive powers to regulate digital communications (Article 19 2018; Laws.Africa Legislation Commons 2018). Although the Rwandan constitution guarantees the right to privacy, this right has not been fully operationalised through comprehensive data protection legislation (Article 19 2018). The current ICT laws primarily emphasise user consent and opt-in mechanisms, without establishing robust safeguards for data handling and oversight (Article 19 2018).

Test 2. Do the laws define a legitimate aim for public space surveillance?

Rwanda enacted significant data protection legislation in 2021. On 15 October 2021, law 058/2021 of 13/10/2021 relating to the protection of personal data and privacy was officially gazetted. The law protects personal data and ensures the privacy of individual users with one of the tenets of this law being the clear and unambiguous consent of an individual to the collection, storage, and processing of personal data, which is a fundamental right. While the 2021 Data Protection Law provides general privacy protections, there is no specific legislation that establishes a clear legal mandate for mass surveillance of public spaces or provides explicit limits against abuse. The surveillance programme appears to

operate under general police powers and presidential directives rather than specific statutory authorisation that defines scope and limitations.

Article 19's 2018 analysis of Rwanda's 2016 Law Governing Information and Communication Technologies found that while the law purports to provide a regulatory framework, it fails to comply with international human rights standards (Article 19 2018). This suggests that even Rwanda's primary ICT legislation, which could potentially cover surveillance technologies, falls short of international standards for rights protection. The surveillance infrastructure deployment appears to rely primarily on executive directives and police operational authority rather than specific legislative authorisation, or without clear reference to specific enabling legislation.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

Rwanda has established a Data Protection and Privacy Office (DPO) as an oversight mechanism. The DPO operates to ensure understanding of rights and obligations, but its specific mandate, resources, and authority to oversee public space surveillance remain unclear. Despite these legal provisions, significant gaps remain in Rwanda's data protection regime.

Rwanda justifies its tight control over media freedom, suppression of dissent, and hostility toward opposition as matters of national unity and security, indicating that surveillance justifications are broadly framed around security and public order rather than specific, limited purposes. The absence of specific legislative limits on surveillance purposes creates a significant risk for mission creep. Without clear statutory definitions of when and why surveillance can be employed, the system can expand beyond its stated traffic safety and crime prevention objectives to encompass broader social monitoring and political control (Laibuta *et al.* 2024). While traffic cameras are officially justified for road safety purposes, the broader surveillance infrastructure lacks clear limitations that would prevent its use for other purposes such as political monitoring, protest surveillance, or general population tracking (Muvunyi and Gitinywa 2021).

Also, the lack of a legal framework provides insufficient clarity regarding who can access surveillance data and what types of data can be collected. As already indicated above, the surveillance system appears to operate under police authority, but specific provisions governing data access, sharing protocols, and interagency coordination are not publicly documented. This lack of legal clarity creates risks for unauthorised access and misuse of surveillance data. Without clear statutory definitions of data collection limits, the surveillance system potentially collects and processes personal data without adequate legal constraints. The

integration of facial recognition technology and automated number plate recognition systems suggests extensive personal data processing, but the legal framework governing such activities remains opaque.

Test 4. Does the law define mechanisms for citizen redress and accountability?

Rwanda's surveillance framework demonstrates significant deficiencies in accountability and redress mechanisms. These regulatory shortcomings have raised concerns among civil society organisations about the potential for surveillance overreach and the erosion of digital rights (Article 19 2018). The absence of specific legislation on personal data protection has given Rwandan security services unfettered discretion to collect data with impunity. Rwanda ratified the International Covenant on Civil and Political Rights, and is therefore bound by article 17, which states: 'No one shall be subjected to arbitrary or unlawful interference with his privacy, family, home or correspondence', creating international legal obligations for privacy protection. However, the domestic implementation of these obligations remains inadequate. But law 60/2013 regulating the interception of communications authorises the prosecutor to issue a 'verbal warrant' to the national security and intelligence service to watch, intercept, and decrypt any information generated, transmitted, received, or stored in any computer resource for reasons of national security according to the police and state authorities (Gitinywa 2020). The absence of specific surveillance oversight bodies means that citizens have limited avenues for challenging surveillance decisions or seeking redress for privacy violations.

6. Digital rights

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

Rwanda's smart city surveillance expansion presents a complex landscape of digital rights where technological advancement intersects with fundamental privacy and civil liberties concerns. The deployment of comprehensive surveillance systems raises critical questions about who holds digital rights, who is affected by surveillance expansion, and who ultimately benefits from these technological interventions.

Test 6. Does the law control who has access to surveillance data?

The surveillance expansion affects all citizens moving through public spaces, but certain populations face disproportionate impact. Urban dwellers, particularly in Kigali, experience the most intensive surveillance due to the concentration of CCTV systems in the capital. Young people and digital-native populations are particularly affected as they represent the primary users of digital services and urban spaces where surveillance is most concentrated. Additionally, civil society organisations, journalists, and political activists face heightened risks due to the potential for surveillance systems to monitor political activities and public gatherings (Anstis and Deibert 2025).

Anstis and Deibert (2025) report that Carine Kanimba, a Rwandan activist living in the USA and Belgium, discovered that her mobile phone had been infected with Pegasus spyware, a powerful commercial surveillance tool that provides its operator with complete access to the targeted device. The primary beneficiaries of Rwanda's surveillance expansion include state or government institutions. Government agencies gain enhanced monitoring capabilities that extend administrative reach and control over public spaces. The Rwanda National Police and security agencies benefit from improved investigative tools and real-time monitoring capabilities.

Test 7. Does the law create an independent oversight body?

The primary digital rights affected by Rwanda's smart city surveillance include the fundamental right to privacy, data protection, and freedom of movement. Rwanda's law on the protection of personal data and privacy was officially gazetted on 15 October 2021, with one of the tenets being the clear and unambiguous consent of an individual to the collection, storage, and processing of personal data, which is a fundamental right. This legal framework theoretically protects citizens' data rights, but its application to government surveillance systems remains ambiguous.

7. Conclusion

Rwanda's smart city surveillance agenda represents a complex convergence of security, political, and economic drivers, supported by significant international funding and advanced technological capabilities. While officially justified primarily on security grounds, the surveillance infrastructure serves multiple state objectives including investment attraction, digital transformation acceleration, and regime stability maintenance. The initiative involves substantial international partnerships for surveillance-enabled governance models. The rapid deployment, significant financial commitment, and high-level political backing suggest surveillance capabilities are viewed as essential infrastructure for Rwanda's development ambitions. However, the expansion raises critical concerns about privacy rights, democratic oversight, and surveillance overreach, particularly given inadequate legal frameworks for data protection. The analysis reveals that while security concerns provide public justification, the surveillance rollout ultimately serves broader political and economic objectives that extend beyond traditional law enforcement functions, reflecting Rwanda's strategy of leveraging technology for development while maintaining centralised control over digital infrastructure and social monitoring.

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Smart city surveillance in Senegal

Assane Sy

1. Introduction

The surveillance of public spaces in Senegal is undergoing an unprecedented change with the rise of digital technologies and the adoption of large-scale control tools. While these devices are presented as essential instruments to strengthen security, prevent crime, and facilitate urban management, they raise serious questions in terms of data governance, respect for privacy and the protection of fundamental rights. The safe city project, led by Huawei in partnership with the Senegalese authorities, is emblematic of this. By combining thousands of smart cameras, facial recognition systems, and centralised command centres, it aims to transform public safety through advanced technological solutions. However, the central question remains: to what extent does this project comply with the principles of legality, necessity, and proportionality, and what safeguards exist to protect citizens' digital rights?

This issue is part of a context marked by the implementation of intrusive devices such as mandatory biometric identification and the mandatory registration of SIM cards, which facilitate the centralisation of personal data. According to the Paradigm Initiative report (Paradigm Initiative 2024), these measures provide authorities with tools that can be used for widespread surveillance, although they are presented as responding to the public interest. The rise of mobile telephony, the popularisation of the internet and the generalisation of cameras in public space have introduced a new security paradigm where the state can, if necessary, require the collaboration of telecom operators to access citizens' data.

The antecedents reinforce these concerns: on the eve of the Court for the Repression of Illicit Enrichment's verdict against Karim Wade in 2015, massive telephone tapping was denounced by the press (Senenews 2015), referring to 'meticulous' surveillance aimed at preventing any mobilisation. Similar practices were reported in 2021 during events related to the Ousmane Sonko case, where several citizens claimed to have been subjected to telephone interceptions, confirmed by telecommunications experts. These cases illustrate the reality of state control that can be exercised beyond the legal frameworks.

In this context, the central question of this report is the following: does the safe city project in Senegal respect the fundamental principles governing surveillance (legality, necessity, and proportionality) and what guarantees exist for the protection of digital rights? To answer this question, the study will follow an approach combining: (1) the analysis of the project and its technical characteristics, (2) the examination of the key actors and their

roles, (3) the regulatory landscape of online surveillance, and (4) the study of the impacts on privacy, freedom of expression, and data protection.

2. Case study description

The safe city project in Senegal is part of the government's Smart Senegal strategy led by the state IT agency (ADIE, now Sénégal Numérique SA) and coordinated by Cheikh Bakhoun, former director of ADIE (Seneweb 2020). This safe city project, although not new, has been carried out since 2019 (CIO-MAG 2020), and was part of a dynamic of modernisation of public action in the field of security with a very particular touch focused on the use of digital technologies. Aware of the growing concerns in terms of security, the Senegalese government has placed this strategic aspect of the 'safe city' at the heart of its policy:

Because security is one of his priorities, the Head of State Macky Sall continues to deploy resources for the benefit of the defense and security forces. In addition to traditional means, technology has broadened the scope of what is possible in strengthening the security of our territories. In a context where populations are increasingly concerned about their safety and that of their property, the State of Senegal is implementing the Smart Senegal Program with a Safe City component. (Sénégal Numérique SA n.d.)

In practice, the safe city project was officially launched in 2019 during a visit by senior Senegalese officials to Shenzhen (China), a model smart city, where the delegation was able to draw inspiration from the advanced surveillance system model, focused on digital technologies, with components such as Huawei, intelligent command centre. This visit (14–15 May 2019) brought together the director general of ADIE, the national police, and also a representative of the Ministry of the Interior, testifying to the strong involvement of the state authorities in the safe city project.

The safe city project is deployed in some cities in the country, namely Dakar, Thiès, Mbour, Saly, Touba, Mbacké, Saint Louis, and Kaolack (Osiris 2020). According to Cheikh Bakhoun, the safe city project is a major step forward in strengthening urban security in Senegal, by providing their defense and security forces with an innovative and highly secure communication system based on e-LTE (4G dedicated) technologies. These smart cameras are interconnected to latest-generation command centres, real hubs of coordination and analysis, which ensure the centralisation of information flows in real time, and were received on 10 September 2020 by Aly Ngouille Ndiaye (former minister of the interior of Senegal under the regime of Macky Sall) (PressAfrik 2020). As a reminder, a smart camera is an advanced video surveillance device capable not only of continuous filming,

like a conventional CCTV camera, but also of automatically recognising faces or number plates thanks to AI algorithms. This makes it possible to identify specific individuals or vehicles among massive video streams. A command centre, on the other hand, is a centralised control centre where all the images coming from these smart cameras can be viewed simultaneously on several screens. In this centre, the data collected by the cameras is analysed in real time using AI tools, which makes it possible, for example, to quickly trace the movements of a person or a private vehicle.

2.1 Technology used

Some projects, such as the safe city programme in Senegal, rely on technologies such as U-Safety, a major product of the Chinese company ZTE, which is a major player in the field of surveillance alongside Huawei. The U-Safety device is present in more than forty countries around the world, including China, Ghana, Mauritius, Sierra Leone, and Senegal (Artigas 2019). It also integrates software for analysing the data collected for efficient processing of information flows, as well as an interoperability system with the Ministry of the Interior's existing systems. In addition, the Defence and Security Forces are equipped with smartphones running on eLTE technology, offering them the ability to film incidents, exchange multimedia content in real time, geolocate, and communicate via a secure network, thus strengthening their operational capabilities in the field (Notrecontinent 2020). Currently, Huawei is working on the development of the state's digital infrastructure (datacentres, Senegalese service centres, fibre optics, smart city, safe city, etc.).

2.2 Number of cameras installed

During a follow-up meeting on the project (March 2020), the managing director of Sénégal Numérique SA indicated that 473 cameras had already been installed out of the 500 initially planned (CIO-Mag 2020). This equipment is distributed in several areas of the national territory, forming an essential basis of the urban video surveillance system and meeting the requirements of prevention, supervision, and intervention of the security forces.

2.3 Technology partnerships

Although the project relies heavily on strategic cooperation with China through the company Huawei, particularly in the context of the development of datacenters, Senegal's service centres, fibre optics, smart city and

safe city, it should be noted that the equipment, software and licences deployed are not exclusively of Chinese origin (socialnetlink 2023).

2.4 Project funding

The safe city project is part of a strategic economic partnership between the People’s Republic of China and the Senegalese government. On 8 February 2018, during an official meeting between the Chinese ambassador to Senegal, Zhang Xun, and the minister of economy, finance and planning, Amadou Ba, the two parties signed a concessional loan agreement as well as a grant agreement to finance the first phase of the Smart Senegal programme, of which the safe city project is a key component (AIDDATA 2018).

This financing, granted by the China Eximbank, takes the form of a turnkey project covering all stages of implementation: engineering, supply and installation of equipment, training of local staff, and maintenance of infrastructure. This is direct bilateral support, aimed at providing Senegal with integrated and centralised smart surveillance capabilities, with advanced surveillance cameras and a command centre for real-time data analysis (AIDDATA 2018).

According to Cheikh Bakhoun (former director of ADIE),

The overall financing of Smart Senegal is 150 million dollars (about 83 billion CFA francs) with the five components. It is a financing agreement with the People’s Republic of China. During the last visit of Chinese President Xi Jinping to Dakar, China granted a concessional loan of 150 million dollars to the State of Senegal to finance this first phase of the “Smart Senegal” program. (Dakaractu 2019).

Figure 8: Chronological stages of the project

Year	Key milestones
2019	Project start-up Visit of Senegal to the Shenzhen smart city forum in China
2020	Installation of the 473 cameras out of the 500 Command centre reception
2021	End of the project

Source: Author’s own based on information from Sénégal Numérique (n.d.).

3. Key actors

The safe city project in Senegal has been carried out by several key players, from the negotiation to the execution of the Smart Senegal programme, which includes the safe city project. At the central level, we have the Senegalese state, which is the main leader of the project. It was the then president, Macky Sall, who negotiated the agreement with China, and the steering was entrusted by ADIE (now Sénégal Numérique SA), institutionally attached to the Ministry of Digital Economy and Telecommunications, which is responsible for the technical project management of Smart Senegal (including the safe city project) and its coordination (Dakaractu 2019).

The Ministry of the Interior and Public Security (led by Aly Ngouille Ndiaye at the time) was the main beneficiary on the security side and within the framework of operational partnerships. So it is this ministry that has created an interministerial committee (police, gendarmerie, Emergency Medical Assistance Service (SAMU), fire brigade, etc.) for the establishment of command centres (CIO-MAG 2020). The director general of the national police and the chief of staff of the gendarmerie were involved in the work of the visit to Shenzhen, China in 2019. At the government level, other ministries were also involved in the implementation of this project. The Ministry of Finance validated the Chinese loan, the Ministry of Justice had taken into account the legal aspects and the Operational Office for the Monitoring of the Emerging Senegal Plan (BOS PSE) monitored the implementation of the Smart Senegal programme.

Some pilot cities have also logically participated in the deployment. The mayor of Dakar and other cities concerned, such as Thies, Kaolack, Touba, and Saint Louis, facilitated the installation of cameras on the roads (occupation of the public highway). However, surveillance remains coordinated by the state, via ADIE and the Ministry of the Interior. Among the actors are also the defense and security forces because they are the main users of the safe city technologies.

The safe city project plans to:

...provide the defense and security forces with an innovative and secure communication system based on e-LTE (4G) technology, the deployment of an intelligent video surveillance system in strategic points and the establishment of Command centers for the centralization of information in real time and decision-making assistance.
(Socialnetlink 2020)

In relation to technological players and partners, it is important to emphasise that the Smart Senegal programme (including the safe city project) relies heavily on the company Huawei. Huawei and ZTE are cited as likely suppliers (Huawei built Senegal's national data centre and provided its interconnection network (JICA 2022), and the ZTE U-Safety solution is deployed in several African countries including Senegal (Artigas 2019). To date, the acquisition scenario (public procurement or grant) is not fully transparent and specified, but it is documented that Huawei provided the infrastructure for the Smart Senegal programme (Jili n.d.).

The official motivations are security and technological. The government stresses that 'security is one of its priorities' and that technology offers new ways to protect citizens (Dakaractu 2019). In practice, the project responds as much to political logics (showing an image of a strong state in the face of insecurity) as to commercial issues (technology companies finding a large market in African cities) and technological issues. The main beneficiaries are the police and the state, which now has a modern surveillance and communication infrastructure. Companies in the digital ecosystem are winning lucrative contracts.

However, the monitored populations themselves were hardly consulted. The human rights defenders (ASUTIC) point out that no human rights impact assessment was made public before the launch of the project. Local digital rights organisations deplore this. According to them, the project was launched without a public assessment of its consequences on privacy, freedom of movement, and expression. For example, ASUTIC points out that these digital policies are born 'in a very highly regulated environment fraught with threats to human rights online', explicitly referring to the safe city project as a source of 'questions about the protection of personal data and the risks of mass surveillance' (ASUTIC 2020). The Africa Center report notes that the introduction of these surveillance technologies was done without much public debate, which reflects the absence of a democratic analysis mechanism.

4. Regulatory landscape

Test 1. Are there validly enacted laws governing public space surveillance?

Senegal does not have a specific law governing safe city or smart city type projects. Video surveillance is therefore based on a set of general texts, the main one being law 2008-12 of 25 January 2008 on the protection of personal data. This law guarantees the right to privacy and creates the Commission for the Protection of Personal Data (CDP), an independent administrative authority responsible for ensuring that this right is respected. It imposes obligations on data controllers in terms of purpose, proportionality, transparency and data security.

Other texts complete this legal framework:

- Law 2016-29 of 8 November 2016 on the Criminal Code, which includes provisions relating to cybercrime.
- Law 2018-28 on the Electronic Communications Code, which provides a framework for the digital ecosystem.
- Law 2016-33 on intelligence, which in theory limits intrusive surveillance 'as a last resort' for national security reasons (Paradigm Initiative 2024).

The 2019 bill (different from the 2008-12 law) on the protection of personal data proposed to explicitly recognise the lawfulness of data processing, including those related to video surveillance, when it is 'necessary for the exercise of official authority or the performance of a mission in the public interest' (CIPESA 2020). The safe city project is therefore legally justified by the mission of maintaining order and public safety. However, there is no provision for independent prior control (judicial or administrative) to authorise the installation of the cameras, apart from a simple advisory opinion from the CDP.

At the international level, Senegal is a party to several instruments for the protection of fundamental rights:

- Article 12 of the Universal Declaration of Human Rights: 'No one shall be subjected to arbitrary interference with his or her private life, family, home or correspondence [...].'
- Article 17 of the International Covenant on Civil and Political Rights: 'No one shall be subjected to arbitrary or unlawful

interference with his privacy, family, home or correspondence, or to unlawful attacks on his honour and reputation.'

- Article 19 of the Universal Declaration of Human Rights: 'Everyone has the right to freedom of opinion and expression; This right includes the freedom to seek, receive and impart information and ideas, regardless of frontiers, through any media whatsoever.'
- Article 13 of the Senegalese constitution of 2001: 'The secrecy of correspondence and postal, telegraphic, telephone and electronic communications is inviolable. This inviolability is subject only to such restrictions as may be made applicable by law.'

These standards require that any surveillance measures must be lawful, necessary and proportionate, which remains questionable in the context of the safe city project.

Test 2. Do the laws define a legitimate aim for public space surveillance?

In law, the stated objective of the safe city project is to strengthen public security and fight crime, which corresponds to a legitimate objective. This purpose is supported by the Ministry of the Interior and Public Security, as well as by the national police, which consider this system as a response to the security emergency (CIO-MAG 2017). However, this apparent legitimacy is strongly qualified by a major lack of transparency regarding the implementation of the project.

To date, no one knows where the data collected by the cameras is stored, who has access to it, or what the exact purpose of its use is. This lack of clarity raises major concerns about the protection of personal data and respect of the right to privacy.

This concern has been expressed by several civil society actors. During a workshop organised on 24 August 2025 by Jonction Senegal, in partnership with Internews, on the theme of violence and cyberbullying, Abdoulaye Ndiaye, head of programs, publicly questioned the authorities: 'What guarantee do we have that the data from the safe city project's video surveillance will not be used for purposes other than public safety? What measures are being taken to protect citizens' privacy?'

These questions reflect a growing fear of mass surveillance and human rights violations. In the absence of independent oversight mechanisms, there is a real risk of 'mission extension', which could turn a public safety project into a tool for widespread oversight.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

Law 2008–12 requires that all data processing be limited to legitimate purposes and determines the obligations of data controllers. It also guarantees a right of access for data subjects (Article 62). However, this right is rarely effective, as it is exercised mainly in the context of judicial proceedings, which makes it inaccessible to the majority of citizens.

In practice, law enforcement and the judiciary have direct access to the recordings for the purposes of investigation and criminal prevention, but there is no public transparency on the list of authorised actors, nor on the conditions of access. The citizen has no idea who is viewing the images and for what purposes.

Regarding the duration of data retention, the law imposes a limitation obligation, but there are no clear sectoral regulations setting a maximum period. The 2019 bill provides that data must be kept 'only for the period necessary' (CIPESA 2020), but this provision remains unbinding. Thus, there is no mechanism requiring the automatic deletion of data after use, paving the way for indefinite retention.

Lawyer Demba Ciré Bathily believes that the legality of video surveillance depends on its proportionality and its scope:

The fact of collecting images allows for more surveillance of citizens. If this is limited to certain public road spaces, no problem should arise. But if it covers private spaces, there are certain proportionalities to be maintained. It should not be a kind of permanent surveillance. If it is in certain public spaces, it can be justified. So from the point of view of the law, this is not a problem.
(Socialnetlink 2020)

Test 4. Does the law define mechanisms for citizen redress and accountability?

The CDP, created by law 2008–12, is the competent authority to ensure that privacy rights are respected. However, its actions within the framework of the safe city project are limited by several factors:

- Limited jurisdiction: the law excludes from its scope files relating to national defence, which reduces its power of control over systems related to public security (CIO-MAG 2017).
- Insufficient resources: the CDP has limited financial and human resources, making it difficult to effectively control a project of this magnitude.

- Lack of binding power: the CDP can issue recommendations and carry out inspections, but it cannot block or suspend a government project that has already been launched.

In 2015, the CDP issued a receipt to the Ministry of the Interior for the implementation of video surveillance (CIO-MAG 2017).

According to Mamoudou Niane, director of legal affairs and cooperation at the CDP, 'For reasons announced by the police and the Ministry of the Interior and Public Security, strengthening security and fighting crime has become an emergency. On the CDP side, officials reassure citizens about this deployment.' (CIO-MAG 2017).

However, this validation is not accompanied by a robust mechanism for ongoing monitoring, nor are there effective sanctions in the event of abuse. Remedies exist in theory, in particular the possibility of going to court to assert one's rights, but they remain poorly accessible, little known, and not very effective in the face of an opaque and centralised system.

5. Digital rights

The safe city project aims to cover all the major urban centres of large cities, so that the entire urban population is potentially filmed, without explicit distinction of age or sex, as long as they circulate on the public highway or in public places equipped with cameras. The most exposed are the city dwellers in these areas: pedestrians, drivers of cars, motorcycles or taxis, street vendors, etc. There is still a fear that the disadvantaged socioeconomic categories who often live in very dense peripheral neighbourhoods or carry out informal activities in cities will not be monitored in the same way as the wealthiest users. The real beneficiaries in terms of 'security' should be ordinary citizens (reinforcement of crime prevention, responsiveness in case of emergency). On the other hand, it is regrettable that neither the public nor the legislator has clear information on how this data is used and processed, which mainly benefits the users of the system (forces and authorities) rather than the people filmed.

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

The safe city project aims to cover all major urban centres in Senegal, involving the potential surveillance of all people circulating in public spaces equipped with cameras. This generalised surveillance affects all users of the city indiscriminately: pedestrians, vehicle drivers, taxis, motorcycles, street vendors, residents of peripheral neighborhoods and low-income populations, often involved in informal activities. Affluent citizens and more visible populations in city centres may indirectly benefit from increased security, but the disproportionate exposure of vulnerable populations is a major social justice issue.

The official objective of the project is to strengthen public security, prevent delinquency, and improve the responsiveness of law enforcement in the event of an emergency. However, the proportionality of these measures remains questionable: the massive and continuous surveillance seems to go beyond what is strictly necessary to achieve these objectives. As CIPESA (2025) points out, the lack of transparency and accountability mechanisms turns these systems into potential tools for human rights abuses rather than instruments for improving security.

The safe city project directly affects several fundamental rights. The right to privacy, protected by article 17 of the International Covenant on Civil and Political Rights (OHCHR n.d.), is jeopardised by the capture of biometric data, videos, behaviours, and movements of citizens. Without a clear legal basis and procedural safeguards such as judicial

authorisation, limited duration of recordings, and data security, these measures may constitute disproportionate interference with privacy.

Freedom of expression is also affected: widespread video surveillance can generate a chilling effect, where citizens censor themselves for fear of being filmed or recognised in the public space. This situation is worrying in a Senegalese context where freedom of expression and demonstration is protected by the constitution and international instruments. In addition, press freedom is vulnerable, as journalists or activists can be identified or filmed during the collection of information, thus compromising the confidentiality of sources and the safety of journalists in the field (CIPESA 2020).

The right to freedom of movement, guaranteed by article 14 of the Senegalese constitution, is also threatened. Cameras with facial recognition and ANPR systems enable accurate electronic tracking of movements. Without a strict legal framework, this monitoring could be used to draw up mobility profiles, to monitor compliance with restrictive measures such as curfews or driving bans, or to identify certain categories of citizens for targeted control.

Test 6. Does the law control who has access to surveillance data?

Access control to the data collected by the safe city project is currently insufficient and not very transparent. Videos, biometric data, and travel information can potentially be accessed by various public actors, and in some cases by private entities, without clear limitation or traceability of use. The lack of strict rules on access increases the risk of data being diverted or misused.

Senegalese law 2008–12 on the protection of personal data requires that any invasion of privacy be governed by guarantees, such as the security of processing, the limitation of storage, the restriction of access, and the possibility of judicial control. In the context of the safe city project, no specific information has been provided on the implementation of these safeguards. This loophole poses a serious risk to digital rights, as it opens up the possibility of arbitrary or discriminatory use of the data collected.

In several countries, such as China, video surveillance data can only be accessed by specifically authorised authorities and under regulated conditions, with regular audits. The absence of such measures in Senegal increases the risk of disproportionate invasion of privacy.

Test 7. Does the law create an independent oversight body?

The safe city project does not provide for an independent and adequately funded oversight body to ensure that the surveillance of public spaces

complies with the law and international human rights commitments. The lack of external and autonomous oversight limits the accountability of actors and prevents the implementation of effective remedies in cases of rights violations.

In other African contexts, CIPESA (2025) notes that invoking 'national security' allows authorities to circumvent fundamental rights protections. Without an effective oversight mechanism, the safe city project risks becoming an example of unregulated mass surveillance, incompatible with Senegal's international obligations, including the ICCPR and the African Charter on Human and Peoples' Rights.

Test 8. Has a human rights impact assessment been carried out?

To date, there is no information on the existence of a prior human rights impact assessment before the project was deployed. The lack of a differentiated analysis according to gender, social class, or the type of use of public space prevents the risks related to privacy, freedom of expression, freedom of movement, and non-discrimination from being measured objectively.

Identified risks include:

- The systematic tracking of citizens' journeys, with the possibility of creating individual or collective profiles;
- The chilling effect on public expression and participation in demonstrations;
- The amplification of social or racial discrimination through algorithmic biases.

International recommendations, taken up by CIPESA (2025), insist on the need for a prior human rights impact assessment for any smart video surveillance project, in order to put in place appropriate and proportionate mitigation measures. The absence of this assessment is a major shortcoming and reinforces the potential for infringement of citizens' rights.

6. Conclusion and recommendations

The purpose of this study was to understand how Senegal's safe city project, based on intelligent video surveillance (cameras with facial recognition and number plate reading), impacts human and digital rights. We have seen that this programme, launched in 2019 as part of the Smart Senegal programme, provides for the installation of about 500 cameras in Dakar and five other cities (Thies, Saly, Touba, Saint-Louis, and Kaolack). The system is linked to a national command centre and several regional centres, allowing law enforcement to view public places in real time and intervene quickly thanks to an interconnection.

According to the authorities, the expected benefits are the prevention of delinquency (thefts, burglaries, and assaults), improved crowd management, and the coordination of emergency response. In theory, continuous image collection can increase citizens' sense of security and help solve crimes more effectively. This project places Senegal in a smart city logic, aligned with its digital strategy and its ambition to modernise.

On the other hand, our analysis highlighted serious challenges. On a technical level, the reliability of the algorithms (facial recognition and ANPR) in a local context remains to be validated: frequent errors could lead to erroneous identifications or missing data, disrupting the promised efficiency. On the financial level, the investment is significant (hundreds of millions of CFA francs) and maintenance costs will have to be covered in the long term. On the political level, the lack of transparency on the data processed or shared with other agencies (migration, internal security) is a cause for concern.

Above all, in terms of fundamental rights, the draft raises major contradictions. Our review of legal and human rights revealed that the protection of privacy is formally guaranteed by the constitution (secrecy of correspondence, Article 13) and by the Data Protection Act. Freedom of expression and demonstration are also among the constitutional rights (articles 8 and 10), as is the right to freedom of movement (Article 14). However, safe city cameras carry out continuous surveillance in public spaces without a clear threshold of authorisation or proportionate requirements. According to international standards (UN Declaration on Privacy), any limitation of these rights must be provided for by law, strictly necessary, and subject to independent oversight. On these points, the project has not provided any concrete elements: no implementing decree or impact study has been made public to justify this mass surveillance or to assess its effects.

However, Senegalese domestic law has binding principles: law 2008-12 requires, in principle, that the processing of personal data be based on consent or a clearly defined public interest purpose. The safe city project, financed and supported by the state itself, could theoretically fall within this 'public interest' exception, but this clause does not exempt the requirement for guarantees (data minimisation, transparency, limited retention period, etc.). The Senegalese penal code also punishes invasion of privacy and illegal collection of information, but so far no effective monitoring of these provisions has been reported for this project.

To reconcile the security of public spaces and the protection of individual freedoms, several concrete measures are necessary:

- Transparency around the safe city project must be strengthened by publishing the regulations and decrees that govern it, the precise purposes of the collection of images (law enforcement, search for missing persons, etc.), as well as the access protocols and the duration of data retention. Civil society and the media should be consulted, and an annual report should be presented to Parliament and made public, including statistics on interventions, arrests, and privacy violations.
- Independent monitoring is also essential. It could involve strengthening the prerogatives of the CDP or the creation of a specific regulatory body, including representatives of civil society, magistrates, and NGOs to issue, regulate, and audit surveillance authorisations. The experience of other African countries and the recommendations of CIPESA argue for the establishment of ethics or parliamentary oversight committees.
- Citizen participation must be encouraged through public consultations, citizens' juries, or round tables with NGOs and residents, in order to present the benefits, risks, and costs of the system, and thus promote its ownership.
- From a legal point of view, it is necessary to adapt the law to explicitly regulate intelligent video surveillance (facial recognition, ANPR, drones), to set maximum data retention periods, to impose privacy impact assessments, and to provide for dissuasive sanctions against abuse.
- Finally, the protection of digital rights must be prioritised by training law enforcement, prohibiting the use of cameras to target political opinions or specific groups, and ensuring that citizens have redress mechanisms to access or have their data deleted, in accordance with law 2008-12.

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Smart city surveillance in Uganda

Richard Ngamita

1. Introduction

Over the past two decades, Uganda has undergone a rapid and unprecedented transformation in its surveillance infrastructure. Once reliant on human informants, physical observation, and limited analogue technologies (Kalyegira 2005; West 2012), the country has shifted towards a digitally integrated model in which almost every major urban intersection, public space, and national highway falls under the gaze of high-definition cameras and automated monitoring systems (Unwanted Witness 2020; *Daily Monitor* 2022). The government's flagship safe city programme embodies this transformation by deploying thousands of CCTV cameras, facial recognition software, ANPR systems, and, most recently, digital number plates connected to an intelligent transport monitoring network (Human Rights Watch 2023; Omollo and Ajuna 2025).

While the official narrative frames these developments as essential tools for improving public safety, combating terrorism, and modernising law enforcement (Uganda Radio Network 2018), their implementation raises fundamental questions about privacy, political freedoms, and state accountability (Jili 2023a; Staden 2024). Uganda's push for increased surveillance aligns with a global trend in which digital monitoring is utilised not only for crime prevention but also, in some cases, to strengthen political control (Hilman and McCalpin 2019; Woodhams 2022). Collaborations with Chinese technology companies such as Huawei and ZTE, as well as newer partnerships with Russian contractors, have expedited the implementation of advanced surveillance technologies. This has integrated foreign technical expertise into Uganda's domestic security framework (Passalacqua and Polito 2024; Security Vision 2022).

This report examines the technological, legal, political, and human rights dimensions of Uganda's surveillance system, with a focus on the safe city initiative and its subsequent expansions. It traces the historical evolution from colonial-era intelligence practices to contemporary digital monitoring, identifies the key actors shaping the system, and evaluates the implications of current laws and policies for citizens' rights. A detailed case study of the safe city programme provides concrete evidence of both its operational scope and its controversies, most notably, the 2025 rollout and suspension of the Electronic Penalty System (EPS) (The Independent 2025).

The research is centred around the key question: to what extent does Uganda's safe city surveillance framework improve public safety, and what implications does it have for privacy, accountability, and civic freedoms? By exploring this question through historical analysis, comparative

studies, legal evaluation, and stakeholder mapping, the report seeks to contribute to ongoing discussions about the balance between security and individual liberties in an age of increasing state surveillance.

The remaining sections of the report include the following: section 2 provides historical background, tracing the evolution of surveillance in Uganda from colonial-era intelligence practices to contemporary digital monitoring systems; section 3 examines Uganda's safe city programme in a comparative context with broader trends across Africa and details the three phases of its implementation; section 4 identifies and analyses the key actors involved in the surveillance ecosystem, including state institutions, corporate partners, funders, and civil society organisations; section 5 evaluates the legal and regulatory framework governing surveillance through four critical tests that assess the lawfulness, legitimacy, and scope of monitoring in public spaces; section 6 explores the implications for human rights and governance through four rights-based tests, focusing on necessity, proportionality, access controls, and oversight mechanisms; And section 7 synthesises the findings, considers the cost-benefit implications, and provides recommendations for reforming Uganda's surveillance governance framework.

2. Background

Surveillance in Uganda predates the modern digital era and can be traced back to British colonial administration practices. During colonial rule, intelligence gathering relied heavily on human informants, local chiefs who compiled reports on political activities, and rudimentary interception methods, such as telephone tapping (Kalyegira 2005). The colonial Special Branch, an intelligence unit, targeted nationalist movements, dissidents, and community leaders through infiltration and covert observation. This created a precedent for framing surveillance primarily as an instrument of state security and political control rather than as a means of safeguarding public safety.

Following independence in 1962, postcolonial governments in Uganda inherited and adapted these intelligence structures (West 2012). State agencies such as the Special Branch and Military Intelligence sustained informant networks, continued wiretapping of telephone lines, and engaged in physical monitoring of opposition figures, journalists, and civil society actors. However, these systems remained limited in scale, constrained by the analogue infrastructure and the financial resources of the state.

A major shift occurred in the early 2000s with the deepening of Uganda's economic and diplomatic ties with China. Beijing's Belt and Road Initiative, and in particular its Digital Silk Road component, promoted partnerships in telecommunications, infrastructure, and digital connectivity across Africa (Norton Rose Fulbright 2016). Uganda became a key site for these engagements. By 2023, more than 800 Chinese companies operated in Uganda, contributing to sectors such as manufacturing, construction, energy, and information technology (JCCOCC 2024).

The telecommunications sector was especially significant for surveillance. Huawei and ZTE, two Chinese technology giants, became central players in building Uganda's digital backbone, constructing base stations, laying fibre-optic networks, and supplying core ICT infrastructure (You Tube 2023). This expansion created the conditions for integrated surveillance by enabling the rapid transmission and storage of large volumes of data. Huawei, in particular, extended its role beyond commercial telecommunications into the security sector, aligning with a broader continental pattern of Chinese technology companies providing turnkey safe city solutions (Jili 2023b; Privacy International 2020).

These partnerships facilitated the transformation of Uganda's surveillance capacity from an analogue system dependent on human observation into a digitally networked architecture capable of real-time monitoring.

Crucially, the technological infrastructure built by Huawei and ZTE provided the foundation for later projects such as CCTV coverage, facial recognition software, vehicle tracking systems, and automated monitoring centres. The shift from surveillance as an improvised practice of intelligence agencies to surveillance as a comprehensive state infrastructure represents a decisive turning point in Uganda's security governance.

This trajectory, from colonial intelligence gathering to post-independence monitoring, and finally to the digital surveillance systems of the 21st century, demonstrates continuity in the use of surveillance as a mechanism of political control, while also marking a significant expansion in scale and technical sophistication. It is against this historical and structural backdrop that Uganda's safe city programme, launched in 2018, must be understood. The programme represents the institutionalisation of these technological shifts into a national framework for public space surveillance, promising security gains but raising profound questions about privacy, accountability, and state–citizen relations.

3. Case study description

Uganda's safe city programme is one of the most ambitious public surveillance projects in East Africa, combining phased CCTV deployment, AI-enhanced monitoring, and integrated transport enforcement systems. This case study examines the programme's three main phases, its operational features, and the controversies surrounding its most recent component – the Electronic Penalty System (EPS).

The foundations of Uganda's safe city programme were laid more than a decade before its formal launch in 2018. The initial phase was not presented as a comprehensive surveillance strategy but rather as a series of discrete projects that gradually expanded state capacity to monitor public space.

Originally entering Uganda in the early 2000s, Huawei quickly became a pivotal player in the country's digital infrastructure. The company secured its first major government contract in 2007. The US\$50 million deal was signed with Uganda Telecom to expand the country's GSM network (Nabyama 2007; Odomel and Kironde 2007), which laid the foundation for the safe city programme.

Huawei's role deepened when the company donated 20 surveillance cameras worth US\$750,000 in 2014 as part of the initial phase of a broader digital surveillance initiative. These cameras were installed primarily in Kampala, including on major roads and urban intersections, to monitor public spaces and enhance law enforcement capabilities. The donation marked the start of a larger-scale safe city project, which was formalised in 2018 when authorities in Kampala signed a safe city project with Huawei worth US\$126 million. This expanded initiative later included about 5,000 CCTV cameras equipped with facial recognition technology, keeping an eye on movements across Ugandan roads (Security Vision 2022).

Huawei's donation also led to broader cooperation with Uganda in the ICT sector, including the launch of its Seeds for the Future programme in 2016 to develop local ICT talent through training initiatives. As the safe city project progressed, Huawei continued to support Uganda's digital infrastructure, including telecommunications networks. Authorities later deployed facial and number plate recognition systems as part of the nationwide surveillance rollout (Woodhams 2022).

This telecommunications infrastructure established by Huawei and ZTE enabled the rapid rollout of Uganda's safe city programme, which was significantly accelerated following the March 2017 assassination of assistant inspector general of police Andrew Felix Kaweesi, a high-profile murder

that prompted President Museveni to call for a comprehensive national surveillance system and led to the official launch in October 2018, marking Uganda’s shift from basic CCTV monitoring to AI-enabled digital surveillance systems and the beginning of a phased expansion (Watchdog Uganda 2017; Techjaja 2019).

The official launch in October 2018 marked the beginning of a phased expansion. The earlier phases established the surveillance infrastructure through Huawei’s implementation, with Huawei training thousands of police officers in video analytics, facial recognition, AI capabilities (Kazibwe 2018), and centralised monitoring capacity in the Naguru National CCTV Monitoring Centre (Uganda Radio Network 2018). These systems directly enabled the phase III expansion into vehicle tracking and automated traffic enforcement in 2025. Without the nationwide fibre-optic networks, high-capacity data storage, and operational command centres built between 2015 and 2022, the integration of digital number plates and the EPS would not have been technically feasible (Emukule 2025; *The Independent* 2025). This demonstrates a clear continuity from pre-2023 surveillance efforts – beginning with 3,233 cameras in phase I covering Kampala Metropolitan area – to the more sophisticated, AI-linked systems of today, with Uganda now planning a phase III expansion requiring US\$62 million to eventually reach the goal of 20,000 cameras nationwide (Parliament of Uganda 2025).

Figure 9: Phases of Uganda’s Huawei-supported safe city surveillance project (2018–present)

Phase	Year	Key milestones
Phase I	2018–2019	Pilot and Kampala metropolitan coverage
Phase II	2019–2022	Nationwide rollout
Phase III	2023– present	AI integration and transport surveillance

Source: Author’s own.

3.1 Phase I (2018–2019): Pilot and Kampala metropolitan coverage

The initial deployment commenced with camera installations in central Kampala before expanding to surrounding suburbs, creating a comprehensive urban monitoring network. The surveillance system incorporated advanced pan-tilt-zoom units equipped with night vision

capabilities, ANPR, facial recognition technology, and live audio broadcast functions (Kazibwe 2018).

Operations initially ran through a temporary command centre at Nateete Police Station until the establishment of the permanent National CCTV Monitoring Centre at Naguru. The system achieved substantial coverage across the metropolitan area, with President Museveni formally commissioning the Naguru facility as the central command hub for nationwide surveillance operations (Ampurire, 2018; Uganda Radio Network 2018).

3.2 Phase II (2019–2022): nationwide roll-out

The subsequent phase extended surveillance coverage beyond urban centres to encompass municipalities, major towns, and strategic transportation corridors nationwide. The network expanded to encompass major regional centres and smaller towns across the country's various districts and regions. Implementation reached substantial completion levels across both phases, with thousands of cameras becoming operational across the national territory (Hilman and McCalpin 2019).

During this expansion, the Uganda Police Force enhanced training programmes in video analytics, biometric data processing, and AI-based threat detection capabilities. Government officials cited the system's effectiveness in criminal investigations, particularly in identifying vehicles and suspects in serious crimes. Officials reported that the surveillance infrastructure has facilitated the handling of tens of thousands of operational and intelligence incidents, with thousands of cases utilising recorded footage for comprehensive investigations (UBC 2025).

However, civil society organisations expressed concerns that the absence of adequate legal frameworks governing the technology's use could enable mass surveillance targeting political opposition, journalists, and vulnerable communities (Unwanted Witness 2020).

3.3 Phase III (2023–present): AI integration and transport surveillance

Phase III, announced in March 2023, sought to fill blind spots in earlier phases and broaden geographic reach. The government announced that phase III aims to address shortcomings from earlier phases by increasing camera coverage to eliminate blind spots, introducing body-worn cameras for officers, and enhancing the data centre's capacity to accommodate rising storage and processing needs (UBC 2025)

The Intelligent Transport Monitoring System (ITMS) linking road traffic enforcement to the national CCTV network via microchipped digital number plates was also introduced during this phase (Omollo and Ajuna 2025).

The ITMS, jointly managed by the Ministry of Works and Transport and the Ministry of Internal Affairs, aimed to improve traffic safety and curb vehicle-related crimes. The system's ability to track vehicle movements nationwide represented a major leap in Uganda's surveillance capacity, but also intensified privacy concerns (Human Rights Watch 2023).

On 12 March 2025, the Uganda Police Force announced that it requires an additional US\$62 million to implement phase III of the national CCTV project (UBC 2025).

3.4 The Electronic Penalty System (EPS) controversy (2025)

On 7 June 2025, the Ministry of Works and Transport launched the EPS, integrating ANPR cameras into the ITMS to automatically detect speeding and red-light violations (SoftPower 2025). The system imposed fines ranging from UGX200,000 for minor infractions to UGX600,000 for serious offences, with a 72-hour payment window.

Within days, public backlash was intense. Many drivers, especially in the informal transport sector, found the fines punitive relative to average incomes. Reports emerged of duplicate fines, penalties for non-existent violations, and mismatches between vehicle data and number plates (Daily Monitor 2025). Drivers complained of missing or poorly positioned speed limit signs, leading to unintentional violations. Critics accused the government of prioritising revenue generation over road safety.

The Association of Taxi Drivers, joined by bus and truck operators, threatened a nationwide strike. Meetings between transport unions and government officials failed to resolve disputes, and on 12 June 2025, the ministry suspended the EPS for a one-month review (*The Independent* 2025).

3.5 Public perceptions and political implications

Public opinion on the safe city programme is sharply divided. Supporters argue the system deters crime and aids investigations, citing examples where CCTV footage led to quick arrests. Opponents, however, view it as a politically motivated tool, referencing reports that Huawei technicians assisted security agencies in intercepting opposition communications in 2018–2019 (*The Wall Street Journal* 2019; Security Vision 2022).

However, empirical evidence supporting the system's crime reduction effectiveness remains mixed. The only comprehensive academic study of Uganda's CCTV system, conducted by EGAP (Evidence in Governance and Politics), found that while areas with cameras experienced 'a statistically significant decrease in reported crime', this may have simply resulted in 'crime displacement' to areas without surveillance coverage rather than a genuine reduction (EGAP 2021). Civil society organisations note 'there is no compelling evidence to show that hardcore criminals have been netted as a result of installing cameras' (Unwanted Witness 2019).

The absence of systematic evaluation studies measuring crime reduction rates or comprehensive analysis of court cases utilising surveillance footage undermines claims that the surveillance system represents a necessary and proportionate response to Uganda's security challenges, particularly given the hundreds of millions of dollars invested in the infrastructure.

The EPS episode has further polarised the debate: for many, it confirmed fears that surveillance infrastructure could be repurposed for revenue extraction or political ends, especially in the absence of strong oversight and transparent governance.

3.6 Linking phases I–III

The progression from static urban cameras to AI-enabled vehicle tracking demonstrates the government's strategic layering of capabilities. Early investments in fibre-optic backbones, data storage, and command centres made phase III's advanced features possible. However, the same infrastructure that facilitates efficient law enforcement also enables pervasive, real-time monitoring of the population, a dual-use reality that continues to fuel Uganda's surveillance controversy.

4. Key actors

The development and operation of Uganda's safe city surveillance framework involve a complex network of state institutions, private companies, foreign contractors, and civil society organisations. Understanding these actors, their roles, and their motivations is essential to evaluating the programme's governance, accountability, and long-term sustainability.

4.1 State institutions

Ministry of Internal Affairs

The Ministry of Internal Affairs provides policy oversight for internal security, including law enforcement and surveillance. It works closely with the Uganda Police Force to coordinate surveillance deployments, approve procurement contracts, and manage interagency collaboration (Uganda Radio Network 2018). The ministry also plays a role in drafting or implementing legislation relevant to surveillance operations, such as the Regulation of Interception of Communications Act (2010).

Uganda Police Force

The Uganda Police Force is the primary operator of the safe city network. It manages the National CCTV Monitoring Centre at Naguru and oversees regional command centres. The Uganda Police Force's responsibilities include camera monitoring, video analysis, facial recognition operations, and coordinating field responses (Kazibwe 2018). Since 2018, thousands of officers have been trained in CCTV operation, data analytics, and AI-assisted threat detection.

National Information Technology Authority–Uganda (NITA-U)

NITA-U manages the country's national data backbone and provides technical support for ICT systems in government. Its role in the safe city programme includes integrating the surveillance network into existing ICT infrastructure, ensuring connectivity, and overseeing cybersecurity measures for the storage and transmission of sensitive data (NITA-U 2019).

Ministry of Works and Transport

The Ministry of Works and Transport became a key stakeholder during phase III with the launch of the Intelligent Transport Monitoring System (ITMS) and the EPS. It supervises the integration

of digital number plates and oversees vehicle-related data flows into the safe city network (Omollo and Ajuna 2025).

4.2 Corporate actors

Huawei

Huawei is the central private-sector partner in Uganda's safe city programme. Since its first major contract in 2007, the company has built telecom infrastructure, supplied CCTV equipment, integrated command centres, and trained Ugandan technicians and police officers (Nabyama 2007; Stumo 2019). Huawei's exclusive ICT partner status since 2015 has made it indispensable to Uganda's digital security architecture (JCCOCCC 2024). Huawei's network of CCTV cameras in Uganda has since expanded into a comprehensive system estimated at approximately 5,000 in number (Security Vision 2022), equipped with facial recognition technology (*South China Morning Post* 2019; Biometric Update 2019).

These cameras feed live footage through dedicated fibre-optic cables to multiple monitoring centres and a central police data hub (Security Vision 2022; Woodhams 2022). Investigations have implicated Huawei employees in assisting surveillance efforts against political opponents, heightening concerns over digital authoritarianism and human rights abuses facilitated by the technology (Unwanted Witness 2020; Security Vision 2025; Kisakye 2025).

ZTE

While Huawei has dominated Uganda's safe city implementation through its US\$126 million facial recognition surveillance system, ZTE has contributed to Uganda's broader telecommunications infrastructure. ZTE has been contracted to build Uganda's national fibre backbone, replacing infrastructure installed in 2003 with dense wavelength division multiplexing (DWDM) equipment (Capacity Media 2022). This telecommunications infrastructure development indirectly supports the capacity required for surveillance systems by providing the underlying network foundation (Norton Rose Fulbright 2016).

M/S Joint Stock Company Global Security (Russia)

This Russian contractor is responsible for manufacturing and managing the microchipped digital number plates used in the ITMS. Operating from a facility in Kawempe North, Kampala, the company has played a central role in expanding vehicle surveillance capabilities (Omollo and Ajuna 2025).

4.3 Funders and financing structures

Safe city implementation appears to have been financed primarily through direct Ugandan government expenditure rather than Chinese loans. The Uganda Police Force announced it had purchased facial recognition cameras from Huawei at a cost of US\$126 million as part of a safe city agreement (Biryabarema 2019), with the system being described as a purchase rather than a loan-funded project. The initial deployment began with Huawei donating 20 units worth US\$750,000 to the Ugandan government around 2014 (Security Vision 2022), representing direct corporate investment by Huawei in the form of equipment donations.

4.4 Civil society and oversight actors

Unwanted Witness

Unwanted Witness is a Ugandan digital rights organisation that has consistently criticised the safe city programme for lacking transparency, oversight, and adequate data protection safeguards. It warns of mass surveillance risks and political misuse of technology (Unwanted Witness 2018, 2020).

Human Rights Watch

Human Rights Watch has documented the potential for abuse in Uganda's surveillance programmes, particularly the privacy risks posed by vehicle tracking systems and the absence of strong legal protections against state overreach (Human Rights Watch 2023).

Collaboration on International ICT Policy in East and Southern Africa (CIPESA)

CIPESA has conducted research on the implications of Uganda's surveillance initiatives for digital rights, drawing attention to foreign influence, compelled assistance from service providers, and the lack of effective legal remedies (CIPESA 2024a, 2024b, 2024c).

Media and watchdog groups

National and international media outlets such as *The Daily Monitor*, *The Independent*, *Reuters* and many more have investigated the safe city programme, exposing operational flaws (for example, the EPS technical errors), procurement controversies, and possible political targeting (Biryabarema 2019; *The Independent* 2025).

**Table 19: Key actors in Uganda's
safe city programme**

#	Actor	Role in safe city programme	Influence on policy/operations	Position on surveillance
1	Ministry of Internal Affairs	Policy oversight, interagency coordination	High – sets security priorities	Supports expansion for national security
2	Uganda Police Force	Operates CCTV network, analyses data, coordinates responses	High – primary operational control	Strongly supports expansion
3	NITA-U	ICT integration, data backbone management	Medium – technical role	Supports integration but is less vocal on rights issues
4	Ministry of Works and Transport	Oversees ITMS, EPS, and vehicle data	Medium – transport-specific policy	Supports traffic enforcement role
5	Huawei	Builds infrastructure, supplies hardware, trains personnel	Very high – exclusive ICT partner	Promotes tech for safety, silent on rights concerns
6	ZTE	Telecom infrastructure support	Low-medium – indirect support	Neutral public stance
7	M/S Joint Stock Company Global Security	Supplies/manages digital number plates	Medium – key to ITMS	Silent on governance issues
8	Chinese Policy Banks	Loan financing	Medium – shapes procurement capacity	No public stance
9	Unwanted Witness	Digital rights advocacy	Medium – influences public debate	Opposes the current implementation without reforms
10	Human Rights Watch	Rights monitoring and advocacy	Medium – international advocacy	Calls for safeguards and oversight
11	CIPESA	Regional digital policy research	Medium – provides evidence-based critiques	Advocates for stronger privacy protections
12	Media (Daily Monitor, Independent, Reuters)	Investigative reporting	Medium – shapes public perception	Mixed coverage, highlighting flaws and benefits

Source: Author's own.

5. Regulatory landscape

Uganda's safe city programme and associated surveillance activities operate within a legal environment that grants broad powers to the state in the name of national security while offering limited enforceable safeguards for privacy and data protection. This section examines the legal framework through four critical legal tests that assess compliance with international standards for lawful surveillance.

Test 1. Are there validly enacted laws governing public space surveillance?

Uganda's public space surveillance operates under validly enacted laws but with a broadly defined scope and purpose. The Constitution of the Republic of Uganda (1995) provides the foundational framework through article 27, which guarantees the right to privacy of a person, home, and property and explicitly prohibits unlawful searches and interference with correspondence. Article 29 protects freedom of expression, assembly, and association, while article 43 permits the limitation of rights in the public interest, including national security, stipulating that such limitations must be demonstrably justifiable in a free and democratic society.

The statutory legal basis includes several key acts. The Anti-Terrorism Act (2002) authorises extensive investigative powers to prevent and respond to terrorism, including surveillance and interception of communications to gather evidence or prevent attacks, compelled cooperation from telecommunications service providers, and criminalisation of refusal to comply with lawful surveillance orders. The Regulation of Interception of Communications Act (2010) governs the lawful interception of communications by authorised security agencies, allowing interception of postal, telephonic, and electronic communications and permitting real-time monitoring of targeted individuals.

The Data Protection and Privacy Act (2019) provides a statutory framework for the collection, processing, and storage of personal data, with key provisions including consent requirements for data processing except where exemptions apply, and section 6, which exempts national security and public order activities from several obligations, including individual consent.

In practice, article 43's broad 'public interest' clause has often been interpreted expansively, allowing the state to justify extensive surveillance measures with minimal judicial scrutiny (Human Rights Watch 2023). While validly enacted laws exist, the legal foundation lacks specific definitions of surveillance scope and purpose.

Test 2. Do the laws define a legitimate aim for public space surveillance?

The legal framework articulates several stated legitimate aims, including national security and counterterrorism under the Anti-Terrorism Act (2002), public safety and crime prevention through the constitution article 43, and public order maintenance via the Data Protection and Privacy Act (2019). However, these laws lack clear definitions of what constitutes legitimate security threats warranting surveillance deployment.

The national security exemption in the Data Protection and Privacy Act has effectively allowed safe city operators to process vast amounts of personal data (including biometric identifiers) without individual consent or independent oversight. The Anti-Terrorism Act has been a primary legal basis for deploying advanced surveillance technology, including facial recognition, under the justification of counterterrorism (Passalacqua and Polito 2024), but it lacks clear tests for proportionality and necessity in surveillance orders.

Evidence of mission creep is documented in multiple cases. In 2018–2019, Huawei technicians, working with Ugandan security forces, helped intercept communications of opposition leader Bobi Wine, actions that were reportedly justified under national security exemptions (The Wall Street Journal 2019). Civil society groups have reported instances where antiterrorism powers were misused to monitor political activists and journalists (Human Rights Watch 2023).

Further evidence emerged in 2025 when the EPS's ANPR capabilities were defended by officials as lawful under the Data Protection and Privacy Act's national security provisions and the Anti-Terrorism Act's public safety mandates (SoftPower 2025), despite the system being used for automated traffic enforcement rather than security purposes. The broad legal definitions enable surveillance beyond stated security purposes, into political monitoring and revenue generation.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

The legal framework grants surveillance powers to multiple agencies, including the Uganda Police Force as the primary safe city operator, security agencies under ministerial authority, and the NITA-U for technical oversight. However, there is no clear delineation of roles among these agencies or clear protocols governing interagency data sharing.

The Regulation of Interception of Communications Act permits real-time monitoring of targeted individuals but critically allows warrants to be issued by the minister responsible for security rather than by a court. This ministerial warrant system enables the rapid deployment of interception

measures in safe city operations, particularly for monitoring 'persons of interest' in real time, but undermines the separation of powers and increases the risk of politically motivated surveillance (Unwanted Witness 2018).

The framework contains several critical gaps regarding data collection limitations. There is no clear mechanism for individuals to challenge unlawful surveillance or seek remedies for misuse. No statutory provisions define data retention periods or deletion requirements. The Data Protection and Privacy Act mandates data controllers to ensure data security and confidentiality, but provides broad national security exemptions that permit the unlimited collection of biometric and personal data.

The framework inadequately defines authorised actors and fails to limit the nature and scope of data collection, particularly through broad national security exemptions that have been applied in ways that blur the line between legitimate security functions and political monitoring.

Test 4. Does the law define mechanisms for citizen redress and accountability?

Uganda lacks independent oversight mechanisms for surveillance operations. Uganda does not have an independent oversight body for surveillance, similar to privacy commissioners found in Canada or the European Union. NITA-U, tasked with oversight under the Data Protection and Privacy Act, is a government agency, raising concerns about independence (Unwanted Witness 2020).

Parliamentary oversight is minimal, as the existing institutional checks, such as parliamentary committees, seldom provide consistent scrutiny of security agency operations beyond budgetary approvals (CIPESA 2024a). The legal framework provides inadequate remedies for citizens, with no clear mechanism for individuals to challenge unlawful surveillance or seek remedies for misuse.

Although judicial review is theoretically possible, very few cases have been brought to court. This is partly due to a lack of public awareness regarding available legal remedies. No statutory compensation schemes exist for surveillance misuse, and national security exemptions create barriers preventing individuals from knowing they have been surveilled. The legal framework lacks independent oversight mechanisms and provides inadequate redress options for citizens affected by surveillance operations.

Regional and international standards

Uganda is a signatory to relevant international instruments, but has not aligned domestic law with stronger privacy protections. Uganda is a signatory to the African Union Convention on Cyber Security and Personal

Data Protection (Malabo Convention), which sets minimum standards for data protection, including independent oversight mechanisms. However, Uganda has not fully aligned its domestic laws with the Convention’s stronger privacy guarantees (CIPESA 2024b).

Under the International Covenant on Civil and Political Rights (ICCPR), to which Uganda is a party, restrictions on privacy must be lawful, necessary, and proportionate. The United Nations Office of the High Commissioner for Human Rights has recommended that states establish independent oversight of surveillance systems and ensure transparent public reporting. These measures are not currently in place for Uganda’s safe city programme (United Nations 2022).

Table 20: Summary of enabling vs rights-limiting provisions

#	Law/provision	Enables surveillance by...	Limits rights by...
1	Constitution (article 43)	Allowing rights limits for national security	Broad ‘public interest’ definition enabling wide surveillance scope
2	Data Protection and Privacy Act (2019)	Exempting national security operations from consent	No independent oversight; no effective remedies for citizens
3	Anti-Terrorism Act (2002)	Authorising broad interception powers	Weak proportionality tests; potential use against political actors
4	Regulation of Interception of Communications Act (2010)	Allowing ministerial warrants for interception	No judicial authorisation; high potential for abuse

Source: Author’s own

In summary, Uganda’s current legal framework emphasises the operational efficiency of surveillance systems at the expense of individual rights. Although constitutional provisions nominally guarantee privacy, statutory exemptions and the lack of independent oversight mechanisms have fostered a permissive environment for extensive state surveillance, resulting in limited accountability.

6. Digital rights

Uganda's safe city surveillance framework has sparked intense debate regarding its impact on civil liberties, democratic governance, and public accountability. While the government promotes it as a modern tool for crime prevention and security enhancement, human rights organisations argue that its design and implementation have fostered an environment conducive to political abuse and privacy erosion. This section examines the surveillance system through four critical rights-based tests.

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

Are the privacy violations from surveillance data use necessary and proportionate to achieve legitimate aims? The evidence suggests Uganda's surveillance system fails this test due to extensive overreach beyond stated security purposes.

Reports by *The Wall Street Journal* (2019) and Security Vision (2022) document how Huawei technicians assisted Ugandan security agencies in intercepting communications of opposition leader Bobi Wine and his associates. These activities, justified under national security exemptions in the Data Protection and Privacy Act (2019) and Anti-Terrorism Act (2002), reveal the blurred boundary between counterterrorism efforts and political intelligence gathering.

Public space surveillance, particularly when combined with facial recognition and vehicle tracking, produces what civil society actors describe as a 'chilling effect' on freedom of assembly and expression, leading citizens to self-censor or avoid participating in protests, community meetings, or political events out of fear of being identified, monitored, and ultimately arrested (Unwanted Witness 2018).

The 2024 March to Parliament protest exemplifies disproportionate surveillance use. The CCTV networks and real-time vehicle tracking through number plate recognition were reportedly used to monitor the movements of protesters and identify key organisers who were then arrested before they could even march (*Daily Monitor* 2024). This demonstrates surveillance measures that exceed proportionate responses to legitimate security concerns.

Test 6. Does the law control who has access to surveillance data?

Are there protections limiting access to personally identifiable information created through surveillance data by both state and private actors?

Uganda's framework provides inadequate access controls with significant gaps in protecting citizens' digital trails.

The legal framework fails to establish clear limitations on data access, with multiple agencies having surveillance powers but no clear delineation of roles or protocols governing interagency data sharing. Data transparency and citizen access rights are virtually non-existent. Currently, the public has no legal right to access personal surveillance data held by the Uganda Police Force or NITA-U, nor is there a publicly available audit on the system's usage.

The lack of transparency regarding how data is stored, who has access to it, and how long it is retained exacerbates privacy violations and further diminishes citizens' trust and undermines democratic accountability. The documented involvement of Huawei technicians in surveillance operations demonstrates inadequate controls over private actors' access to sensitive surveillance data and systems.

Test 7. Does the law create an independent oversight body?

Does the law provide for an adequately funded and fully independent oversight body to ensure surveillance operates according to law and human rights commitments? Uganda fails this test comprehensively.

Uganda currently lacks a dedicated, non-governmental authority to monitor compliance with privacy and data protection standards. Instead, oversight is relegated to NITA-U, an agency that operates under direct government policy influence (CIPESA 2024a). Parliamentary oversight is minimal, as parliamentary committees have seldom scrutinised safe city operations beyond mere budgetary approvals, and there are no statutory requirements for public reporting on surveillance activities or data breaches.

This absence of independent oversight fuels suspicions that the governance of the system is more accountable to political leadership than to the public, further intensifying the divide in public sentiment regarding the safe city programme.

Test 8. Has a human rights impact assessment been carried out?

Was a human rights impact assessment carried out prior to implementation? The available evidence in the documentation reviewed contains no reference to pre-implementation human rights assessments evaluating potential impacts on privacy, freedom of expression, assembly, or association.

The subsequent controversies and rights violations documented throughout the system's operation suggest implementation proceeded without proper evaluation of rights implications, as evidenced by the documented political surveillance and chilling effects on civil liberties.

Public accountability and trust

The failures in human rights have resulted in significant divisions within the public. Supporters of surveillance argue that instances where CCTV footage has helped solve serious crimes, such as murders and robberies, demonstrate its value. For example, cameras played a crucial role in identifying suspects in a high-profile murder case in Kampala in 2019 (Uganda Radio Network 2019). However, critics emphasise concerns about systematic overreach in surveillance practices.

The 2025 EPS controversy demonstrates how surveillance infrastructure can be misappropriated for revenue generation or punitive enforcement. The mass issuance of erroneous fines and the absence of clear appeal mechanisms have significantly eroded public trust, particularly among commercial drivers and low-income earners.

7. Conclusion

7.1 Synthesis of findings

Uganda's safe city programme might be a cutting-edge initiative in public surveillance, showcasing a level of technological sophistication and scale that is rarely matched by neighbouring countries. The programme has been introduced in phases, starting with the deployment of CCTV cameras throughout the capital city of Kampala. It is now advancing towards a comprehensive, nationwide network that integrates AI for real-time transport monitoring. This ambitious endeavour reflects the Ugandan government's determination to harness modern digital infrastructure to enhance public security.

Despite its innovative approach, the trajectory of the safe city programme reveals several significant governance challenges:

- Legal overreach: The legal framework surrounding the programme includes broad statutory exemptions that allow for extensive surveillance activities. This lack of stringent judicial oversight raises concerns about potential abuses of power and the erosion of individual privacy rights.
- Operational opacity: There is a troubling lack of transparency regarding the handling of surveillance data. Citizens have no access to information about how this data is collected, stored, or shared, leading to unease about who controls this sensitive information and for what purposes.
- Potential for political misuse: Allegations have emerged suggesting that the surveillance infrastructure is being used to monitor political opponents, which significantly undermines public trust in the impartiality of the system. These reports contribute to a perception that the surveillance apparatus may be weaponised for political gain (*The Wall Street Journal* 2019).
- Public backlash: The controversy surrounding the EPS in 2025 is illustrative of the public's growing discontent with enforcement technologies that are perceived as ineffective or invasive. Many citizens expressed concerns that such systems prioritise revenue collection over genuine safety improvements, leading to widespread scepticism about their legitimacy (*The Independent* 2025).

In summary, while the safe city programme has the potential to be effective in enhancing security measures, its execution raises critical questions regarding civil liberties, accountability, and the balance between safety and privacy. As the programme evolves, addressing these governance issues will be essential to maintain public confidence and ensure that the implementation of advanced surveillance technologies serves the interests of all Ugandans.

7.2 Cost–benefit considerations

Proponents of the safe city initiative emphasise several key advantages, which include:

- Faster resolution of criminal cases: advanced surveillance technology enables quicker investigations and evidence collection, leading to faster case resolutions and more effective crime-solving by law enforcement.
- Enhanced situational awareness: a network of cameras and sensors provides real-time insights into urban areas, allowing law enforcement to respond proactively to incidents and maintain a more immediate presence.
- Deterrence of crime: visible surveillance can dissuade certain criminal activities, such as vandalism and theft, as potential offenders may think twice when aware of being monitored.

However, these benefits come with significant costs, which include:

- Financial burden: establishing a safe city involves multimillion-dollar contracts for technology and infrastructure, with ongoing maintenance often requiring external financing (Biryabarema 2019).
- Social trust issues: increased surveillance may erode trust between citizens and the state if it is perceived as politically motivated, leading to concerns about alienation and safety.
- Rights-based concerns: extensive surveillance can create a chilling effect on free speech and political participation, as individuals may fear being monitored (Unwanted Witness 2018).

In conclusion, while the safe city initiative offers promise for urban safety, it is essential to consider the financial, social, and rights-based implications that accompany its implementation.

7.3 Recommendations for reform

- Strengthen legal safeguards
 - Amend the Data Protection and Privacy Act (2019) to narrow national security exemptions and require independent authorisation for biometric data processing.
 - Revise the Regulation of Interception of Communications Act (2010) to replace ministerial warrants with judicial warrants, in line with international standards (United Nations 2022).
 - Introduce explicit necessity and proportionality tests for all surveillance authorisations.
- Establish independent oversight
 - Create an independent surveillance commissioner or authority, separate from government ministries, with powers to audit surveillance operations and investigate complaints.
 - Mandate annual public reporting on surveillance scope, purpose, and data handling practices.
- Improve transparency and public engagement
 - Develop clear, accessible policies on data retention, access rights, and deletion procedures.
 - Engage civil society groups like Unwanted Witness and CIPESA in consultative forums to review surveillance policies and technical upgrades.
- Enhance technical reliability and accountability
 - Conduct regular independent audits of system accuracy, especially for automated enforcement systems like EPS, to prevent wrongful penalties.
 - Ensure clear and consistent signage for all traffic enforcement zones to improve compliance and reduce disputes.
- Diversify technology partnerships
 - Reduce reliance on a single foreign supplier (Huawei) by diversifying procurement and building local technical capacity. This would mitigate dependency risks and encourage competitive pricing.

7.4 Future outlook

The next phase of the safe city programme's development is expected to feature deeper integration of AI, predictive policing tools, and possibly an expansion into rural areas. However, without simultaneous investment in legal reforms, oversight bodies, and efforts to build public trust, Uganda risks establishing a surveillance system that is technologically advanced yet democratically vulnerable. On the other hand, adopting a framework that respects rights could position Uganda as a regional leader in responsible surveillance governance.

Uganda's safe city programme embodies both the promise and peril of rapid surveillance expansion in the digital age. In less than a decade, the country has transitioned from a limited analogue intelligence apparatus to a nationwide, AI-enhanced surveillance network encompassing CCTV cameras, facial recognition, vehicle tracking, and automated traffic enforcement.

The system has demonstrated value in solving crimes, improving situational awareness, and modernising law enforcement operations. Yet, it has also magnified longstanding governance weaknesses, including legal loopholes, a lack of independent oversight, and the potential for political misuse. The 2025 EPS controversy served as a stark reminder that public acceptance of surveillance depends not only on its technical capabilities but also on fairness, transparency, and proportionality in its application.

The central challenge ahead is not merely technological, but institutional: building a surveillance governance framework that safeguards rights, ensures accountability, and fosters public trust. Without these safeguards, Uganda's surveillance infrastructure risks deepening political polarisation and eroding civic freedoms, even as it delivers measurable security gains. With targeted reforms, however, Uganda could demonstrate that robust public safety and the protection of fundamental rights need not be mutually exclusive.

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Smart city surveillance in Zambia

Juliet Tembo and Bulanda T. Nkhowani

1. Introduction

This report documents the developments in the roll-out of surveillance of public spaces across Zambia using digital technologies as part of the safe city initiatives at an initial cost of US\$210 million. Starting with a historic overview of surveillance in Zambia, the report then outlines advancements in the deployment of surveillance using 'digital technologies'. This report provides the most comprehensive documentation to date of Zambia's safe city project, including its timeline, main features, key actors, the companies that supply the technologies, and civil society organisations critiquing it. In addition, the report details the volume and type of technology used, project spending, prevailing legal and regulatory environment, and the impact on citizens' rights. The report concludes by making recommendations arising out of this analysis on how Zambia can deploy the safe city project in a safe and rights-respecting manner. The key research question that this report explores is what interests and motivations explain the huge investment in the Zambia safe city initiative and whether the project contributes to enhancing public safety and reducing crime.

2. Background

2.1 Historical context of surveillance in Zambia

Public surveillance is not new in Zambia. It has evolved from pre-independence times in distinct phases, each reflecting its sociopolitical dynamics. During the colonial era, Zambia, or Northern Rhodesia as it was formally known, was under British rule (1888–1964), and surveillance was a critical tool for maintaining control. Led by the British Colonial Office, the administration focused on monitoring and controlling the African population, particularly men of working age and anticolonial activists in the Copperbelt province, which was a mining hub. During this time, surveillance methods included informant networks, police patrols, and administrative systems to track people's movements and labour and political activities essential for resource extraction and suppressing resistance (Ogasawara 2019). These practices were part of a broader colonial strategy, with lasting impacts on modern surveillance systems, as noted in studies on colonial legacies (Roberts *et al.* 2023).

Zambia secured political independence in 1964 and transitioned to a one-party state under the leadership of Kenneth Kaunda and the United National Independence Party (UNIP). The government retained and adapted colonial surveillance structures, with the police Special Branch actively monitoring UNIP's political opponents, journalists, and dissidents. This period of one-party rule, lasting until 1991, saw surveillance justified as necessary for national unity and security, often stifling free expression, especially during economic crises in the 1980s (Chinyere and Hamauswa 2016). Gagliardone (2020), highlights how these state surveillance practices continued colonial control mechanisms, adapted to a postcolonial context.

In 1991, Zambia transitioned into a multiparty political democracy era. The Movement for Multi-Party Democracy (MMD) and subsequent administrations, including the Patriotic Front (PF) under Michael Sata (2011–2015) and Edgar Lungu (2015–2021), expanded digital surveillance capabilities such as mobile phone and internet monitoring, often citing national security reasons. Research documents the rapid expansion of digital surveillance under Lungu's presidency following the procurement of Chinese-supplied surveillance technologies to track opposition figures and journalists (Roberts *et al.* 2023). In 2015, two journalists, Clayson Hamasaka and Thomas Zgambo believed to have been in the United Party for National Development (UPND) media team, sued one of the largest mobile network providers for wiretapping their phones and diverting messages to unknown people,

thereby infringing on their fundamental rights to private communications and freedom of expression (Lusaka Voice 2015). While the directors of the mobile network provider faced significant criminal proceedings, as of 2016, it was unclear how the case concluded. The 2016 elections saw allegations of social media monitoring to suppress dissent, with critics arguing that surveillance entrenched authoritarian tendencies (Freedom House 2017).

Since 2021, under President Hakainde Hichilema's UPND government, Zambia has pursued democratic reforms, including repealing the colonial-era defamation law in 2022 (US Department of State 2024). However, investment in smart city surveillance systems has continued, despite vehement criticism of his predecessor Edgar Lungu for wasteful expenditure on the safe city project, often framed as enhancing public safety and urban efficiency (Ministry of Home Affairs and Internal Security 2023). These initiatives, primarily supported by Chinese firms like Huawei, reflect a global trend of integrating advanced technologies into public space monitoring.

Freedom House (2022) and Phiri and Abraham (2023) note cases of social media monitoring during elections, raising concerns about privacy violations. In March 2020, the government warned online users that it had installed equipment that would enable the police and Zambia Information, Communication and Technology Authority (ZICTA) to monitor social media.

The most significant recent development is the Public Security Safety Network, dubbed the safe city project, commissioned in 2022, with support from China, reflecting global trends in smart city surveillance (Lusaka Times 2022). Prior to this, the state relied on traditional state surveillance mechanisms and penal code laws to prevent dissent and protests. This historical continuum – from colonial informant networks to post-liberation state control and modern digital systems – sets the stage for Zambia's current safe city surveillance framework, raising questions about the balance between security and civil liberties (Gagliardone 2020).

3. Case study description

Zambia's safe city project represents a significant shift in mass public surveillance driven by technology, resulting from rapid urbanisation and the need for efficiency in security service delivery by government agencies. Technology now makes it possible to electronically survey a thousand locations. Incorporation of AI now enables automation of the search for specific people's faces, car number plates, or specific movements in target locations using algorithmic surveillance, using facial recognition and car number plate recognition technologies.

Zambia's safe city initiative encompasses 900 sites, equipped with approximately 1,600 cameras, procured at a cost of US\$210 million, enabling the police to monitor any and all activities happening where cameras are mounted. The initiative has been peddled as a solution for enhancing public security and efficiency by law enforcement, with the capacity to integrate biometric data systems. However, lessons from pre-independence Zambia indicate that surveillance has been used by the state to exert power over constitutionally protected rights of citizens, and limit the agency of critical voices and political opponents (Phiri and Abrahams 2025), a pattern that still exists to date, supported by digital technologies.

Safe city surveillance in Zambia began gaining traction in the mid-2010s under the Patriotic Front government, which ruled Zambia from 2011–2021. Initial discussions regarding digital surveillance with China started in 2013 (Tumfweko News 2014) when a contract of US\$13.5 million was awarded to ZTE as the main contractor. The single-sourcing procurement process raised outcries from various stakeholders. According to Tumfweko News, in a letter dated 29 August 2013 written to then minister of home affairs Edgar Lungu, African Parliamentarians Network Against Corruption (APNAC) chairperson, Cornelius Mweetwa, accused the government of corruption and demanded to know how the contract had been signed without calling for bids in the procurement process. There was also an outcry from Transparency International Zambia (TIZ), which submitted a 57-page dossier to the Anti-Corruption Commission (ACC) on the alleged irregular signing of the contract. The 2013 contract was subsequently cancelled but was revived in 2015 with the government receiving fresh bids and awarding the contract to ZTE at a cost of US\$210 million, split into two parts, US\$65,310,967 for technical, engineering, and training services, and US\$144,689,033 for the equipment. However, the new contract in 2015 was procured under the State Security Act, CAP 111 of the Laws of Zambia, which deems the safe city project as one of national security. The safe city initiative was launched

in September 2015 after Zambia's President Edgar Lungu's first official visit to China in March 2015. President Lungu and Chinese President Xi Jinping witnessed the signing of the framework and loan financing agreement of Phase 1 of the Smart Zambia Project under unknown borrowing terms.

The Public Security Safety Network Phase III project was funded by the Development Bank of China through a commercial contract and loan agreement of US\$210 million, indicating a significant jump from the initial US\$13.5 million contract (William & Mary's Global Research Institute n.d.). The primary financiers were China Development Bank, which provided US\$178.5 million, and City Bank Zambia Limited, which contributed US\$63 million (Office of the Auditor General 2024). In 2022, Honourable Jack Mwiimbu, minister of home affairs and internal security from the ruling UPND party confirmed that the government would be proceeding to complete the project, despite halting all other digital projects to avert the debt crisis (IT Web Africa 2022), however, he lamented the costs, stating that if his administration had been in government at the time, the contract would not have been approved (News Diggers 2023). The project was commissioned by the minister of home affairs and internal security, Jack Mwiimbu, and the Chinese ambassador to Zambia Du Xiaohui, on 31 August 2022 (*Lusaka Times* 2022). Some have questioned the necessity of the safe city project, particularly the cost, which many citizens believe could have been used on other key developmental areas like road infrastructure, sanitation, and agriculture, among others (*Lusaka Times* 2015). As of December 2023, US\$63 million, which had been obtained from City Bank Zambia Limited, had been repaid in full, leaving an outstanding balance of US\$178.5 million yet to be repaid to the China Development Bank (Office of the Auditor General 2024).

ZTE is the primary project partner, awarded US\$178.5 million for the supply and implementation of the majority of the technology and systems, while Huawei was involved through the smart government component and supplied equipment and built the national data centre for US\$75 million (Huawei 2022). While construction of the safe city project commenced in December 2017, a report by the Parliamentary Committee on National Security and Foreign Affairs confirmed that the project was still in the initial implementation phase in June 2019. As of August 2022, the project was 98 per cent complete (IT Web Africa 2022).

The overall project encompasses smart healthcare, smart education, smart agriculture, smart government, smart economy, smart environment, smart infrastructure, and safe city. Currently, Zambia is implementing electronic payment systems that enable electronic billing and revenue collection. The electronic health monitoring systems capture and consolidate health data from various sites for enhanced and

efficient decision making. Communication platforms and websites are also being integrated to enhance interaction between businesses and individuals (Tumfweko News 2014; *Lusaka Times* 2022).

Specifically, the safe city project entails setting up surveillance cameras in public places such as markets, along highways, critical infrastructure areas, and water works. Its main components include; e-policing – an integrated system for supporting law enforcement departments for specific or joint tasks, investigations and information sharing; a coalition emergency response system – a unified emergency response communication system for rapid responses to emergencies; a video conferencing system – for electronic meetings to enhance internal collaboration with the provincial command centres; a video surveillance system – for video surveillance and traffic management; and an intelligent traffic system – for monitoring of mobile road users for security reasons. The components are being implemented across various departments of the Ministry of Home Affairs and Internal Security, which include the Zambia Police Service, Immigration, Drug Enforcement Commission, Zambia Correctional Services, and the Anti-Terror Unit (Office of the Auditor General 2024).

The safe city surveillance system is currently operational in nine out of ten provincial command centres situated in Chinsali, Chipata, Kabwe, Kasama, Mansa, Mongu, Ndola, Solwezi, and Choma, along with two regional command centres located in Kitwe and Livingstone.

The system encompasses approximately 1,600 cameras, and all cameras are connected to a data centre in the province. The provincial data centre feeds into the national data centre, located in Lusaka, which is complemented by two disaster recovery sites located in Kitwe and Livingstone (Office of the Auditor General 2024). The national data centre can monitor all the surveillance cameras, including the intelligent traffic system, which has 58 sites integrated with the Road Transport and Safety Agency (Office of the Auditor General 2024).

The project entailed the construction of thirty-six communication towers, the installation of radio communication systems, intelligence traffic systems, a 4G wireless broadband trucking network, and a video surveillance system. The technologies in the safe city project include high-resolution CCTV cameras with online, daylight and night vision, infrared vehicle detection and number plate recognition, and 24-hour real-time streaming and offline upload capabilities from the backbone of the network; the cameras are fitted with AI-powered facial recognition that enables identification in public spaces (Office of the Auditor General 2024).

While ZTE was the major implementation partner and supplied the majority of the technology, Huawei supplied the AI-powered vehicle number plate recognition (Phiri and Abraham 2023) and data centres. In 2018, the then Ministry of Home Affairs and Road Transport and Safety Agency undertook an exercise to harmonise the advanced road safety management system and the intelligent mobility solutions (IMS), which both proposed to deploy surveillance cameras across the country. The institutions agreed to have a unified command and control centre, and improve the capacity of and increase the number of surveillance cameras to ensure maximum geographical coverage and the ability to use big data and analytics to detect threats collaboratively. The two institutions also agreed to adopt interoperable systems that can share information in real time (Smart Zambia 2018).

As of August 2024, ZTE had transferred the safe city project to the Ministry of Home Affairs and Internal Security and delivered a comprehensive training for the relevant national officers on the system's operations in October 2023. However, despite the handover and the training, ZTE Zambia Services Limited (a local subsidiary of ZTE) continued to control the technical administration of the IT infrastructure, including network applications and database systems, indicating that the ministry may not have full control of the systems (Office of the Auditor General 2024). This also brings into question the capacity and skills of local partners to independently manage the safe city project, especially in light of changing technology.

Table 21: Breakdown of safe city spending in Zambia

Date	Donor/company	Value
2015	China Development Bank, Ministry of Home Affairs and Internal Security, ZTE - China Development Bank - City Bank Zambia Limited	Amended contract amount US\$210 million (committed)
		• US\$65,310,967 for technical, engineering and training services
		• US\$144,689,033 for equipment
		• US\$178.5 million (contribution)
		• US\$63 million (contribution)
2015	ZTE	US\$178.5 million (awarded)
2016	Huawei	US\$75 million (awarded)

Source: Authors' own based on data from Office of the Auditor General (2024).

Figure 10: Timeline of national safe city roll-out in Zambia

Year	Key milestones
2013	Bid process disputed • Opposition MPs and Transparency International Zambia dispute tender process due to single sourcing. Announcement • First agreement signed between the government of Zambia and China through commercial loan agreement of US\$ 13.5 million
2015	Contract Amendment • Second contract signed and awarded to ZTE at cost of US\$210 million.
2017	Project Starts • Project development commenced with ZTE as primary project partner for infrastructure, systems, capacity building and national command centre. Huawei built the National data centre and oversees national ID systems development.
2021	National Data Centre • Data Centre launched in partnership with Huawei
2022	Project Commissioned • Project commissioned at 98% complete. • Command centre launched.
2023	Project Handover • Project handed over to the Ministry of Home Affairs and Internal Security. Officers trained on systems operations.

Source: Authors' own.

4. Key actors

The Chinese government, through the support of the Development Bank of China, has emerged as a key supporter of Zambia's smart city initiative. The Development Bank of China was a key financier of the project and committed US\$178.5 million of the proposed US\$210 million for the implementation of the entire project, while City Bank Zambia Limited, the co-project financier, committed US\$63 million. ZTE is the primary project partner and was awarded the majority of the project funds for supplying of the key project infrastructure, systems, and capacity building to locals, including setting up of the national command centre, while Huawei was involved through the smart government component of the project and built the national data centre, supplied technical equipment, and is offering support for a biometric digital identity system under a separate contract (*Lusaka Times* 2022).

Zambia's Ministry of Home Affairs and Internal Security is the lead ministry in charge of managing and implementing the project, alongside other government departments, and is set to take over full control of the systems. The Zambia Police Service is one of the lead project agencies and oversees the e-policing aspect of the project, handling monitoring, investigations, and data sharing. Other lead government agencies involved include the Department of Immigration, the Drug Enforcement Commission, the Zambia Correctional Services, the Zambia Correctional Services, and the Anti-Terror Unit that leads on project implementation and investigations (Office of the Auditor General 2023).

Meanwhile, the Road, Transport and Safety Agency (RTSA) under the Ministry of Local Government focuses on implementing the intelligent road safety management system (IMS), which supports monitoring and management of traffic and maintaining road safety. The IMS was launched in 2018, following an exercise to synchronise the deployment of cameras and access to the command centres by the RTSA and the Ministry of Home Affairs and Internal Security to avoid duplication of efforts (Smart Zambia 2018).

On the other hand, the Parliamentary Committee on National Security and Foreign Affairs has provided oversight of the project implementation since inception and will play a crucial role in the passage of the Closed-Circuit Television Public Protection Bill 2025 and other related safe city legislation (Zambia Law Development Commission 2022).

China's footprint in the financing and building of Africa's digital technology infrastructure is significant, with neighbouring countries including Botswana

and Zimbabwe adopting similar smart city initiatives (Gagliardone 2020). Chinese technologies have gained prominence in Africa due to low price points compared to other Western markets. The Zambian case, however, shows that the technology was overpriced due to alleged corruption in the local procurement process (National Assembly of Zambia 2022). Arcesati argues that Chinese firms, which are often backed by the state and operate under Chinese law, are susceptible to state influence, blurring the lines between commercial interests and the state objectives on foreign policy, resulting in economic and political influences on Africa’s digital architectures, rules, and norms (Arcesati 2021). Arcesati further warns that China’s digital footprint in Africa may pose future competition and foreign policy challenges if African governments don’t weigh the benefits of these technologies with potential risks on sovereignty, security, civil liberties, and emerging technological capabilities (Arcesati 2021). While marketed as security solutions, the smart city initiative has the potential to open doors for malicious acts, including the potential for unauthorised access to citizen data by third parties and technology suppliers.

From the discussion above, it is clear that ultimately, and aside from legitimate security concerns and commercial interests, safe city projects in Africa are used by states as tools for spying, furthering political and diplomatic interests, and neo-colonialism.

Table 22: Key actors involved in the safe city initiative in Zambia

Actor	Roles
Ministry of Home Affairs and Internal Security: • Zambia Police Service • Department of Immigration • Drug Enforcement Commission • Zambia Correctional Services • Anti-Terror Unit	Lead ministry
	Lead project agency/manager e-policing – investigations, monitoring, data sharing
	Lead project agency/manager
	Lead project agency/manager – investigations
	Lead project agency/manager – investigations
Parliamentary Committee on National Security and Foreign Affairs	Lead project agency/manager
	Oversight and scrutiny of project implementation
Road, Transport and Safety Agency	Traffic surveillance

Actor	Roles
ZTE	Key implementing partner, supplied majority of the technology and systems
Huawei	Technology partner, supplies cameras, data centres
Transparency International Zambia	Whistleblower on the initial tender process

Source: Authors' own based on data from Ministry of Home Affairs and Internal Security and Office of the Auditor General (2024).

5. Regulatory landscape

While the safe city project promises improved security, it raises significant concerns about the protection of citizens' human rights, particularly privacy, freedom of expression, and protection from arbitrary state actions. This section examines the adequacy of Zambia's legislative framework in safeguarding these rights, addressing whether human rights impact assessments were conducted, and the existence of laws governing surveillance mandates, data access, data protection, retention periods, and independent oversight.

Article 17 of the Zambian constitution (2016) protects the right to privacy including protection from unlawful searches, seizure of possessions, and unauthorised disclosure of personal communications. Furthermore, the national constitution affirms protection of the right to free expression under article 20, the right to movement under article 22, and the right to participate in political activities under article 45.

Test 1. Are there validly enacted laws governing public space surveillance?

Despite the advanced state of safe city development in Zambia, the space remains unregulated; there is no law backing defining the purpose and scope of surveillance in Zambia. The project was procured under the State Security Act, CAP 111 of the Laws of Zambia, which provides a framework for dealing with espionage, sabotage, and other activities prejudicial to the interests of the state.

The Data Protection Act 2021 reinforces the right to privacy and provides a framework for accessing, processing, and storage of personal data; however, it does not cover smart CCTV surveillance data, and it is silent on data relating to video and audio footage and recordings (CCTV footage). The Act is currently under review, and the cabinet approved its repeal in April 2025. The repeal process aims to address the inclusion of a broader range of data categories, including non-personal like industrial and public data, AI, machine learning, and data analytics (*Zambian Observer* 2025).

There has been notable resistance by civil society, human rights watchdogs, and lawyers over the lack of legislation to govern surveillance technology or CCTV usage in the country, citing data protection and privacy concerns as provided for by the Data Protection Act 2021 and article 17 of the national constitution (Sakala and Mulenga 2024). In addition, critics have accused the government of trying to quietly pass CCTV legislation that will see Zambia become a surveillance and police state (TechMag 2025).

In 2020 the Ministry of Home Affairs and Internal Security engaged the Zambia Law Development Commission, to scope and develop a legislative framework for the protection of fundamental rights in the usage of CCTV in public spaces and private spaces accessible to the public, management of data by law enforcement agencies, and to prevent abuse, violation of privacy, and compromise of national security (Zambia Law Development Commission 2024). Subsequently, the Bill was handed over to the Ministry of Home Affairs and Internal Security in December 2022. Following several internal processes, the Closed-Circuit Television Public Protection Bill 2025 was introduced to Parliament in March 2025 and came up for third reading in July 2025 (Zambia Monitor 2025). The Bill aims to provide for crime prevention and public safety, regulation of surveillance practices, protection of privacy and data security, enhancing law enforcement and judicial processes, encouraging investment and smart CCTV development, and compliance with regional and international standards (Ministry of Home Affairs and Internal Security 2025). This law, if well drafted and implemented, will set Zambia apart from other countries that lack a dedicated instrument to enable video surveillance while providing human rights protections and oversight.

Test 2. Do the laws define a legitimate aim for public space surveillance?

The Bill proposes the creation of a centralised national public security command centre, complete with an implementing committee. The committee will comprise of officers appointed and supervised by the minister of home affairs and internal security, from the deputy inspector general of police, who shall be the chairperson, a representative of the Ministry of Home Affairs and Internal Security, Zambia Correctional Service, Immigration Department, Drug Enforcement Commission, Attorney General, Zambia Information and Communications Technology Authority, Zambia Security Intelligence Service, and the Anti-Terror Unit. This composition brings into question the agency's independence and increases the risk of political interference in its operations. A local civil society, the Movement for Good Governance, has criticised the proposed law for granting sweeping powers to government agencies to install, monitor, and access CCTV footage across Zambia, with little to no independent oversight (Zambian Monitor 2025). Additionally, the law does not provide sufficient judicial safeguards for command centre officers to adhere to while enforcing the law. For instance, an officer at any reasonable time and without prior notice, may enter and inspect CCTV or mobile CCTV footage, documents, and equipment at any premises that they have reasonable grounds to believe is used for the commission of an offence, and upon completion of the inspection activity, the officer is required to submit a written report to the executive director of the command centre. This grants sweeping powers to the command centre officers to arbitrarily access data

and potentially install unwanted monitoring software. The proposed law loosely defines what data can be collected and severely fails to address what purpose actors such as the command centre officers can use it for.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

The Closed-Circuit Television Public Protection Bill 2025 provides for the regulation of CCTV systems and the protection of data collected through CCTV and mobile CCTV, including public security surveillance for the protection of members of the public. However, it does not provide a well-defined legitimate aim for public surveillance. The proposed law further provides for mandatory licensing for anyone operating a CCTV system, within or outside their home or business premises. The mandatory licensing of private CCTV systems in homes further raises questions about what should be considered a public space or what are public interests in a private space. The Bill further provides for government agents to inspect premises without judicial oversight and gives them powers to arrest individuals suspected of non-compliance, thereby making room for arbitrary arrests of CCTV users and breaching constitutional privacy rights. The Bill prescribes harsh penalties for violators, and vast powers for government agents to search, view, and collect footage and data without judicial oversight. There are no limits as to how long collected footage and data that is part of ongoing investigations or legal proceedings can be kept.

On the other hand, the recently updated Cybersecurity Act 2025 has provisions that may facilitate surveillance. Notably, section 39 of the Act requires electronic communications service providers to install systems that can facilitate real-time interception of communications, including individuals' private communications, thereby giving room to misuse and surveillance of private communications by the government, unscrupulous individuals, and other unauthorised persons (CIPESA 2025). Similarly, the Act establishes the Zambia Cyber Security Agency, which has been moved from an independent state agency and brought under the supervision of the Office of the President, thereby affecting the agency's independence and increasing the risk of political interference in its operations. In addition, section 21 establishes the central monitoring and coordination centre for authorised interceptions, including forwarding of all intercepted communication and call-related information of targets.

Locally, the unregulated safe city developments have brought into question the issue of citizens' civil liberties and sparked digital rights concerns among human rights defenders and the public, specifically concerns about the potential use of the technology to spy on citizens and persons of interest by state actors, thereby infringing on their right to privacy (Nkhowani

2021). Similarly, legislators have cautioned against the selective use and application of surveillance cameras to fight crime, stating that citizens had yet to see the benefits of implementing the project (National Assembly of Zambia 2024). Chiumbu further notes that the lack of public sensitisation and consultation on the initiative leaves citizens unaware of the rights violation, and as they cannot hold authorities accountable (Chiumbu 2021).

Test 4. Does the law define mechanisms for citizen redress and accountability?

In its project report, the Zambia Law Development Commission had recommended that the national command centre committee require representation from the government and the Human Rights Commission, to provide oversight and represent neutral institutions that cannot sit on the committee due to the sensitive nature of national security matters. Other proposals included having civil society and other neutral stakeholders sit on non-sensitive sub-committees (Zambia Law Development Commission 2022). The creation of oversight subcommittees would ensure accountability and redress ensuring that the activities and decisions carried out by the implementing committee are as intended in the legislation and in alignment with Zambia's commitments to human rights law. However, the proposed law fails to provide adequate redress and oversight mechanisms. Other concerns include the appointment of the command centre's executive director by the president; this move severely undermines the command centre's independence from political interference in discharging its duties and scrutiny, and positions the centre as a vehicle for facilitating targeted surveillance on political opponents, activists, and individuals.

As Zambia progresses with the safe city implementation, the lack of transparency and public participation and awareness in the development of CCTV legislation and technology deployment, including robust judicial oversight and accountability mechanisms, raises alarms about privacy violations, which would contravene Zambia's constitution and international human rights instruments.

6. Digital rights

The implementation of the Zambia safe city project marks a significant shift to Zambia's approach to public security and surveillance with the use of advanced surveillance technologies. It can be argued that these innovations aim to reduce crime and enhance public safety and efficiency in responding to security incidents; on the other hand, they raise critical concerns about digital rights and, in particular, threaten the right to privacy, freedom of movement, and association, which in turn infringe on freedom of expression and protection from arbitrary state actions. This section examines the project's implications for digital rights through four lenses: whether surveillance is necessary and proportionate, the adequacy of access control measures, the existence of independent oversight mechanisms, and whether a human rights impact assessment was conducted prior to implementation. It integrates the project's technical, financial, and legal details, as well as concerns about foreign involvement by ZTE and Huawei, to provide a comprehensive analysis aligned with Zambia's sociopolitical dynamics.

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

The safe city project's surveillance, involving mass data collection through facial recognition and biometric systems, raises questions about whether the resulting privacy violations are necessary and proportionate to achieve its legitimate aims of crime prevention and public safety, as articulated in the Closed-Circuit Television Public Protection Bill 2025 (Ministry of Home Affairs and Internal Security 2023). While these aims align with Zambia's need to address urban crime amid rapid urbanisation, the project's scope – 1,600 cameras across public spaces with real-time monitoring and indefinite data retention – appears disproportionate. Privacy International (2020) emphasises that surveillance must be strictly necessary, yet no publicly available evidence suggests a human rights impact assessment (HRIA) was conducted to justify this scale, a gap consistent with the legal/regulatory analysis. The potential for facial recognition to misidentify individuals, particularly those with darker skin tones, risks unjust outcomes, as documented globally (Buolamwini and Gebru 2018). Historical precedents, such as the Special Branch's targeting of dissidents during the one-party state era, suggest surveillance has often served control rather than public safety (Chinyere and Hamauswa 2016). For Zambians, the blanket surveillance of daily activities – shopping in markets or travelling on highways – exceeds what is required, creating a chilling effect on freedom of expression, as seen during the 2016 elections when social media monitoring suppressed dissent (Freedom House

2017). Without clear limits on data use or judicial oversight, the project's privacy intrusions are neither necessary nor proportionate, undermining constitutional protections under article 17 (Sakala and Mulenga 2024).

Test 6. Does the law control who has access to surveillance data?

Protections to limit access to the digital trail of personally identifiable information (PII) generated by the safe city project are woefully inadequate, posing significant risks to privacy and security. The Data Protection Act of 2021 provides general principles for data processing but is silent on video surveillance, facial recognition, and biometric data, leaving no specific restrictions on who can access CCTV footage or linked biometric IDs (Sakala and Mulenga 2024). The Closed-Circuit Television Public Protection Bill 2025 grants broad access to government entities, including the Zambia Police Service, Immigration Department, and Drug Enforcement Commission, without requiring judicial authorisation (Zambia Monitor 2025). This lack of access control is particularly concerning given the involvement of foreign firms like ZTE and Huawei, which retain technical administration over IT infrastructure, including the national data centre (Office of the Auditor General 2024). Arcesati (2021) notes that Chinese firms may be obligated to share data with their government, a risk exemplified by the 2018 African Union data breach, where surveillance data was transferred to Chinese servers (Kuo 2018). For Zambian citizens, this means sensitive PII – captured through facial recognition or biometric IDs – could be accessed by state or private actors, including foreign entities, without accountability. The historical use of surveillance to target journalists, as in the 2015 wiretapping case, underscores the potential for misuse, leaving citizens vulnerable to privacy violations (Lusaka Voice 2015).

Test 7. Does the law create an independent oversight body?

The safe city project lacks an adequately funded and fully independent oversight body to ensure that public space surveillance complies with legal and human rights standards. The Closed-Circuit Television Public Protection Bill 2025 establishes a national public security command centre committee, but its composition – dominated by government appointees under the minister of home affairs and internal security, including the Zambia Police Service and Anti-Terror Unit – lacks independence and risks political interference (Zambia Monitor 2025). The Zambia Law Development Commission (2022) recommended including neutral stakeholders like the Human Rights Commission, but these were excluded, mirroring the government-controlled oversight structures of the colonial and post-independence eras (Chinyere and Hamauswa 2016). The Cybersecurity Act of 2025 further undermines oversight by placing

the Zambia Cyber Security Agency under the Office of the President, eroding its impartiality (CIPESA 2025). Without an independent body, as advocated by Privacy International (2020), there is no assurance that surveillance is conducted lawfully or proportionately. For Zambians, this lack of oversight means there is no mechanism to challenge abuses, such as unauthorised data access or profiling, perpetuating a legacy of unchecked state power (Phiri and Abrahams 2025).

Test 8. Has a human rights impact assessment been carried out?

There is no publicly available evidence that a HRIA was conducted prior to the safe city project's implementation, a critical oversight given its potential to infringe on digital rights. The project's procurement under the State Security Act in 2015, following corruption allegations in 2013, prioritised secrecy over transparency, as noted in the case study (Tumfweko News 2014; Office of the Auditor General 2024). Privacy International (2020) emphasises that HRIAs are essential to evaluate risks to privacy, freedom of expression, and non-discrimination, yet the absence of such an assessment leaves unaddressed the project's impact on vulnerable groups, such as activists or minority communities. The integration of facial recognition, known for biases against darker skin tones, heightens these risks, as does the link to Zambia's national digital ID framework, which could enable mass profiling (Buolamwini and Gebru 2018; Gagliardone 2020). Historical patterns of surveillance targeting dissenters, such as during the 2016 elections, suggest that without an HRIA, the project may replicate past abuses (Freedom House 2017). For citizens, this gap means their rights are at risk without prior evaluation, undermining trust in the project's governance.

7. Recommendations

To safeguard digital rights, Zambia must adopt a rights-based approach to surveillance governance. First, a mandatory HRIA should be conducted to assess the project's impact on privacy and other rights (Privacy International 2020). Second, the Data Protection Act of 2021 should be amended to include specific provisions for video surveillance, addressing facial recognition and retention limits. Third, the Closed-Circuit Television Public Protection Bill 2025 should mandate judicial oversight, especially for implementing activities of the command centre officers, limit data access to authorised actors, and establish clear retention periods. Fourth, an independent oversight body, incorporating civil society and the Human Rights Commission, should be established with adequate funding to ensure accountability. Finally, public awareness campaigns and consultations should empower citizens to understand and assert their digital rights, aligning with Zambia's democratic reforms (US Department of State 2024).

8. Conclusion

Zambia's safe city project, while aimed at enhancing public security through AI-driven surveillance, perpetuates a historical legacy of state control that threatens digital rights and civil liberties. From colonial informant networks to post-independence monitoring and modern surveillance, as detailed in the historical and case study sections, the project risks undermining privacy, freedom of expression, and due process. The absence of a robust legal framework, inadequate access controls, lack of independent oversight, and failure to conduct a human rights impact assessment amplify these risks, as explored in the legal and digital rights analyses. Foreign involvement by ZTE and Huawei raises concerns about digital sovereignty, echoing historical external influences. To address these challenges, Zambia must enact specific legislation, mandate judicial oversight, establish independent oversight mechanisms, and prioritise public engagement. By aligning the project with constitutional protections under article 17 and international human rights standards, Zambia can ensure that security advancements enhance, rather than erode, citizens' fundamental freedoms.

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Smart city surveillance in Zimbabwe

Thobekile Matimbe

1. Introduction

This report provides the most comprehensive account to date of smart city public space surveillance in Zimbabwe. The report explores how public space surveillance and smart city initiatives can affect human rights in Zimbabwe. Surveillance technology has the potential to advance development and public safety. However, it also has the potential to violate human rights. The surveillance of public spaces in Zimbabwe is not new and dates back to precolonial times. Adopting a deep dive into the deployment of public space surveillance, this report leverages desk research to unpack the extent of digital surveillance deployment in public spaces and its incidence. Zimbabwe was a testbed for facial recognition technology in Africa and an early digital public space surveillance adopter. The report provides a detailed, descriptive account of the smart city project's key dates and core elements. An account is provided of the key actors in government, as well as the private sector companies supplying surveillance tools and the main contract awarded. Analysis of the legal framework for smart city surveillance and its implications for digital rights is provided before the conclusion, offering pertinent recommendations based on a human rights-based approach.

2. Background

Zimbabwe is a Southern African country with an estimated population of 17,000,088 people in 2024 (Worldometer 2025). The country has 27 main cities with Harare as the capital city, Bulawayo as the second largest and Mutare as the third largest of 10 provinces (Government of Zimbabwe n.d.). Zimbabwe gained independence from British colonial rule on 18 April 1980 and was formerly known as Rhodesia in colonial times. It was colonised by the British in 1890, during which time, urban segregation was established by Rhodesian policies and society was divided based on race. The British settlers occupied specific urban settings to the exclusion of black Africans. To enforce segregationist policies, the movement of black Africans and anticolonial activists was subject to surveillance by the police Special Branch, which was retained after Zimbabwe gained independence in 1980 (*Zim Eye* 2020).

The Land Apportionment Act of 1930 was a segregationist law that brought about four distinct areas, namely the native purchase or African areas, the European urban areas, the forest areas, and unassigned land, with Africans treated as travellers and passers-by in European urban areas. This naturalised the surveillance of black Africans in European urban areas. A shift in policy in 1969 saw the Land Tenure Act (55 of 1969) passed. It birthed the creation of African townships by reinforcing segregation of races to their own areas (Colonia Relic n.d.).

The Municipal Amendment Act of 1964 occasioned the discrimination in public spaces. A black African describes this as 'petty' discrimination, highlighting that facilities were not open for public use (International Commission of Jurists 1976). Still, where there were no posters in place, there would be an employed person in designated areas whose job was to surveil movements and keep the Africans away. Surveillance of public spaces was present during this time, occasioned by legislation and human supervision, targeting black Africans based on their race. This aligns with perceptions that 'Blackness has been at the epicenter of developing strategies, technologies, and infrastructures of surveillance' in colonial times (Browne 2015).

Public space surveillance in Zimbabwe continued after independence, especially during protests, elections and political infighting. Surveillance is recorded across different periods in President Robert Gabriel Mugabe's era. Video footage has been recorded, presented in court, or discussed in allegations against the late (then) leader Morgan Tsvangirai of the Movement for Democratic Change (Voice of America 2004) and in accusations against former Vice President Joyce Mujuru and others concerning threats against Robert Gabriel Mugabe respectively. In these cases, Military Intelligence and

the Central Intelligence Organisation are implicated in using both covert and overt surveillance, and using gendered surveillance against former Vice President Joyce Mujuru from 2001–2014 (Tendi n.d.). Surveillance was noted in Zimbabwe's 2013 presidential elections, where heightened political activity was followed by heavy state surveillance (Human Rights Watch 2013).

Smart city initiatives are also gaining traction in Zimbabwe. This aligns with the African Union Digital Transformation Strategy for Africa (African Union, 2020), which includes digital infrastructure development as one of its key pillars. Zimbabwe's National Development Strategy for 2021–2025 (NDS1 2021) became operational from January 2021 to December 2025 and provided the first formal launchpad of several smart city initiatives. Zimbabwe's interest in smart city initiatives that address crime and promote development has been mainly noted since 2014, with most surveillance cameras installed in major cities since 2018. Smart city initiatives in Zimbabwe are primarily in the form of city council projects installing technology to enhance their functions or through development projects that leverage technology in their public spaces and functions. These initiatives are powered by features that have facial recognition capabilities and enhanced by artificial intelligence (Travers n.d.).

The first comprehensive smart city development project, which is distinct from the deployment of other smart city initiatives such as the deployment of public space surveillance technologies, was the Zim Cyber City Project, a real estate project launched in 2023 (Smart Cities World 2023). Guided by the Smart Zimbabwe 2030 Master Plan (Smart Zimbabwe n.d.), Zimbabwe seeks to leverage ICTs to launch Zimbabwe to the status of an upper-middle-income economy by 2030. Other development projects have been launched by private actors, such as the Pomona development project, launched in July 2023 in Mutare (Westprop Holdings Limited 2023).

The Smart Zimbabwe 2030 Master Plan, as a guiding framework, envisages smart cities as the impetus for development, promoting their uptake. Zimbabwe's smart city initiatives are aligned with the Zimbabwe National Information and Communication Technology Policy (2022–2027) that articulates key priorities, among others, infrastructure development and management and a regulatory framework on ICTs and institutional mechanisms. Smart city projects are being implemented against a legislative backdrop of the Cyber and Data Protection Act 2021 (CDPA 2021), which presents data protection and cybersecurity provisions. Zimbabwe has the constitution of 2013, the supreme law guiding the protection of human rights.

The government's inclination to roll out public space technologies as reported since 2014 (TechZim 2014) is evident in government public pronouncements, such as an announcement in October 2023 by the

minister of information communication technologies, postal, and courier services, echoing commitment to the Smart Zimbabwe 2030 Master Plan. City councils lay out the infrastructure in partnership with private actors, with standard features that include surveillance cameras in public spaces, usually mounted at traffic lights and city lighting systems.

3. Case study description

While smart city initiatives advance development and public space surveillance may have benefits of addressing crime, the surveillance threats they pose to individuals are of concern to Zimbabwean citizens, who expressed a lack of trust in these initiatives (Paradigm Initiative 2024). Interventions have been deployed in urban settings in Zimbabwe, focusing on Harare, Bulawayo, Mutare, and Victoria Falls as major cities where projects exist, with the first smart city initiative in public spaces deployed in 2014 according to accessible reports, in Harare. Implementing innovative city initiatives in Zimbabwe poses threats to the Zimbabwean citizenry, as seen over the past five years, especially with a lack of transparency in the procurement processes. Tech-enabled surveillance in public spaces in Zimbabwe has occurred mainly since 2014, over a series of fragmented implementations in different cities. The chronology of installations cannot be located in one source of information, with various details available, especially from media sources. There is no readily available information on the number of surveillance cameras deployed in all recorded cases. This demonstrates a lack of proactive disclosure by government actors deploying surveillance in public spaces.

The first CCTV surveillance cameras were reportedly installed in Harare in 2014 by the City of Harare at the intersection of Julius Nyerere Way and Jason Moyo Avenue (*Herald Online* 2014). The Bulawayo City Council installed its first city traffic cameras in 2018, along Leopold Takawira Avenue and Samuel Parirenyatwa Street in Bulawayo, with the Zimbabwe Republic Police (ZRP) (*Herald Online* 2018). There is no readily accessible information on the cost of the project or the number of cameras installed.

In 2018, Cloudwalk, a Chinese company, announced that it would export facial recognition technologies to Zimbabwe as part of the Belt and Road Initiative, cited as China's pioneer deployment of AI technology in Africa. Cloudwalk reportedly signed a strategic cooperation framework agreement with the Zimbabwean government for the facial recognition project (Security Vision 2024). The project objectives were to assist the government in establishing a smart financial service network and provide intelligent security-surveillance in public spaces, including airports, railway stations, and bus stations. This project has been cited as deploying three-dimensional light facial recognition technologies from CloudWalk to develop a large dataset and train an AI system with datasets of darker skin tones (Security Vision 2024).

The government reportedly announced that President Emmerson Mnangagwa, Ambassador Huang Ping, and Zongnian Chen, chair of the board of Chinese company Hikvision, signed a memorandum of understanding in 2018 to foster cooperation between Hikvision and Zimbabwe with the objectives of growing the Zimbabwean ICT sector with software integrated into the facial recognition hardware produced locally by Zimbabweans for security, disaster response, and traffic control (TechZim 2018).

The government of Zimbabwe, through media reports, also announced that it was working with Hikvision to pilot an innovative city project in Mutare with the Chinese government having a controlling stake in Hikvision with surveillance technologies (The Herald 2019). The partnership was deemed a donation by the former Zimbabwean ambassador to China, Christopher Mutsvangwa, in 2019, highlighting that Zimbabwe had approached China to support the AI revolution in Zimbabwe (The Herald 2019). Reportedly, in March 2020, Huawei received US\$20 million to start installing a grid of public surveillance cameras as part of a five-year smart city project of US\$100 million, citing Hikvision as the supplier of facial recognition software (Privacy International 2021).

A media report indicates that the government also installed an undisclosed number of surveillance cameras in the Harare central business district in 2021, including at Market Square (*Zim News* 2021). In August 2022, Zimbabwe's tourist town, Victoria Falls, began installing smart city lights with an initial project target of 100 streetlights to move towards smart city status. The streetlights have cameras that have surveillance capabilities. A Chinese company, Satewave Technologies (Satewave Technologies n.d.), implemented the project in Harare and Victoria Falls (Paradigm Initiative 2024). Smart street lighting initiatives in Victoria Falls have been rolled out since 2022, aimed at addressing human–animal conflict. These projects were still pending in 2024, with only Melfort having a disclosed budget of US\$4.5 billion. All were promoted by the Ministry of National Housing and Social Amenities (Zimbabwe Investment Development Agency 2024) and were not yet completed.

In 2022, the City of Bulawayo launched the Parking Management System (2021–2027) with an awarded contract to Tendy Three Investments, a South African company. The initiative involved placing parking terminals and installing CCTV cameras with software designed to count vehicles, increase visibility, and enhance security in Bulawayo, suggesting smart capabilities (CITE 2024). Information on the capacity of surveillance technologies deployed is not readily accessible from the City of Bulawayo, showcasing the absence of proactive disclosures of information. The company highlighted

that when the police require any information, they can access the footage and download it for their purposes. There was no indication of any specific adherence to human rights and ensuring that any requests for that footage would be in line with due process. The project has four main objectives: increased revenue for the City of Bulawayo, employment creation for the people of Bulawayo, order and sanity in the central business district (CBD), and a sense of security in the city centre by providing security to the vehicles of the motorists through CCTV and marshals (City of Bulawayo 2021).

The Zimbabwe Cybercity Project, a real estate modern city in Mount Hampden, Harare, is a development project clothed in surveillance features of CCTV footage in public spaces (Mulk Investments n.d.). It had a start date of December 2022 and commenced in 2023. This project is a state-of-the-art cybercity building initiative in Harare owned by Mulk Investments with the government's support. It is estimated at US\$500 million and set on a 2.5 million square foot space. This project is still in progress and the project is reported as falling under the patronage of President Emmerson Mnangagwa, who granted Mulk International an exclusive license to establish a blockchain and digital assets special economic zone and has reportedly mentioned that it will install a surveillance feature linked to law enforcement (*Reuters* 2023). It is feared that the project will unveil state surveillance in the form of facial recognition technologies that rely on AI to match live images of people in public spaces. This project is on land owned by CFI Holdings Limited and the building is financed by the Chinese government (*Newshawks* 2023).

While in August 2024, Tendy Three Investments launched a US\$2.2 million project with the installation of cameras in Bulawayo, in October 2024, a smart surveillance system project was also launched to fight crime in the city. The surveillance system is a collaborative initiative between the Bulawayo Business Against Crime Forum of Zimbabwe (BACFOZ), Safeguard (a security company) and the ZRP. There is no available information on the nature of the collaboration and the source of the technologies, as well as the project budget.

Table 23: Mapping public space initiatives in Zimbabwe

Public space surveillance initiatives	Dates <i>Note: Only the year is provided where the month is unknown</i>	Overview of initiative	City/town	Amount	Company
First city cameras in Harare	September 2014	The first city CCTV with surveillance cameras installed	Harare	Undisclosed	Undisclosed
Purchase of a large-scale facial recognition programme	March 2018	A proposed large-scale facial recognition programme in Zimbabwe	Nationwide	US\$2.2 million	Cloudwalk Technology
Facial recognition technology deployment	June 2018	For border posts, state entry points, and airports	Mutare	Undisclosed	Hikvision (Chinese)
Surveillance camera installation (ZimTech Review 2018)	October 2018	Surveillance cameras at traffic lights in the central business district	Harare	US\$2 million	Undisclosed
Surveillance cameras in central business district (ZimTech Review 2020)	March 2020	Surveillance cameras at traffic lights in the central business district	Harare	Not reported	Not reported
Bulawayo traffic cameras	October 2018	The first traffic cameras were installed in Bulawayo for traffic control	Bulawayo	Undisclosed	Undisclosed
Harare city centre traffic lights	2021	Installation of traffic cameras to detect crime	Harare	Undisclosed	Undisclosed
Bulawayo parking management system (PMS)	February 2022	Purpose to control parking	Bulawayo	US\$2.5 million	Tendy Three Investments (South African company)
Street lighting project with cameras	August 2022	Streetlighting	Victoria Falls and Harare	Unknown	Satewave Technologies
Smart parking project with cameras	August 2024	Installation of cameras and software designed to monitor parking activities	Bulawayo	US\$2.2 million	Tendy Three Investments
surveillance system project	October 2024	Launched to curb crime and reduce vehicle-related crimes	Bulawayo	Undisclosed	Undisclosed

Source: Author's own.

4. Key actors

Civil society organisations and journalists have raised concerns about the safety of citizens when public spaces are under surveillance (Paradigm Initiative 2024). Public space surveillance in Zimbabwe is orchestrated by various actors and is conducted in both overt and covert forms. The key actors promoting public space surveillance are state security agencies, city councils, private companies and government funders. The Zimbabwe Defence Forces (ZDF) has been a major player in deploying surveillance tools in public spaces in the form of surveillance cameras. The ZDF is implicated in cases of surveillance of political activists and how public space surveillance technologies can target individuals prone to it, such as political actors and human rights defenders (*Newsday* 2020). Central intelligence officers, members of the Central Intelligence Organisation, deemed the most powerful arm of the security apparatus, are feared for their overt and covert surveillance tactics (Hanson 2008). They reportedly conduct surveillance of citizens using CCTV in public spaces and keep videos of citizens, especially those of prominence such as human rights defenders and journalists, tracking their movements through surveillance cameras and other forms of covert surveillance (*Zim eye* 2020).

The ZDF has been amassing capabilities in public space and cybercommunication surveillance, supported by Chinese government loans. Chinese companies like Huawei, Hikvision, and CloudWalk are key suppliers of public space surveillance technologies in Zimbabwe (Africa Defense Forum 2023), and Iran and Russia are also new entrants in the supplier chain (Munoyariwa 2021). Munoyariwa highlights that the surveillance capabilities owned by the ZDF include facial recognition devices from CloudWalk, CCTV cameras from Hikvision, and international mobile subscriber identity catchers from Iran used to track political opponents (Munoyariwa 2021). The ZDF was implicated in installing facial recognition technology in Harare in 2018, with cameras installed in Harare's CBD area at the Parliament building areas and opposite Africa Unity Square, where protests usually occur (Munoyariwa 2021). Munoyariwa references that military surveillance in Zimbabwe is domiciled in numerous ZDF units such as the Military Intelligence, the Special Air Service and the Signals Unit, which procure a plethora of surveillance equipment empowering them to surveil public spaces and communications (Munoyariwa 2021).

City councils established by the Urban Councils Act, 1995 are responsible for traffic management and street lighting. They deploy public space surveillance in various main cities to monitor traffic flows. In Harare, for

instance, cameras have been deployed at major intersections in the city. Two leading actors are responsible for installing the cameras: the city council and the ZRP (Harare City 2023). The ZRP has issued warnings that surveillance cameras in central business districts will catch traffic offenders, indicating a camera system that takes videos of offenders from city intersections. Smart traffic cameras possess ANPR capabilities, also known as licence plate recognition coupled with advanced deep-learning algorithms (Hikvision n.d.). However, there is no readily available information on the actual capabilities and data centre management of technologies deployed by city councils.

In Bulawayo, the ZRP collaborates with the City of Bulawayo to deploy CCTV cameras to monitor individuals. All installed cameras are linked to crime prevention, with the ZRP cited as a partner in using cameras to address traffic offences or crime. On 1 October 2024, the City of Bulawayo launched a CCTV surveillance system in the city with the aim of enhancing safety and traffic monitoring, enabling the two actors to monitor streets for criminal activities and regulate traffic at traffic lights (Bulawayo 24 2024).

Distrust of the purposes for which these actors deploy surveillance in public spaces stems from records of misuse of surveillance for political gains and interests. In 2007, for instance, the Central Intelligence Organisation was implicated in exposing a sex scandal of a critic of the late former President Robert Mugabe, Archbishop Pius Ncube through a surveillance camera (*The Guardian* 2007), who later resigned from his role following the surveillance incident and adultery charges against him (Africa Faith and Justice Network 2007). In response to an announcement of the Zim Cyber City project, Ndabaningi Nick Mangwana, a spokesperson, announced that CCTV cameras will be fitted on all public infrastructure, including on traffic lights, building facades, and in underground rail to assist law enforcement (Mangwana 2022). In another post, he expresses that in 2025, all sorts of cameras will be deployed for law enforcement, posing concerns over the exact purpose of procuring surveillance tools in public spaces (Mangwana 2025). Ndabaningi Nick Mangwana is the permanent secretary for the Ministry of Information, Publicity and Broadcasting Services and has reportedly made posts on X suggesting that surveillance cameras are for catching infractions, without stating what infractions are referred to, posing concerns for human rights defenders and political actors.

Civil society organisations are key stakeholders in holding the government accountable and calling for transparency. The Media Institute of Southern Africa has raised privacy concerns over using Cloudwalk's facial recognition technologies, which are being deployed in Zimbabwe, to train the Chinese company's AI systems with African facial features (MISA Zimbabwe 2018). A report was submitted to the United Nations concerning surveillance in public

spaces, calling for the government of Zimbabwe to refrain from violating the right to privacy through surveillance (MISA Zimbabwe 2019). Paradigm Initiative has documented cases of mistrust of public space surveillance technologies, showing how human rights defenders and political actors are untrusting of public space surveillance (Paradigm Initiative 2024).

Section 37(2) of the CDPA repealed section 4 of the Interception of Communications Act (ICA) that provided for a Monitoring of Interception of Communications Centre and replaced it with a Cyber Security and Monitoring of Interception of Communications Centre (Cyber Monitoring Centre) whose responsibility is to provide technical advice on cybersecurity to authorised persons and service providers. This entity is housed in the president’s office, as prescribed in section 37 of the CDPA, amending the ICA. Powers to intercept communications in terms of the ICA are vested in four foremost authorities, mainly the chief of defence intelligence or his or her nominee, the director-general of the president’s department responsible for national security or his or her nominee, the commissioner of the ZRP or his or her nominee, or the commissioner-general of the Zimbabwe Revenue Authority or his or her nominee. While on digital surveillance, this law shows the government’s interest in ensuring state security has oversight over surveillance. The Postal and Telecommunications Regulatory Authority of Zimbabwe (POTRAZ) is the data protection authority with the mandate to ensure that there is lawful processing of data in Zimbabwe and the protection of personal data.

Table 24: Key actors involved in public space surveillance in Zimbabwe

Main Actors	Roles	Details
State security agencies • Zimbabwe Republic Police • Zimbabwe National Army • Central Intelligence Organisation	• Order surveillance or request surveillance data	• Law enforcement • General surveillance requirements
City councils (reporting major city interventions) • Harare City Council • Bulawayo City Council • Mutare • Victoria Falls	• Deploy smart city initiatives	• Street lighting • Traffic control • Parking system

Main Actors	Roles	Details
Private companies • Mulk Investments • Huawei • HikVision • CloudWalk	• Manage development project • Supply surveillance technologies	• Business requirements
Government funders/actors • China • Iran • Russia	• Finance public surveillance initiatives • Partnership agreements	• Sponsorship • Partnership

Source: Author’s own.

5. Regulatory landscape

Test 1. Are there validly enacted laws governing public space surveillance?

The Constitution of 2013 is the supreme law guiding the protection of human rights. Section 57 provides as follows:

Every person has the right to privacy, which includes the right not to have:

- Their home, premises or property entered without their permission;
- Their person, home, premises or property searched;
- Their possessions seized;
- The privacy of their communications infringed; or
- Their health condition disclosed.

This right to privacy is protected as well and any departure from it is in terms of section 87(2), which provides that fundamental rights and freedoms may be limited in terms of a law of general application and to the extent that the limitation is fair, reasonable, necessary, and justifiable in a democratic society based on openness, justice, human dignity, equality, and freedom, taking into account all relevant factors listed in that part. The starting point is legality, where an existing law should exist.

There is no specific law in Zimbabwe that gives a general authorisation for public space surveillance, like in cases of digital surveillance enabled by the Interception of Communications Act (ICA) 2007 or in cases of cybercrimes provided for by the CDPA. However, the president may effect any form of public surveillance through the Presidential Powers (Temporary Measures) Act 1986 a law that empowers the president, in terms of section 2, to make regulations in cases where it is considered necessary to do so. This law has been criticised as it disregards the separation of powers and gives the president of the country law-making powers. This law can provide for the enactment of regulations allowing public space surveillance, but it has not been used.

Another instance where public space surveillance may be ordered is in terms of the Emergency Powers Act 2007, which empowers the president with broad powers to pass regulations in a state of emergency to address public safety. Regarding protests, the Maintenance of Peace and Order Act (MOPO) 2019 provides for actions that the police can take in maintaining public order; the powers do not extend to deploying surveillance tools to monitor protests.

In the ICA, clearly, the interception of communications is enabled by the unfettered discretion of the minister of transport and communications or any other minister to whom the president may, from time to time, assign the administration of the Act. With public space surveillance, there is a generalised acceptance that, in law enforcement and addressing national security, it is permissible. The ZRP extend their law enforcement function to deploy cameras generally without legal backing for a broader mandate to deploy surveillance capacities in public spaces. They are therefore not compliant with the human rights test of limiting civil and political rights, mainly of legality, necessity, and proportionality (Privacy International n.d.). In the absence of a law, the test of legality is not passed. However, national security can empower the ZRP to act on any orders as stipulated in the laws mentioned below.

The CDPA 2021 has the objective to increase cybersecurity in order to build confidence and trust in the secure use of information and communication technologies by data controllers, their representatives and data subjects. It prescribes what should happen with regard to data collected using digital technologies. For purposes of public space surveillance, the CDPA prescribes in section 9(1) that the data controller uses data collected only for purposes for which the data is collected and not for any unlawful means deemed illegitimate. This law governs how data collected should be processed, offering some safeguards on data collected through surveillance technologies. However, biometric data is sensitive data and can be processed in furtherance of national security in section 11(4) of the CDPA.

Test 2. Do the laws define a legitimate aim for public space surveillance?

While there is no specific law authorising public space surveillance, other laws that refer to the interception of communications, cyber, and data protection mentioned in this part refer to national security as key component, the legitimate aim pursued by the state. The legitimate aim addressed in the ICA is mainly national security, preventing and detecting a serious offence or the interests of the economic well-being of Zimbabwe. The CDPA provides for several legitimate purposes for processing personal data including if the processing is necessary to comply with national security laws, is necessary for the exercise or defence of legal claims.

The Presidential Powers (Temporary Measures) Act 1986 provides for the interests of defence, public safety, and public order if there is no other law to deal with the situation while the Emergency Powers Act 2007 empowers the president to pass regulations in a state of emergency to address public safety, order, essential services, and even detention or deportation of individuals. The MOPO (2019) provides for actions that the police can take in maintaining public order, which is the legitimate purpose pursued by the Act.

Installing public space surveillance tools in the form of cameras and other tracking systems is not explicitly provided for in Zimbabwe. However, the Urban Councils Act 1995 gives city councils the mandate of traffic management and installing street lighting, among other functions. Through this function, city councils adopt a general authority to install cameras in city centres but have no mandate to prosecute crime in the cities. Cameras are installed in public spaces under the auspices of controlling traffic, with cameras installed at traffic lights and in street lighting systems, as seen in Bulawayo and Victoria Falls, respectively. There are no readily accessible by-laws that entrench specific authority to deploy surveillance cameras for the management of crime. There is no judicial oversight for deploying these surveillance technologies in public spaces, no accountability for using the cameras, and no transparency regarding managing data collected and how long it is stored.

Test 3. Does the law limit the legitimate actors and purpose of public space surveillance?

There are no particular laws governing public space surveillance. As such, guardrails are established within the Cyber and Data Protection Act, which provides for exceptional cases in pursuance of the legitimate purposes outlined in section 11(5) of the Act. The Cyber and Data Protection Authority (DPA) establishes the Postal and Telecommunications Regulatory Authority (POTRAZ) as the designated actor that regulates the way personal information may be processed, ensuring that is lawful. For purposes of addressing data breaches, the DPA must be notified within 24 hours (CDPA 2021).

6. Digital rights

The attendant impact of public space surveillance on human rights falls mainly on the right to privacy and freedom of movement. The African Commission on Human and Peoples' Rights stipulates in principle 40.1 that everyone has the right to privacy, including the confidentiality of their communications and the protection of their personal information (African Commission on Human and Peoples' Rights 2019). Data collected through public space surveillance must be protected and comply with data protection principles that call for lawful processing (CDPA 2021). The rights to privacy and freedom of movement are entrenched in the International Covenant on Civil and Political Rights, to which Zimbabwe is a state party, in sections 17 and 12, respectively. Zimbabwe acknowledges these rights as fundamental, as presented in the section on regulatory landscape.

Test 5. Is smart public space surveillance legal, necessary, and proportionate?

The general collection and use of surveillance footage from public spaces has an adverse impact on the right to privacy. In the absence of being proved necessary and proportionate, it violates the right to privacy (United Nations 1984). The right to privacy is imperilled by mass surveillance deployed through tech-enabled surveillance technologies. The recording, analysing, and retaining of facial images of individuals in public spaces without their consent constitutes interference with their right to privacy. It is regarded as occurring on a mass and indiscriminate scale (United Nations 2022). The likely abuse of this is the targeting of specific groups, which is the constant outcry of civil society actors and citizens in Zimbabwe, with existing fears of facial recognition technologies being used to target government critics (Paradigm Initiative 2024).

As required in international law, any violation of fundamental rights should address a concrete legitimate aim, such as averting a specific threat to public safety or security that is significant enough to outweigh their adverse human rights impacts. Such measures need to be limited, focused on specific locations and times, for instance, when evidence indicates that a crime is likely to occur or threats to public safety and security may emerge (United Nations Human Rights Council 2022). The United Nations has articulated that general monitoring of people in public spaces is almost invariably disproportionate (United Nations Human Rights Council 2022). Surveillance measures in public spaces should be narrowly targeted to minimise privacy violations. In view of this, a general monitoring of public spaces is disproportionate.

The United Nations Report of the Special Rapporteur on the promotion and protection of the right to freedom of opinion and expression on surveillance and human rights outlines critical considerations for surveillance and the prioritisation of human rights (United Nations 2019). It emphasises that the unlawful or arbitrary surveillance and/or interception of communications, the unlawful or arbitrary collection of personal data or unlawful or arbitrary hacking, and the unlawful or arbitrary use of biometric technologies, as highly intrusive acts, violate or abuse the right to privacy. Furthermore, they can interfere with other human rights, including the right to freedom of expression and to hold opinions without interference, and the right to freedom of peaceful assembly and association, and may contradict the tenets of a democratic society, including when undertaken extraterritorially or on a mass scale (UN Human Rights Council 2019).

Murray *et al.* (2023) warn that surveillance leads to a 'chilling effect': a well-known phenomenon that arises when individuals or groups modify their behaviour due to a fear of the consequences of being observed. Public space surveillance has the same chilling effect on freedom of movement and how people conduct themselves, affecting human rights. This is deemed surveillance-induced self-censorship (Murray *et al.* 2023). This chilling effect mimics the time of the pass laws, when freedom of movement was curtailed and citizens placed under surveillance to monitor their movements.

Test 6. Does the law control who has access to surveillance data?

The deployment of surveillance in Zimbabwe occurs without adequate safeguards in place to limit access to the digital trail of personally identifiable information created through the use of surveillance data by both state and private actors. Most surveillance technologies deployed by city councils are owned by private companies that collect and store data. Yet, there is no transparency from the companies on how they respond to government or third-party requests for data collected by their surveillance technologies (Paradigm Initiative 2024).

Test 7. Does the law create an independent oversight body?

In Zimbabwe, the lack of judicial oversight over surveillance practices creates a scenario where state security agents and city councils violate digital rights. Generally, communications surveillance is authorised by the executive function – the minister responsible for the Act, while judicial authority is not prioritised. In the same way, in public space surveillance, there is not prioritisation of judicial oversight. The government of Zimbabwe must ensure that public space surveillance measures comply with human rights standards, being provided by a law that pursues a legitimate purpose, ensuring that the means used are strictly necessary

and proportionate (African Commission on Human and Peoples' Rights 2024). This test is not currently passed in Zimbabwe, as public space surveillance is generally deployed with no judicial oversight to assess and establish the legitimacy of the deployment and other test elements.

7. Recommendations

The United Nations has articulated how governments often fail to adequately inform the public about their surveillance activities, and even where surveillance tools are initially rolled out for legitimate goals, they can easily be repurposed, often serving ends for which they were not originally intended (United Nations 2022). The human rights-based approach is a conceptual framework for the process of human development that is normatively based on international human rights standards and operationally directed to promoting and protecting human rights (UN Sustainable Development Group n.d.). Applying this approach to public space surveillance ensures human protections for individuals, especially those vulnerable to state surveillance, such as journalists and human rights defenders. Technology can lead to unintended harm if appropriate risk assessments are not carried out, including human rights impact assessments.

The United Nations has well established that as a primary step, governments deploying surveillance tools must ensure that they do so in accordance with a domestic legal framework that meets the standards required by international human rights law. This has also been articulated by Privacy International, which stipulates that use of a technology or system to deliver public functions can only ever be legitimate if it is 'legal', in the sense of falling under an appropriate legal framework that authorises such technology to be used for such purposes (Privacy International, n.d.). Surveillance should be authorised by a competent, independent, and impartial judicial body, with all appropriate limitations on time, manner, place, and scope of the surveillance (UN Human Rights Council 2019). Nevertheless, judicial authorisation of government use of surveillance technologies is necessary but insufficient, as procuring the technologies should also be transparent and subject to meaningful public oversight, consultation and control, more so in law enforcement.

The United Nations reports a case of the city of Oakland in California, which addresses three main steps in procuring surveillance technologies as follows (UN Human Rights Council 2019):

- An approval process, carried out by the relevant departments, that takes into account the state's human rights obligations.
- Public notice of such purchases through regular processes, and public consultations on issues such as the human rights implications of such purchases and whether the technologies at issue will be effective at achieving their intended purposes.

- Regular public reporting on such approvals, purchases and uses. This is also referenced as critical by various surveillance experts (Roberts *et al.* 2021).

The above steps demonstrate due diligence that can be taken by government actors, such as city councils, in promoting digital rights. This approach reflects a human rights-based approach that is critical to technology deployment. Zimbabwe can best adopt a compliance approach with international human rights standards before, during, and after deploying public space surveillance, applying the three-part test of legality, legitimate purpose, and necessity and proportionality.

Private companies supplying surveillance technologies have key responsibilities in terms of their business and human rights, such as promoting human rights and conducting due diligence, including deploying human rights impact assessments to mitigate human rights impacts through their products and services (United Nations 2012). They have a responsibility to demonstrate transparency on how they respond to requests for data where they supply surveillance-enabling technologies. There is no indication that these are being done by any of the actors responsible before deploying tech-enabled surveillance in public spaces in Zimbabwe or at any other point.

8. Conclusion

Public space surveillance in Zimbabwe is not a new phenomenon but an ongoing process dating as far back as colonial times, when pass laws in then Rhodesia imposed restrictions on the movements of black Africans. While in colonial times, technology was not available to enhance surveillance capabilities, the digital age has heightened the scope and depth of surveillance through facial recognition and AI properties. In colonial times, pass laws empowered law enforcement agents to monitor citizens' movements, similar to how state security agents currently rely on various public space projects to monitor the movements of individuals in public spaces through technology. This has a chilling effect on the ability of individuals to move freely while under surveillance, causing self-censorship and potential avoidance of public spaces for those aware of the deployment of various technologies such as surveillance cameras. Opaque procurement processes towards the purchase of surveillance-enabling technologies are concerning, calling for more transparency by state actors on selecting technology suppliers for public space surveillance. Ultimately, a human rights-based approach requires due diligence practices, ensuring that individuals are aware of the deployment of tech-enabled surveillance.

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Glossary of terms

State surveillance refers to state measures aimed at monitoring and supervising activities of the population (Ramasoota 2000). Government surveillance is often regarded as important for maintaining security and safety in both stable and unstable democracies, particularly where there are high levels of insecurity, such as terrorism, robberies, extra-judicial killings and general high crime rates (Brandon and Farrington 2009).

CCTV: Closed-circuit television is a monitoring and surveillance tool for security and other reasons such as monitoring of traffic and observing behaviour of dangerous patients and incarcerated inmates (Larsen 2011).

Mass surveillance involves the acquisition, processing, generation, analysis, use, retention, or storage of information about large numbers of people, without any regard to whether they are suspected of wrongdoing (Privacy International n.d.).

Public space surveillance refers to the monitoring of public streets, squares, and roads. Public space surveillance is a type of mass surveillance when it involves hundreds or thousands of cameras.

Smart city surveillance refers to the use of artificial intelligence to automate the capture of real-time video or digital sensor surveillance data. This study focuses on specific types of data capture: CCTV cameras using facial recognition technology and automatic number plate recognition (ANPR) technology centralised in dedicated 'data centres' that provide a command and control centre for police and security operatives.

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